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Master's Thesis

Asymmetric Influence and Interventions in Climate Change Belief Systems

Author:
Henry Zwart

Examiner:
Johan Bollen

Supervisor:
Vítor Vasconcelos

Assessor:
Sara Constantino

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Declaration of Authorship

I, Henry Zwart, declare that this thesis, entitled 'Asymmetric Influence and Interventions in Climate Change Belief Systems' and the work presented in it are my own. I confirm that:

- This work was done wholly or mainly while in candidature for a research degree at the University of Amsterdam.
- Where any part of this thesis has previously been submitted for a degree or any other qualification at this University or any other institution, this has been clearly stated.
- Where I have consulted the published work of others, this is always clearly attributed.
- Where I have quoted from the work of others, the source is always given. With the exception of such quotations, this thesis is entirely my own work.
- No part of the intellectual or written content of this thesis was created using generative AI.
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- Where the thesis is based on work done by myself jointly with others, I have made clear exactly what was done by others and what I have contributed myself.

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Abstract

Faculty of Science
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Master of Science in Computational Science

Asymmetric Influence and Interventions in Climate Change Belief Systems

by Henry ZWART

Increased recent polarisation on urgent issues such as climate change and public health has highlighted the critical role of individuals' beliefs in shaping collective attitudes and behaviour. A central challenge in the study of belief change and behavioural dynamics is understanding the structure and internal forces between beliefs. To this end, computational belief system models—the most prevalent of which treat belief systems as interrelated networks of beliefs—have emerged as a valuable lens for understanding belief dynamics in terms of interactions within a larger belief system. Most authors in this field acknowledge that some beliefs may display asymmetric influence, for instance, strongly influencing other beliefs while being relatively insensitive themselves. However, current research focuses almost exclusively on models which assume symmetric influence between beliefs. In theory, this assumption directly affects belief dynamics and common methods for assessing belief importance. In practice, a scarcity of theoretical and empirical studies on the existence and dynamic impacts of asymmetric influence leaves the actual impacts largely unknown.

Here, we address both theoretical and empirical shortcomings. In the first half of this study, we present the Kinetic Belief System as a computational model of belief system dynamics that supports both symmetric and asymmetric influence assumptions. We then derive a parameter estimation method that we use to calibrate the model to a longitudinal dataset of climate-related beliefs in the US ($n = 1693$). In the second half of the study, we use the calibrated models to investigate the prevalence of asymmetric influence among climate-related beliefs, and the resulting dynamic impacts for simulated belief-level interventions, compared with the corresponding symmetric-influence model. For the asymmetric model, we also investigate two secondary aims: examining the conditions under which individual-level interventions targeting attitudes on climate action are likely to be effective and comparing the structural features of conservative and liberal belief systems. Our findings support the existence of asymmetric influence among beliefs, demonstrating that these appear to be structured around a small set of highly 'influential' beliefs that are difficult to influence. We show that when such beliefs are present, models that assume symmetric influence may lead to spurious conclusions about the relative importance of beliefs and different conclusions regarding intervention strategy.

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Notation

- **Scalars** are denoted using regular weight lowercase variables, e.g., x.
- **Vectors** are denoted using boldface lowercase variables, e.g., $\boldsymbol{x}$.
- **Matrices** are denoted using boldface uppercase variables, e.g., $\boldsymbol{X}$.
- **Sets** are denotes using regular weight uppercase variables, e.g., X.
- **Random variables** are denoted using the uppercase version of the sample variable, or the script typeface when the sample variable is uppercase. e.g., for a scalar a or vector $\boldsymbol{a}$, we denote the random variable as A or $\boldsymbol{A}$, respectively, whereas for a set B we denote the random variable as $\mathcal{B}$.
- Unless otherwise stated, log refers to the natural logarithm.
- For $n \in \mathbb{N}$, $[n]$ denotes the **index set**, such that $[n] \coloneqq \{1, ..., n\}$.

We adopt specific notation for the context of belief systems and beliefs. In particular:

- **Models** are referred to using variants of the variable $\mathcal{M}$, e.g., $\mathcal{M}_{\text{asymmetric}}$.
- **Generic beliefs**, in the generic concept sense, as discussed at the start of Section 2.1, are denoted using the uppercase letter S, possibly with a subscript, e.g., S_i.
- **Particular beliefs**, in the specific instance sense, as discussed at the start of Section 2.1, are denoted using the lowercase letter s, possibly with a subscript, e.g., s_i.
- **Belief random variables** are denoted using the symbol σ, possibly with a subscript.
- **Belief data variables**, as measured by a survey (for instance), and corresponding to a measurements of particular belief state, are denoted using the symbol x.

For instance, the generic concept 'belief regarding the contents of the box' would be denoted as S. This takes on specific values, such as 'the box is empty', denoted as s, which are realisations of the random variable σ. If we were measure an individual's belief about the contents of the box, this data variable would be denoted as x.

Chapter 1

Introduction

Our subjective interpretations of the world, natural phenomena, and those around us depend on a collection of beliefs about how things are and how they work.[1] These beliefs are highly interdependent, related by logical and psychological associations in what are often referred to as belief systems (Converse, 1964; Fishbein & Ajzen, 1977). Beliefs are also subject to social dynamics (Galesic et al., 2021), as evidenced by observed geographic segregation of political attitudes in the United States (Brown & Enos, 2021). Moreover, an individual's beliefs have external consequences for their behavioural decisions (Granovetter, 1978) and, in turn, are influenced by these decisions (Fishbein & Ajzen, 1977; Olson & Stone, 2005), often through reinforcing feedback loops. Put plainly, the beliefs that allow us to make sense of the world internally inevitably shape our collective impact on it.

Distinct beliefs behave differently within a belief system. Some, such as political ideology (e.g., conservativism or liberalism), are highly stable (Green & Platzman, 2024; Kiley & Vaisey, 2020; Osborne & Sibley, 2020). Others appear to be more malleable (e.g., attitudes toward particular political candidates). In addition, some beliefs appear to be more influential than others. For instance, one might intuitively expect political alignment to be dependent on individuals' policy preferences. However, Unsworth & Fielding (2014) demonstrate that causal influence may actually flow primarily in the opposite direction. When shown identical policies with different partisan framings, individuals' support for those policies shifts in line with their political alignment, suggesting that partisan identity may shape policy preferences more so than the other way around.

[1]We use the inclusive definition of beliefs proposed by Galesic et al. (2021), as also adopted by Dalege et al. (2025). This includes "beliefs as assumptions about states of the world, ... views on moral and political issues, ... evaluations or cognitive aspects of attitudes or as own preferences".

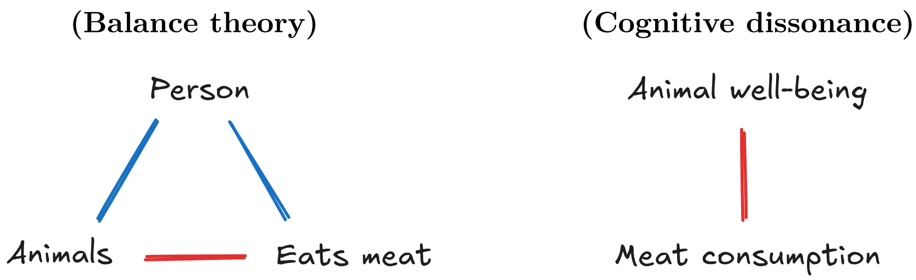
Given belief systems' interdependent nature, understanding the behaviour of any single belief requires consideration not only of how it interacts with other beliefs, but also how *those beliefs* interact with the broader belief system. This has motivated empirical approaches to studying belief system structure based on network science, which treat belief systems as undirected networks describing the pairwise correlational structure (edges) between distinct beliefs (nodes) (Boutyline & Vaisey, 2017; Costantini et al., 2015; Dalege et al., 2017; Epskamp et al., 2012; Epskamp, Borsboom, et al., 2018).

Analysis of *belief system networks* using network science methods has proven valuable for understanding belief system structure and the roles of specific beliefs in topic-specific belief systems. Using cross-national surveys, several studies have examined geographic variation in belief system structure on topics including politics (Keskintürk, 2022; van Noord et al., 2025) and climate change (Lee et al., 2025), comparing both specific relations and whole-network features such as density (the proportion of realised connections) and inconsistency (the number of negative correlations). Similarly, Chambon et al. (2023) examine how beliefs relating to COVID-19 changed over time in the Netherlands at the start of the global pandemic. Despite ongoing debate about its usefulness and applicability to belief system networks (Bringmann et al., 2019), node centrality is frequently used to assess belief position within a belief system (Borsboom et al., 2021; Brandt et al., 2019), relative influence (Robinaugh et al., 2016), or structural importance (Dalege et al., 2017; Fonseca-Pedrero, 2018; Hevey, 2018). Chambon et al. (2022) demonstrate that node centrality may also contribute to the propagation of belief-level interventions, in the context of a belief system on COVID-19.

Although belief system networks characterise the statistical associations between beliefs, they make no claims about the nature of these associations, how they arise, or their implications for belief dynamics. Several explanatory models have been proposed to address these shortcomings, generally considering endogenous belief system dynamics as driven by individuals' efforts to achieve consistent sets of beliefs; that is, which are perceived as mutually compatible. This perspective is particularly amenable to formalisation within the framework of complex systems, treating belief system dynamics as arising from micro-level attempts to resolve perceived inconsistencies between beliefs.

Current theories (and models) of belief system dynamics have been especially influenced by two classical theories of belief and attitude consistency: Heider's *balance theory* (1946) and Festinger's *theory of cognitive dissonance* (1962). Heider's balance theory interprets attitudes as signed, affective associations between concepts. Inconsistency arises when three concepts are related in a triangle for which the product of edge weights is negative. Festinger's theory of cognitive dissonance, on the other hand, takes the view that beliefs (and by extension, attitudes) can be more or less compatible with one another, and inconsistency arises when an individual holds incompatible beliefs.

The difference between these theories is best clarified with an example (illustrated below). Consider a person who cares deeply about animal well-being but who also eats meat. Heider (*left*) considers the cognitive inconsistency as arising from the negative association between animals and meat consumption (interpreted as 'harm'). To resolve this, the individual could adjust their attitudes toward animal well-being, change their dietary habits, or limit the perceived association between animals and meat consumption. Festinger (*right*) relates the individual's attitudes directly rather than via their conceptual referents, but arrives at the same conclusion, that the person will tend to change one of the two attitudes, or reduce their perceived negative association.



Balance theory and the theory of cognitive dissonance have given rise to two families of belief system models based on the statistical-physics notion of energy minimisation (Ising, 1925). The present study primarily considers models inspired by Festinger's theory of cognitive dissonance, which is more concerned with relations *between* beliefs than beliefs as relating concepts. For models based on Heider's balance theory, we refer the reader to Greenwald et al. (2002), Rodriguez et al. (2016), and Aiyappa et al. (2024).

The *Causal Attitude Network* (CAN) model (Dalege et al., 2016) formalises Festinger's theory of cognitive dissonance as an Ising-style model, also drawing on the more recently proposed connectionist perspective on attitude change (Monroe & Read, 2008). It can be viewed as a theory-driven variant of the belief network approach described above. While the Ising model was originally presented as a model of ferromagnetic behaviour (Ising, 1925), it has since been variously applied to describe diverse systems of interacting variables which attract or repel one another (Nguyen et al., 2017). In the CAN model, beliefs and attitudes are represented as spins that take on values in the set $\{-1, +1\}$, corresponding to two opposing states (e.g., 'climate change is happening' versus 'climate change is *not* happening' or 'support' versus 'oppose'). These are related by reinforcing (+) or opposing (−) interaction effects, based on the notion of cognitive dissonance. The model dynamics are described in terms of energy (read 'inconsistency') minimisation. That is, individuals tend to reduce cognitive dissonance, so that positively associated beliefs generally agree and negatively associated beliefs generally disagree.

Importantly, the CAN model defines interaction relations as symmetric, such that related beliefs exert equal reinforcing or opposing influence on one another. Under this assumption, the underlying Ising model satisfies detailed balance and can be considered an equilibrium system (Christensen & Moloney, 2005). This is mathematically convenient, as it allows for implicit definition of the CAN model dynamics using the Boltzmann distribution, such that each belief system state is observed with constant probability as a decreasing function of that state's inconsistency (i.e., energy). Highly consistent states are frequently observed, whereas highly inconsistent states are rarely observed.

Despite its simple foundations, the CAN model and recent variants have proven effective at capturing observed phenomena and demonstrated value for predicting and testing hypotheses about belief change. For instance, both Dalege et al. (2018) and van der Maas et al. (2020) show that such models may explain how merely thinking about a topic can induce more extreme views. In several cases, the CAN model (or an analogous model) has been used to explore, via simulation, the potential effects of interventions or persuasion attempts on belief system dynamics (Bertero & Franetovic, 2025; Dalege et al., 2017; Lunansky et al., 2022; Schlicht-Schmälzle et al., 2018). Furthermore, van der Maas et al. (2020) and Dalege et al. (2025) demonstrate that extensions of the CAN model that incorporate social influences on belief dynamics are also highly expressive. In both cases, the authors model social network influences via theoretically simple social cohesion mechanisms (i.e., the assumption that individuals who interact tend to align their beliefs). This turns out to be sufficient to capture various phenomena observed in reality, including: (i) attitude polarisation, (ii) how interventions can drive extreme attitudes on topics that had previously been relatively uncontroversial, and (iii) how minority views propagate through a population (Dalege et al., 2025).

However, although mathematically convenient, symmetric interactions are not psychologically necessary. Indeed, this has been broadly acknowledged within the belief systems modelling literature as a limitation of current approaches. Most authors explicitly mention either causal directionality underlying inferred bi-directional associations, or asymmetric/non-reciprocal influences as *plausible* (Brandt, 2022; Brandt & Sleegers, 2021; Epskamp, van Borkulo, et al., 2018; Keskintürk, 2022; Lee et al., 2024; Monroe & Read, 2008; Powell et al., 2023). van der Maas et al. (2020) go so far as to assert that asymmetric influences are *likely*, for instance, in relations linking beliefs and behaviour. Moreover, there is empirical evidence supporting asymmetric interactions. For instance, Chambon et al. (2023) observe directional differences in temporal networks (van der Krieke et al., 2015) used to study belief systems related to COVID-19; however, as identifying asymmetry was not a core aim of this study, they do not test the significance of these differences.

Despite general agreement on the plausibility of asymmetric interactions, current research remains predominantly focused on symmetric models.[2] A handful of authors refer to simulation studies in which asymmetric edges are used and found to have minimal impact on model dynamics (Monroe & Read, 2008; van der Maas et al., 2020). However, these experiments are rarely described in detail, and we are not aware of any instances in which asymmetric models are estimated from data. For mean-field approximations, van der Maas et al. (2026) argue that theoretical results derived for Ising-style belief system models are robust to assumptions of asymmetry or non-reciprocity. However, they note that studies outside the belief system modelling literature show that non-reciprocal Ising models can exhibit considerably more complex behaviour than the symmetric model (see Avni et al. (2025)).

One area in which the effects of asymmetric influence are likely to surface is the assessment of belief importance, often quantified using node centrality indices (as discussed above). While most undirected-network centrality indices have directed-network analogues, these can differ substantially in value (Bringmann et al., 2019). This becomes especially critical when belief centrality is used to predict intervention effectiveness. For instance, beliefs with many, or strong, connections (that is, high degree centrality or strength centrality) are often considered both (i) good targets for interventions intended to propagate to other beliefs (see Chambon et al. (2022)), and (ii) more resistant to such interventions, on account of their received influence from adjacent beliefs (see Brandt & Vallabha (2023)). However, both conclusions rely on the assumption of bidirectional influence. When influence may be asymmetric or non-reciprocal, we may find that some beliefs are characterised almost exclusively by incoming or outgoing interactions, leading to situations where well-connected beliefs have limited influence on other beliefs or little to no resistance to interventions, respectively.

At the time of writing, empirical and theoretical studies into the existence, nature, and potential structure of asymmetric influences among beliefs remain scarce. Of particular concern, the absence of theoretical models of asymmetric influences appears self-perpetuating, with some studies citing historical trends as justification for a continued

[2]Bayesian network models are a notable exception, which consider belief systems as directed acyclic graphs (Cook & Lewandowsky, 2016; Powell et al., 2023). However, these do not allow for reinforcing relationships. Moreover, they fall victim to the same problem as the statistical belief system networks. While they describe the relationships observed in the data, they offer no explanation as to the nature of these relationships.

focus on symmetric models (Brandt, 2022, p. 3). Consequently, it remains unclear whether asymmetric influence—should it exist—meaningfully affects belief system dynamics or the conclusions we draw from models thereof.

In this study, we investigate the prevalence of asymmetric influence among beliefs about climate change in the US, as well as the dynamic implications of symmetric and asymmetric modelling assumptions for belief-level interventions. Using a combination of data-driven and simulation-based methods, we will address the following four research questions:

RQ1. To what extent are belief-level influences *symmetric* or *asymmetric* in climate change belief systems in the United States?

RQ2. How do symmetric-influence assumptions impact expectations regarding population-level intervention strategy and effectiveness in climate change belief systems? To what extent do the impacts depend on where, in the belief system, an intervention is applied?

RQ3. How does individual-level intervention effectiveness, measured as the resulting shift in behaviour toward a desired belief state, depend on an individual's pre-intervention beliefs, in asymmetric climate change belief systems?

RQ4. How do asymmetric climate change belief systems for liberal and conservative subpopulations compare structurally, with respect to sparsity, strength of belief-level interactions, and presence of particular interactions?

This study is divided into two halves. In the first half, we establish a framework for modelling belief systems with asymmetric influence relations and estimating such models from time-series data. In the second half, we use this framework to address the research questions listed above.

In Chapter 2, we introduce the **Kinetic Belief System** model (KBS), a kinetic Ising model formulation (Fredrickson & Andersen, 1984; Glauber, 1963) of the Causal Attitude Network (CAN) model (Dalege et al., 2016). The KBS model represents directed belief influences using distinct parameters, so it is well-suited for studying asymmetry in belief systems. KBS's dynamics are explicitly time-dependent, enabling straightforward analysis of belief system dynamics, including post-intervention behaviour, on an *individual* basis. This contrasts past studies that analyse intervention effects via simulation on the CAN model (Bertero & Franetovic, 2025; Dalege et al., 2017; Lunansky et al., 2022; Schlicht-Schmälzle et al., 2018) or GGM models (Wu et al., 2026), which consider neither individuals' pre-intervention belief states, nor the timescale of model dynamics. Chapter 3 then outlines a parameter estimation method for the KBS model based on maximum likelihood estimation. The proposed method uses knowledge of a predefined soft thresholding function to robustly estimate binary model parameters from survey data that is not necessarily binary, without requiring explicit binarisation. In Chapter 4, we then use this method to calibrate the KBS model using a two-wave longitudinal dataset comprising beliefs about climate change.

We then subsequently use the calibrated model to address our research questions in the second half of the study. Inspired by approaches used in the context of the CAN model, we formalise interventions in Ising-style belief system models as additional influencing variables in Chapter 5. We also outline our experimental methods to: (i) quantifying expected intervention effects on an individual basis and (ii) comparing effects between symmetric and asymmetric models, using Common Random Numbers to ensure result comparability. We assess the existence of asymmetry and its population-level implications for intervention dynamics in Chapter 6, and heterogeneity in intervention outcomes and belief systems in Chapter 7, relating these back to our research questions in Chapter 8, also discussing broader implications for belief system modelling.

The **climate beliefs dataset** used in this study is sourced from the Longitudinal Panel of Perceptions About Climate Change and Covid (**CCCV**), a representative longitudinal survey collected in the US between 2020 and 2023 (Constantino et al., 2022). Comprehensively validating this survey data and constructing the targeted dataset of beliefs relating to climate change used in our experiments constituted substantial components of this investigation, and contribute to future use of the CCCV dataset. We detail both processes in Chapter 11.

In its totality, this study presents a theory-driven approach to studying the structure and dynamics of asymmetric belief systems. We apply this to investigate the existence and prevalence of asymmetry in an observational context regarding beliefs about climate change, and the consequences of symmetry and asymmetry assumptions for reasoning about intervention dynamics.

Chapter 2

The Kinetic Belief System model (KBS)

In this chapter, we define the **Kinetic Belief System** model (**KBS**) as a model of belief system dynamics that supports (but does not require) asymmetric influence relations among beliefs.

The KBS model builds on the Causal Attitude Network (CAN) model by Dalege et al. (2016), extending their formulation of an *attitude network* (which we call a *belief system*) to a kinetic Ising model framework (Fredrickson & Andersen, 1984; Glauber, 1963). Our treatment of belief system dynamics in the KBS model mirrors approaches used by Haslbeck et al. (2021) and Brandt & Sleegers (2021) for modelling belief dynamics in symmetric belief systems (similar to the CAN model).

We begin the chapter by formalising belief system *dynamics* as a modelling problem, after which we present the KBS model as a solution to this problem that permits asymmetric influence relations. Finally, we outline one approach to simulating the KBS model using Glauber dynamics.

> **Note**
>
> In this study, we use the inclusive definition of beliefs proposed by Galesic et al. (2021), also adopted by Dalege et al. (2025). This includes "beliefs as assumptions about states of the world, … views on moral and political issues, … evaluations or cognitive aspects of attitudes or as own preferences".

2.1 Modelling belief system dynamics

Our theory of belief system dynamics rests on three main assumptions. First, we assume that beliefs can be represented by distinct random variables. This allows us to consider beliefs as entities characterised by an instantaneous state. In particular, this assumption is incompatible with the perspective that beliefs are contextual and have no singular internal state of their own (Bendaña & Mandelbaum, 2021; Riemer et al., 2014).

Second, we assume that the state transition probability for a given belief (potentially) depends on the previous states of all beliefs, including itself, formalising the idea that past beliefs influence present ones. Finally, we assume that these conditional distributions are time-invariant. While belief *states* may change over time, the dynamics by which this happens do not.

> **Important**
>
> The term **belief** is presently ambiguous. It can refer to generic concepts (e.g. belief regarding the contents of a box) or specific instances of those concepts (e.g., 'I believe that the box is empty').
>
> For the remainder of this thesis, unless stated otherwise, we adopt the *generic* sense. That is to say that we use the term 'belief' without assuming any particular epistemic state, view, evaluation, or preference.

Consider a set of beliefs $\boldsymbol{S} = \{S_1, S_2, ..., S_N\}$. Given the above assumptions, we can describe the trajectory of a belief S_i with domain Ω_i as a sequence of states

$$\{s_i^t\}_{t=1}^{\infty} \quad \text{where} \quad s_i^t \in \Omega_i \tag{2.1}$$

characterised by the family of conditional distributions $P\left(\sigma_i^{t+1} \mid \boldsymbol{\sigma}^t\right)$ for $1 < t \in \mathbb{N}$ and an initial state $P(\sigma_i^1)$.

Noting that the state of a belief S_i depends only on the previous states of all other beliefs, it follows that any pair of beliefs S_i and S_j are conditionally independent at a given timestep, given all previous belief states. In particular, this means we can describe the evolution of the complete set of beliefs as a Markov process:

$$\{\boldsymbol{s}^t\}_{t=1}^{\infty} \quad \text{where } \boldsymbol{s}^t \in \Omega_1 \times \cdots \times \Omega_N \tag{2.2}$$

with transition probability given by the product of the individual beliefs' transition probabilities:

$$P(\boldsymbol{\sigma}^{t+1} \mid \boldsymbol{\sigma}^{t}) = \prod_{i}^{N} P\big(\sigma_i^{t+1} \mid \boldsymbol{\sigma}^{t}\big) \tag{2.3}$$

and a pre-specified initial state probability $P(\boldsymbol{\sigma}^1) = \prod_{i=1}^{N} P(\sigma_i^1)$.

We use the term **belief system** to refer to the combination of a collection of beliefs $\boldsymbol{S}$ and its associated transition probability distribution $P(\boldsymbol{\sigma}^{t+1} \mid \boldsymbol{\sigma}^{t})$. In this conceptualisation, the task of modelling a belief system reduces to describing how the instantaneous configuration of belief states affects the distribution over possible future states.

2.2 Kinetic Belief Systems

The Causal Attitude Network (CAN) model represents an attitude as an Ising-style network of beliefs (in the inclusive sense defined above), or *spins*, which are related via bidirectional edges reflecting reinforcing (+) or cognitive-dissonance (−) relations (Dalege et al., 2016). Each spin in the network takes on values in the domain $\{-1, +1\}$, representing two opposing states. Since the underlying Ising model contains only symmetric edges, the CAN model satisfies detailed balance and is therefore an equilibrium model of belief system dynamics (Cardy, 1996; Christensen & Moloney, 2005). As such, its associated transition probability is stationary and is described by the Boltzmann distribution (ibid.).

We define the **Kinetic Belief System** model (**KBS**) as a variation on the Causal Attitude Network, based on the kinetic Ising model (Fredrickson & Andersen, 1984; Glauber, 1963). Structurally, the KBS model differs from the CAN model in two key respects: (i) interaction effects between a given pair of beliefs are not necessarily bidirectional or equal in strength, and (ii) self-interaction effects are permitted.

Formally, let $\boldsymbol{S} = \{S_1, ..., S_N\}$ be a collection of beliefs. A kinetic belief system model for $\boldsymbol{S}$ is described by parameters $\boldsymbol{\theta} = \langle \boldsymbol{J}, \boldsymbol{h} \rangle$, where $\boldsymbol{J} \in \mathbb{R}^{N \times N}$ is a weighted adjacency matrix of directed **interaction effects** (or **influence effects**), and $\boldsymbol{h} \in \mathbb{R}^N$ is a vector of **baseline activation** effects for the elements of $\boldsymbol{S}$.

A baseline activation effect, $h_i \in \mathbb{R}$, describes the tendency for the belief S_i to take on the value +1 in the absence of interactions with other beliefs, or, more generally, when pairwise belief interactions have a net effect of zero. S_i tends to adopt the value +1 under these circumstances, if, and only if, $h_i > 0$, and otherwise tends to adopt the value -1. This tendency increases with the magnitude of h_i.

Other beliefs influence S_i's state through alignment or opposition relations. For a belief S_j, we define an interaction effect J_{ji} (read *'influence of j on i'*). When J_{ji} is positive or negative, S_i is influenced to adopt the state S_j^t or $-S_j^t$ respectively. In the special case when $j = i$, we refer to J_{ii} as a *self-influence* or *self-interaction* effect. Positive self-influence effects reflect the inertia of S_i, i.e., the tendency to sustain a particular state irrespective of other beliefs. Negative self-influence effects are not clearly interpretable. In addition to modelling differences in inertia, the inclusion of self-interaction effects allows us to capture the timescale on which the system as a whole evolves.

In contrast with the CAN model, we define the KBS model's dynamics explicitly using the conditional distribution $P(\boldsymbol{\sigma}^{t+1} \mid \boldsymbol{\sigma}^t)$. There are (at least) two reasons for this. First, critically, unlike in the CAN model, we cannot assume that the KBS model will exhibit equilibrium dynamics, as asymmetric interaction effects violate detailed balance, so the state transition probability is not necessarily stationary (Nguyen et al., 2017). Second, by describing the model dynamics explicitly, we are forced to consider how each variable in the belief system responds to the states of all other beliefs (including itself). As such, temporal dynamics are 'baked into' the KBS model, making it straightforward to simulate forward in time to investigate potential model dynamics. We make extensive use of this property during our simulated intervention experiments in the second half of this study.

The conditional distribution $P(\boldsymbol{\sigma}^{t+1} \mid \boldsymbol{\sigma}^t)$ describes the dynamics of the entire KBS model; however, rather than defining this directly, we will instead stitch it together from the dynamics of the individual beliefs using Definition 2.3.

Let $S_i \in \boldsymbol{S}$ be an arbitrary belief, and let $\boldsymbol{s} \in \{-1, +1\}^N$ describe the states of all beliefs in $\boldsymbol{S}$ at time t. We define the **effective baseline activation** of S_i at time t as the net pressure on this belief to adopt the state $+1$, resulting from the combined forces of its own baseline activation and inertia, as well as influence effects received from other beliefs:

$$h_i^{\text{eff}}(\boldsymbol{s}) := h_i + \sum_j J_{ji} s_j \tag{2.4}$$

We then define the conditional probability that S_i takes on the value $+1$ at time $t + 1$

as the logistic transform of the effective baseline activation:[3]

$$P\big(\sigma_i^{t+1} = +1 \mid \boldsymbol{\sigma}^t = \boldsymbol{s}\big) = \text{logistic}\big(2h_i^{\text{eff}}(\boldsymbol{s})\big) \tag{2.5}$$

We obtain the probability that S_i takes on the value -1 directly via the complement.

Notice that the behaviour of Equation 2.5 is consistent with our earlier interpretations of the baseline activation and interaction effects; the probability that S_i adopts the value $+1$ increases with the baseline activation, h_i, through positive (reinforcing) influences from beliefs with state $+1$ or negative (opposing) influences from beliefs with state -1, and when S_i previously had the state $+1$.

Finally, using Definition 2.3, we can now describe the complete conditional transition probability for the KBS model in terms of those belonging to the individual beliefs:

$$P\big(\boldsymbol{\sigma}^{t+1} = \boldsymbol{s}^{t+1} \mid \boldsymbol{\sigma}^t = \boldsymbol{s}^t\big) = \prod_{i=1}^{N} P\big(\boldsymbol{\sigma}_i^{t+1} = s_i^{t+1} \mid \boldsymbol{\sigma}^t = \boldsymbol{s}^t\big) \tag{2.6}$$

Where the initial distribution $P(\boldsymbol{\sigma}^1)$ is specified directly.

2.3 Symmetric and asymmetric belief systems

In the special case where $\boldsymbol{J}$ is symmetric, we say that the kinetic belief system model is **symmetric**; otherwise, it is **asymmetric**. In an asymmetric KBS model, for any pair of distinct beliefs $S_i \neq S_j \in \boldsymbol{S}$, it may be the case that an influence relation exists in only one direction, or that the directed relations differ in sign and/or magnitude. On the other hand, all influence relations in a symmetric KBS model are directionally equivalent.

The symmetric KBS model is a simple extension of the symmetric Ising model with self-interaction effects and temporal dynamics. However, due to the self-interaction effects, the symmetric KBS model still does not satisfy detailed balance, and therefore is not guaranteed to tend toward an equilibrium steady state (Nguyen et al., 2017).

[3]The factor of two here simplifies our derivations of the likelihood and partial derivatives in the next chapter. Note that this form of the probability is equivalent to one in which the factor of two is absorbed into the baseline activation and interaction parameters.

2.4 Simulation via Glauber dynamics

We can use Glauber dynamics to sample trajectories from the KBS model (Glauber, 1963). Given an initial state $\boldsymbol{s}^1 \in \{-1, +1\}^N$, we sample a trajectory of length $T \in \mathbb{N}$ by drawing consecutive samples using the KBS transition probability defined in Equation 2.6:

$$\{\boldsymbol{s}^t\}_{t=1}^T, \quad \text{where } \boldsymbol{s}^t \sim P(\boldsymbol{\sigma}^t \mid \boldsymbol{\sigma}^{t-1} = \boldsymbol{s}^{t-1}) \text{ for } t \in \{2, ..., T\} \tag{2.7}$$

All beliefs can update during every time interval. This update routine is referred to as **synchronous Glauber dynamics**, in contrast with **asynchronous Glauber dynamics**, in which only one belief can update during a given interval (Glauber, 1963; Nguyen et al., 2017).

Asynchronous models are often more realistic, particularly for phenomena which evolve continuously in time, i.e., where the synchronicity assumption is invalid—we discuss this further in Section 8.1 (↪). However, from a parameter estimation perspective, models which assume synchronous dynamics are comparatively simpler to calibrate, e.g., using maximum likelihood estimation. For an extended discussion of this matter, we refer the reader to Nguyen et al. (2017, p. 34).

We note that while an asynchronous formulation of the KBS model is possible, it would require a different formulation of the belief system dynamics modelling problem defined in Section 2.1, to prevent multiple beliefs from updating simultaneously.

Chapter 3

Parameter estimation in the KBS model

In the previous chapter, we introduced the Kinetic Belief System (KBS) as a model of belief system structure and dynamics, that allows for asymmetric influence relations. While the KBS model is theoretically interesting, its *empirical* value hinges on the ability to infer its parameters from observational data.

In this chapter, we provide a method for doing just this, based on maximum likelihood estimation—this turns out to be not so straightforward, due to the mismatch between the binary representation of beliefs assumed by the KBS model and typical approaches for assessing beliefs in surveys. We omit derivation details in this chapter; however, the curious reader can find these in Appendix A.

3.1 The problem with maximum likelihood estimation

Given a dataset D and a parameterisable model with $p \in \mathbb{N}$ parameters, maximum likelihood estimation (MLE) seeks the parameterisation $\boldsymbol{\theta}^* \in \mathbb{R}^p$ which maximises the probability of observing D under the model, also known as the likelihood:

$$\boldsymbol{\theta}^* = \underset{\boldsymbol{\theta} \in \mathbb{R}^p}{\operatorname{argmax}}\, P(D \mid \boldsymbol{\theta}) \tag{3.1}$$

MLE is often used for parameter estimation in similar modelling situations (Lee et al., 2015; Nguyen et al., 2017). It has a significant drawback, however, in that it assumes that the observations in the dataset D are representable within the model. This poses a problem when calibrating binary belief system models—such as the KBS model—

using survey data, which often describes belief states with multi-valued (e.g., Likert) scales. To use MLE in such situations, we must first binarise the observations to the set $\{-1, +1\}$.

There are several approaches to binarising continuous values; the simplest, which we will call **sign-thresholding**, is to map values to $\{-1, +1\}$ according to their sign. Zero must be handled separately, for instance, using a deterministic rule (e.g., $0 \mapsto +1$) or a Bernoulli random variable. This produces discontinuous behaviour around the zero point, which may be unrealistic in this setting. If we are treating zero as an ambivalent or neutral belief state that can be mapped to either -1 or $+1$, one could argue that we should treat small-magnitude positive or negative values similarly.

We can improve on this using **soft thresholding**, in which each data value $x \in \mathbb{R}$ is perturbed by an independent noise term $\varepsilon \sim \mathcal{N}(0, \xi)$ prior to this mapping. Figure 3.1 illustrates this process for a negative value of x and a specified $\xi \in \mathbb{R}_+$.[4]

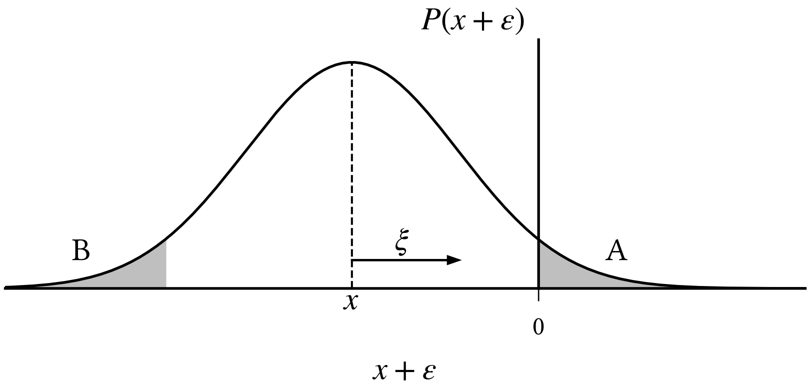


Figure 3.1: Soft thresholding maps $x \in \mathbb{R}$ to $\{-1, +1\}$ by sign-thresholding $x' = (x + \varepsilon)$ where $(x + \varepsilon) \sim \mathcal{N}(x, \xi)$. Negative values are mapped to $+1$ with probability $P(x' > 0) = \mathrm{A} = \mathrm{B} = P(\varepsilon < x)$, which increases with ξ and $|x|^{-1}$.

Observe that x is mapped to $+1$ if, and only if, ε is sufficiently large, such that $x + \varepsilon > 0$ (region A), or equivalently, when $\varepsilon < x$ (region B). The probability that x is mapped to $+1$ is then:[5]

$$P(x \mapsto +1) = \Phi\left(\frac{x}{\xi}\right) \tag{3.2}$$

[4]The specific choice of ξ is important. As $\xi \to 0^+$, the soft thresholding function converges to the simple ('hard') thresholding function where zero is mapped using a Bernoulli(0.5) random variable. As $\xi \to \infty$, we approach a function which maps *all* values randomly in this way. Appropriate values are context-dependent, based on the range of values which are considered 'near-neutral'.

[5]See derivation in Appendix A ($\hookrightarrow$).

where Φ is the standardised normal cumulative distribution function. This probability is small when x has a large magnitude or when ξ is small. With soft thresholding, small-magnitude values are mapped to -1 and $+1$ with nontrivial probability, while the binarisation of large values is effectively deterministic.

However, there remains a deeper issue with binarisation, that soft thresholding does not resolve: *any* form of binarisation necessarily erases information about the magnitude of the original data. For instance, consider a survey question that assesses an individual's attitude toward a particular policy using a 7-point Likert scale.[6] After binarisation, if a value is mapped to $+1$, we can no longer determine whether this originally corresponded to weak or strong support, neutrality, or even opposition.

In short, since it uses a single binary output for each observation, MLE cannot capture ambiguity or neutrality and instead requires that all data be made unambiguous and non-neutral through binarisation.

3.2 Maximising the *expected* likelihood

We now introduce a variation on maximum likelihood estimation which resolves the above issue by avoiding binarisation altogether.

Let D be a dataset with real values, and let $\mathcal{D}_B$ be the random variable representing possible binarisations of D using a thresholding function, b. In standard MLE, we maximise the likelihood for a particular realisation D_B of the random variable $\mathcal{D}_B$. However, when the probability distribution for $\mathcal{D}_B$ is known, we can avoid explicit binarisation by instead maximising the **expected** likelihood, marginalising over the binarisation process:

$$\boldsymbol{\theta}^* = \operatorname*{argmax}_{\boldsymbol{\theta} \in \mathbb{R}^p} \mathbb{E}[P(b(D) \mid \boldsymbol{\theta})] \tag{3.3}$$

If b is the simple thresholding function described above, that maps each value to its sign (and maps zero deterministically), then Equation 3.3 is equivalent to the MLE problem defined in Equation 3.1. However, when b is a (non-constant) probabilistic binarisation function, Equation 3.3 uses information regarding both binarisation possibilities.

Consider the case where $b := b_\xi$ is the soft-thresholding function defined in the above section, with $\xi \in \mathbb{R}_+$. If ξ is chosen appropriately,[4] then data values which are considered

[6]In this case, the Likert-7 scale has response options: strongly oppose $\prec$ oppose $\prec$ weakly oppose $\prec$ neither support nor oppose $\prec$ weakly support $\prec$ support $\prec$ strongly support.

somewhat neutral or ambivalent are mapped to both -1 and $+1$ with nontrivial probability. When using Equation 3.3, the contribution of such values to the objective function is an expectation over possible binarisations, yielding a parameterised model that explains both possibilities in accordance with their probabilities. Therefore, by avoiding explicit binarisation, we can obtain binary models that incorporate the neutrality and varying degrees of certainty present in non-binarised data.

This is far from the only approach that can be used to account for neutrality in belief system models. For instance, van der Maas et al. (2026) suggest using three-valued models, such as the Blume-Capel model (ibid.), that explicitly represent intermediate states using zero-valued spin states. Such approaches provide extra representational capacity—our approach cannot simulate neutral states—but raise the analytical complexity. For the purposes of this study, which has an exploratory nature, we prefer the relative simplicity of binary models.

3.3 Parameter estimation for the KBS model

We now adapt the maximum expected likelihood estimation approach described above to the KBS model context, explicitly defining the expected likelihood and its derivatives, which are used for calibration.

Consider a population of $M \in \mathbb{N}$ individuals with a shared belief system $\mathcal{M}$ comprising $N \in \mathbb{N}$ beliefs. Suppose that we have measured the belief system state of each individual, $m \in [M]$, at each of $T \in \mathbb{N}$ uniformly spaced times:

$$\left\{\boldsymbol{x}_{(m)}^{t}\right\}_{t=1}^{T}, \quad \text{where each } \boldsymbol{x}_{(m)}^{t} \in \mathbb{R}^{N} \tag{3.4}$$

The measurements need not be binary, but are assumed to be real. The collection of measurements across all individuals forms a dataset D, which is the particular observed value of the random variable $\mathcal{D}$ representing possible datasets.

To calibrate the (symmetric or asymmetric) KBS model, we identify the parameterisation that maximises the expected likelihood, marginalising over possible binarisations using a soft thresholding function, b_{ξ}, as defined above. Following Epskamp, Borsboom, et al. (2018), we also use regularisation to reduce the risk of overfitting.

Let $\mathcal{L}_D(\boldsymbol{\theta})$ denote the expected log-likelihood for a given parameterisation $\boldsymbol{\theta} \in \mathbb{R}^p$, where $p \in \mathbb{N}$ is the number of model parameters in the (symmetric or asymmetric)

KBS model:[7]

$$\mathcal{L}_D(\boldsymbol{\theta}) \coloneqq \mathbb{E}\big[\log P\big(b_\xi(D)|\boldsymbol{\theta}\big)\big] = \sum_{D_B} P\big(b_\xi(D) = D_B\big) \cdot \log P(D_B|\boldsymbol{\theta}) \tag{3.5}$$

We select the parameterisation $\boldsymbol{\theta}^* \in \mathbb{R}^p$ which solves the following optimisation problem:

$$\boldsymbol{\theta}^* = \operatorname*{argmax}_{\hat{\boldsymbol{\theta}} \in \mathbb{R}^p} f\big(D; \hat{\boldsymbol{\theta}}\big), \quad \text{where} \quad f\big(D; \hat{\boldsymbol{\theta}}\big) \coloneqq \mathcal{L}_D\big(\hat{\boldsymbol{\theta}}\big) - \lambda \sum_{\theta \in \hat{\boldsymbol{\theta}}} \sqrt{\theta^2 + \varepsilon} \tag{3.6}$$

where the second term is a smooth variant of L1 regularisation (Tibshirani, 1996), which penalises nonzero parameters that do not meaningfully contribute to the expected likelihood. The smoothing hyperparameter, $\varepsilon \in \mathbb{R}_{>0}$, and regularisation strength, $\lambda \in \mathbb{R}_{\geq 0}$, are determined beforehand. We will discuss both the regularisation term and methods for selecting these hyperparameters in Section 3.3.1.

We now give explicit forms for the expected log-likelihood and the partial derivatives of the objective function with respect to the model parameters for the (symmetric and asymmetric) KBS model. The asymmetric and symmetric variants of the KBS model differ only in the partial derivatives with respect to the interaction parameters. We omit derivation details here, but these can be found in Appendix A.

Let $p_{\boldsymbol{s}}(\boldsymbol{x})$ denote the probability that an observation $\boldsymbol{x} \in \mathbb{R}^N$ is binarised to the state $\boldsymbol{s} \in \{-1, +1\}^N$ using the binarisation function b_ξ,

$$p_{\boldsymbol{s}}(\boldsymbol{x}) = P\big(b_\xi(\boldsymbol{x}) = \boldsymbol{s}\big) \tag{3.7}$$

and, for an individual $m \in [M]$ and time $t \leq T$, let $\mathbb{E}\big[\boldsymbol{\sigma}^t_{(m)}\big]$ denote the expected value of the binarised observation $\boldsymbol{x}^t_{(m)}$. The expected log-likelihood for a KBS model parameterisation, $\boldsymbol{\theta} = \langle \boldsymbol{J}, \boldsymbol{h} \rangle$, is then derived as:[8]

$$\mathcal{L}_D(\boldsymbol{\theta}) = \left(\sum_{m=1}^{M} \sum_{t=1}^{T-1} \sum_{i=1}^{N} \mathbb{E}\Big[\sigma^{t+1}_{(m),i} \cdot h^{\text{eff}}_i\Big(\boldsymbol{\sigma}^t_{(m)}\Big)\Big] \right) - Z(D; \boldsymbol{\theta}) \tag{3.8}$$

where $h^{\text{eff}}_i(\boldsymbol{s})$ is the effective baseline activation (Equation 2.4) given the previous state $\boldsymbol{s}$, and Z is the expected log partition function, summed across observations:

[7]For numerical stability, we typically maximise the log-likelihood rather than the likelihood. Since the logarithm is a strictly increasing function, the two approaches yield identical results in principle; however, the multiplication of small probabilities when computing the likelihood can lead to numerical overflow.

[8]See expected log-likelihood derivation in Appendix A (↪).

$$Z(D;\boldsymbol{\theta}) = \sum_{m=1}^{M}\sum_{t=1}^{T-1}\sum_{i=1}^{N}\mathbb{E}\Big[\log 2\cosh h_i^{\text{eff}}\Big(\boldsymbol{\sigma}_{(m)}^{t}\Big)\Big] \tag{3.9}$$

For the parameterisation $\boldsymbol{\theta}^*$ which solves Equation 3.6, the partial derivatives of the objective function with respect to each parameter evaluate to zero, i.e., $\frac{\partial f}{\partial \theta} = 0$ for every parameter $\theta \in \boldsymbol{\theta}^*$.

The partial derivative of f with respect to an arbitrary baseline activation, h_i, for $i \in [N]$, is derived for both the symmetric and asymmetric variants as:[9]

$$\frac{\partial}{\partial h_i} f(D;\boldsymbol{\theta}) = \left(\sum_{m=1}^{M}\sum_{t=1}^{T-1}\mathbb{E}\Big[\sigma_{(m),i}^{t+1} - \tanh h_i^{\text{eff}}\Big(\boldsymbol{\sigma}_{(m)}^{t}\Big)\Big]\right) - \frac{\lambda h_i}{\sqrt{h_i^2 + \varepsilon}} \tag{3.10}$$

For $i, j \in [N]$, the partial derivative of f with respect to the directed interaction parameter J_{ji} in the asymmetric variant (i.e., the influence of j on i), is:

$$\frac{\partial}{\partial J_{ji}} f(D;\boldsymbol{\theta}) = \left(\sum_{m=1}^{M}\sum_{t=1}^{T-1}\mathbb{E}\Big[\Big(\sigma_{(m),i}^{t+1} - \tanh h_i^{\text{eff}}\Big(\boldsymbol{\sigma}_{(m)}^{t}\Big)\Big)\cdot \sigma_{(m),j}^{t}\Big]\right) - \frac{\lambda J_{ji}}{\sqrt{J_{ji}^2 + \varepsilon}} \tag{3.11}$$

In the symmetric model, partial derivatives with respect to self-interaction parameters are also calculated using Equation 3.11. However, for cross-interaction parameters, J_{ij} with $i < j$, we must account for the contributions to both S_i and S_j in the partial derivative. Let $\alpha_{m,t}(i,j)$ denote the inner summand in Equation 3.11:

$$\alpha_{m,t}(i,j) := \Big(\sigma_{(m),i}^{t+1} - \tanh h_i^{\text{eff}}\Big(\boldsymbol{\sigma}_{(m)}^{t}\Big)\Big)\cdot \sigma_{(m),j}^{t} \tag{3.12}$$

Then the partial derivative of f with respect to the bi-directional interaction parameter J_{ij}, for $i < j$, in the symmetric variant is:

$$\frac{\partial}{\partial J_{ij}} f(D;\boldsymbol{\theta}) = \left(\sum_{m=1}^{M}\sum_{t=1}^{T-1}\mathbb{E}\big[\alpha_{m,t}(i,j) + \alpha_{m,t}(j,i)\big]\right) - \frac{\lambda J_{ij}}{\sqrt{J_{ij}^2 + \varepsilon}} \tag{3.13}$$

3.3.1 Smooth L1 regularisation

Since the number of parameters in the KBS model scales quadratically with the number of modelled beliefs, parameter estimation is prone to overfitting—even for relatively small sets of beliefs—due to the typically small size of psychological datasets. Following Epskamp, Borsboom, et al. (2018), we apply a smooth variant of L1 regularisation

[9]See the partial derivative derivations in Appendix A (↪).

(the second term in Equation 3.6) during parameter estimation. This has the effect of penalising nonzero parameters that do not meaningfully contribute to the expected likelihood, instead requiring that parameters reflect relationships which are sufficiently prevalent in the data.

This regularisation term has two hyperparameters: a smoothing constant, $\varepsilon \in \mathbb{R}_{>0}$, and a regularisation strength parameter, $\lambda \in \mathbb{R}_{\geq 0}$. For the objective function (f in Equation 3.6) to be differentiable, the smoothing constant must be positive. The value of ε should be chosen such that the smooth regularisation term is a reasonable approximation to the standard formulation of L1 regularisation, i.e., such that

$$\sqrt{\theta^2 + \varepsilon} \approx |\theta| \tag{3.14}$$

It is not possible to obtain a good approximation for the full range of parameter values due to finite precision. Rather, one should choose ε such that this approximation holds for parameter values $|\theta| > \tau$, where $\tau \in \mathbb{R}_{>0}$ is the minimum parameter considered 'effectively nonzero'. As a rule of thumb, we may take ε to be at least two orders of magnitude smaller than τ^2.

The hyperparameter $\lambda \in \mathbb{R}^+$ controls regularisation strength, with higher values resulting in sparser models. Epskamp, Borsboom, et al. (2018) suggest choosing λ which minimises the Extended Bayesian Information Criterion (EBIC) (Chen & Chen, 2008):

$$\mathrm{EBIC}(\lambda) = \overbrace{k \ln q - 2\mathcal{L}_D(\boldsymbol{\theta}^*(\lambda))}^{\mathrm{BIC}(\lambda)} + 2k \cdot \gamma \ln p \tag{3.15}$$

where $\boldsymbol{\theta}^*(\lambda)$ denotes the solution to the optimisation problem defined in Equation 3.6 for regularisation strength λ, $q \in \mathbb{N}$ is the number of observations in the dataset, $p \in \mathbb{N}$ is the number of model parameters, and k is the number of (effectively) nonzero parameters with $|\theta| < \tau$ in the parameterisation $\boldsymbol{\theta}^*(\lambda)$. Notice that λ is both a parameter in the optimisation problem defined in Equation 3.6, and is itself selected using the solution to this problem.

The point of difference between the EBIC and the standard Bayesian Information Criterion (BIC) (Schwarz, 1978) is in their assumed priors over the space of possible models. While the BIC assumes a uniform prior, the EBIC penalises values of k which permit a large number of models (Barber & Drton, 2015; Foygel & Drton, 2010). For instance, the class of models comprising eight spins (i.e., with 72 parameters) contains considerably fewer models with 10 parameters (5.4×10^{11}) than models with

36 parameters (4.4×10^{20}). In this scenario, by assuming a uniform prior, the BIC implicitly penalises models which are too sparse, or too dense.

The parameter $\gamma \in \mathbb{R}_{\geq 0}$ specifies the degree to which the EBIC penalises such classes of models, with $\gamma = 1$ corresponding to a uniform prior over models with k parameters, for each $k \in 0...p$.[10] Barber & Drton (2015) demonstrate that small, nonzero choices of γ are sufficient to improve the accuracy of Ising model structural inference in various contexts.

[10]Note that this is different from the prior used in the BIC, which is uniform over the entire space of models. The prior in the EBIC, for $\gamma = 1$, is uniform only within each class of models containing k parameters.

Chapter 4

A KBS model of climate beliefs

In this chapter, we calibrate both the symmetric and asymmetric KBS models to the **climate beliefs dataset** described in Chapter 11. We then evaluate the calibrated models with respect to both structural accuracy ('how accurate are the parameter estimates?') and predictive capacity ('how well do the models explain the data?'). We also calibrate two additional asymmetric models to the conservative and liberal subsets of the climate beliefs dataset for later use in Chapter 7.[11]

The climate beliefs dataset comprises eight beliefs relating to climate change (Table 4.1), extracted from the CCCV survey (Constantino et al., 2022). The dataset includes responses from 1693 repeating participants, measured during waves 3 and 4 of the survey. A subset of the variables are indices constructed from sets of variables in the CCCV survey. The marginal distributions for each variable are displayed in Figure 4.1.

Item	Index	Interpretation
CC Real	No	Belief that climate change is/is not real.
CC Human	No	Belief that climate is/is not caused (at least partly) by human activities.
CC Worry	No	Level of worry about current and future climate change.
CC Others Worry	No	Belief regarding *others'* level of worry about current/future climate change.
Weather Worry	No	Level of worry about near-term extreme weather events/natural disasters.
Politics	Yes	Political views, a combination of political alignment and ideology.
CC Impact	Yes	Belief about current climate change impacts' severity, in general.
CC Action	Yes	General attitude toward action on climate change.

Table 4.1: Variables included in the climate beliefs dataset. Index variables are constructed by averaging their constituent columns, after rescaling to the interval $[-1, 1]$. Note that this table is identical to Table 11.2 displayed in Chapter 11.

[11]We do not evaluate the conservative and liberal models here, but return to this in Chapter 7.

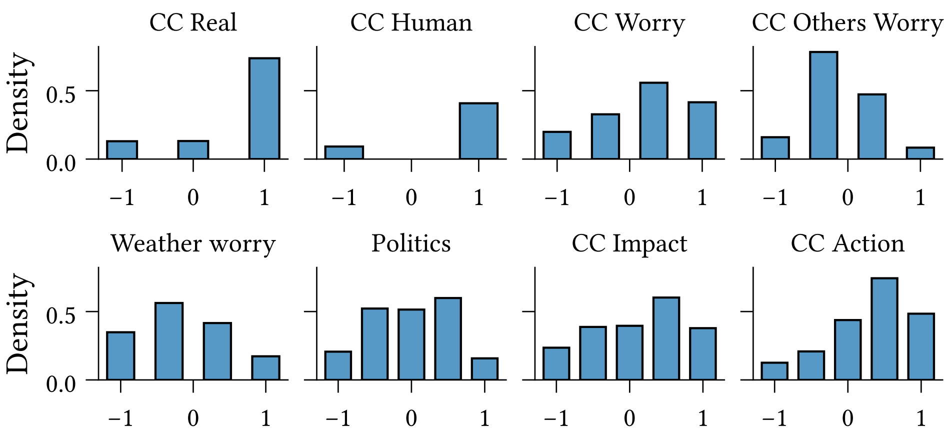


Figure 4.1: Marginal distribution for each of the eight variables in the climate beliefs dataset (Table 4.1). Note that this figure is identical to Figure 11.7 displayed in Chapter 11.

Chapter 11 provides complete details on the CCCV survey, including validation and cleaning, as well as the construction of the climate beliefs dataset.

4.1 Hyperparameters and calibration details

The maximum expected likelihood parameter estimation method described in the previous chapter requires several hyperparameters to be specified, namely: the soft binarisation scale term, $\xi \in \mathbb{R}_{>0}$, the regularisation smoothing and strength hyperparameters, $\varepsilon \in \mathbb{R}_{>0}$ and $\lambda \in \mathbb{R}_{\geq 0}$, and the EBIC prior parameter $\gamma \in \mathbb{R}_{\geq 0}$. The values of these parameters are summarised in Table 4.2.

As discussed in the previous chapter, the soft thresholding scale parameter, $\xi \in \mathbb{R}_{>0}$, is context-specific and should be chosen to reflect the range of measurement values considered 'near-neutral'. To choose this, we first map the minimum and maximum

Parameter	Model type	Dataset	Value
λ	Symmetric	Climate beliefs	1.4×10^{-2}
	Asymmetric	Climate beliefs	1.2×10^{-2}
	Asymmetric	Conservative	3.4×10^{-2}
	Asymmetric	Liberal	4.3×10^{-2}
ξ	All	All	2.03×10^{-1}
ε	All	All	10^{-8}
γ	All	All	2.5×10^{-1}

Table 4.2: Values for regularisation strength (λ) and smoothing (ε) hyperparameters. Regularisation strength model- and dataset-specific; *Conservative* and *Liberal* refer to subsets of the climate beliefs dataset comprising individuals with the specified ideology (Chapter 11).

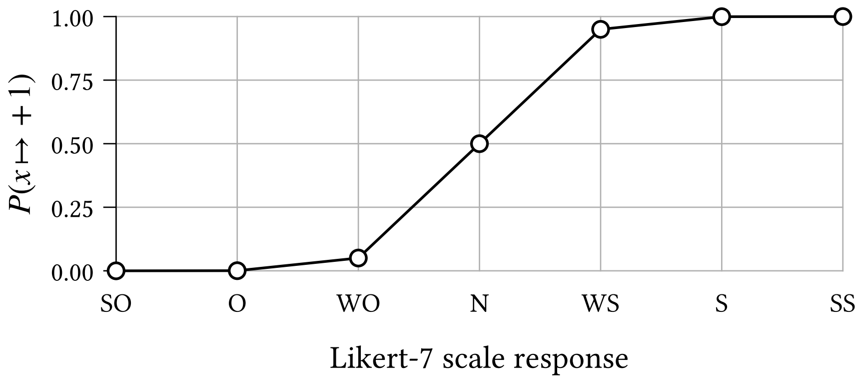


Figure 4.2: The probability of binarisation to +1 for each possible Likert-7 scale survey response,[6] using a soft thresholding function b_ξ as defined in Equation 3.2 with $\xi \approx 0.2$.

(allowable) values for each variable to -1 and $+1$, respectively, such that these reflect the two belief states in the KBS model. We then choose the soft thresholding function b_ξ with $\xi = 0.203$ such that a 'weakly oppose' response to a 7-point Likert scale (i.e., value $-1/3$ in the normalised data) is mapped to $+1$ with probability 0.05. The resulting probability distribution is displayed in Figure 4.2 for a 7-point Likert scale.[6] Under this choice of b_ξ, values to either side of 'neutral' are considered mostly unambiguously positive or negative, with some flexibility for 'weak' responses.

We choose the regularisation hyperparameters in accordance with the discussion in Section 3.3.1 (*Smooth L1 regularisation*). Taking $\tau = 10^{-2}$ as the threshold below which parameters are considered 'effectively zero', we choose the regularisation smoothing hyperparameter to be $\varepsilon = 10^{-8}$, such that $\varepsilon \ll \tau^2$. We set separate regularisation strengths for the symmetric and asymmetric models, as well as the conservative and liberal asymmetric models, taking values $\lambda \in [10^{-4}, 1]$ that minimise the EBIC

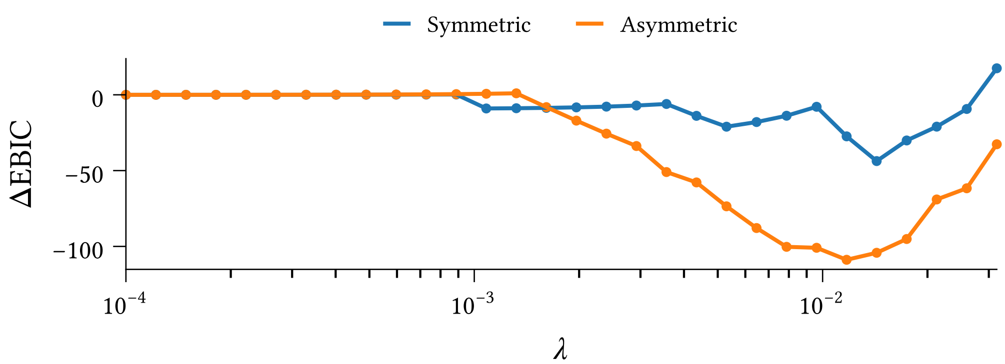


Figure 4.3: Effect of regularisation strength λ on the Extended Bayesian Information Criterion for symmetric and asymmetric belief system models optimised to the climate beliefs dataset using Equation 3.6. The vertical axis measures the difference in EBIC relative to the no-regularisation case ($\lambda = 0$); smaller values are better.

(Equation 3.15). We take $\gamma = 0.25$ when computing the EBIC, i.e., the smallest value tested by Barber & Drton (2015). Figure 4.3 shows the EBIC results for the symmetric and asymmetric models calibrated to the complete climate beliefs dataset.

4.1.1 Implementation details

We solve the parameter estimation problem using the SciPy 1.17.1 implementation of the quasi-Newton BFGS optimisation algorithm (Nocedal & Wright, 2006, p. 136; Virtanen et al., 2020), with the analytic Jacobian comprising the partial derivatives stated in Section 3.3. We take $\boldsymbol{\theta} = \mathbf{0}$ as the initial guess. To ensure numerical stability irrespective of dataset size, we rescale the expected log-likelihood contribution to the objective function and partial derivatives, dividing by the number of observed observations (time intervals) in the dataset. For $M \in \mathbb{N}$ individuals and $T \in \mathbb{N}$ timesteps, this amounts to a scale factor of $\frac{1}{M(T-1)}$.

4.2 Evaluating the calibrated model

As recommended by Epskamp, Borsboom, et al. (2018), we evaluate the accuracy of the inferred interaction parameters by examining the bootstrapped confidence intervals around each parameter estimate. We use bootstrapping to estimate the uncertainty in our parameter estimates due to sampling error. Let $\boldsymbol{D} \in \mathbb{R}^{M \times T \times N}$ be the complete dataset, where M, T, and N denote the number of participants, observations, and beliefs, respectively.[12] We construct 500 bootstrapped datasets by sampling rows (participants) with replacement from $\boldsymbol{D}$, such that each has the same shape as the complete dataset.

For each bootstrapped dataset $\boldsymbol{D}_{(i)}$ we calibrate a model $\mathcal{M}_{(i)}$ with $p \in \mathbb{N}$ parameters:

$$\langle \boldsymbol{J}_{(i)}, \boldsymbol{h}_{(i)} \rangle =: \boldsymbol{\theta}^*_{(i)} \in \mathbb{R}^p \tag{4.1}$$

Figure 4.4 shows the baseline activation parameters, $\boldsymbol{h}$, and interaction effect matrix, $\boldsymbol{J}$, for each model.[13] The two models exhibit very similar baseline activations. The observed values indicate that after accounting for interaction effects, (i) belief in both the existence and human-causes of climate change tend to be high, (ii) concern about extreme weather tends to be low, and (iii) people mostly believe that other individuals are not particularly worried about climate change. Regularisation has pushed some

[12]In the climate beliefs dataset we have $M = 1693$, $T = 2$, and $N = 8$

[13]We only display the upper triangular elements of the symmetric model's interaction matrix, since the matrix is symmetric.

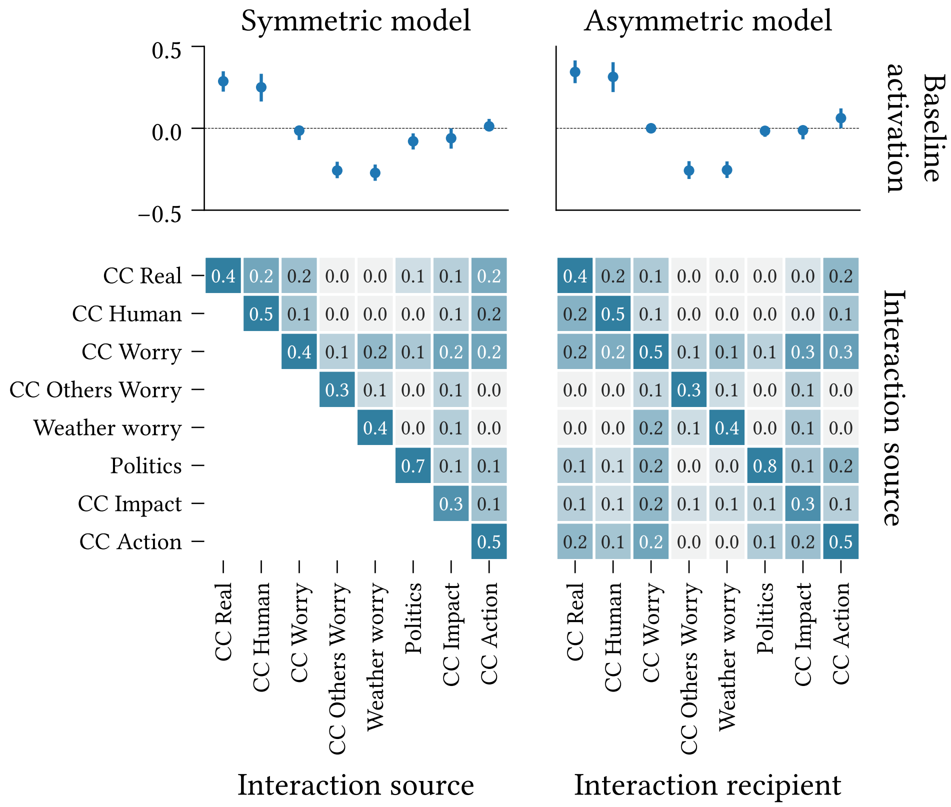


Figure 4.4: (*Top*) Baseline activations, $\boldsymbol{h}$, and (*Bottom*) interaction effect matrices, $\boldsymbol{J}$, for the symmetric and asymmetric belief system model variants, calibrated to the climate beliefs dataset. Error bars on the baseline activations display 95% confidence intervals, calculated over bootstrapped models ($n = 500$) using the percentile method.

values (e.g., worry about climate change) to zero, indicating that the dataset provides limited evidence that these tend to be either positive or negative.

Both models feature a dominant diagonal, indicating that most variables are slow-moving relative to the modelled timescale (they are 'sticky'), which is consistent with prior studies examining the rate of change in beliefs (Green & Platzman, 2024; Kiley & Vaisey, 2020).[14] This is particularly true for `Politics`, and less so for `CC Others Worry` and `CC Impact`.

Observe that all interaction effects are non-negative. This is by design; we have re-coded the dataset variables such that this is the case. The fact that such a recoding exists implies that it is possible within this belief system to hold internally consistent beliefs, i.e., with no cognitive dissonance. We provide a formal proof of this statement in Appendix A.

[14]The timescale is determined by the interval between survey responses, i.e., approximately six months (Chapter 11).

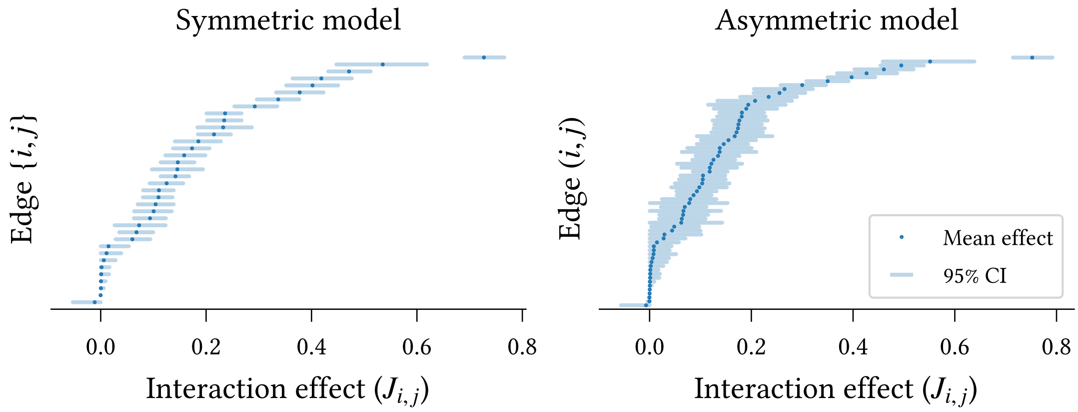


Figure 4.5: Interaction effect parameter accuracy for symmetric and asymmetric models calibrated to the climate beliefs dataset (Chapter 11). Mean parameter values are shown in increasing order. 95% confidence intervals are calculated using the percentile method over models calibrated to bootstrapped datasets (500 repeats).

Figure 4.5 shows the estimated interaction effects, ranked in increasing order, for each model. The confidence intervals are calculated using the percentile method (Rousselet et al., 2021) across the bootstrapped models' parameters. Most parameters display small 95% confidence intervals, indicating an accurate fit for both models. On average, the symmetric model exhibits smaller confidence intervals than the asymmetric model ($0.065 < 0.085$), as expected given the asymmetric model's larger number of parameters. In both models, the self-interaction effect on `Politics` is significantly larger than all other parameters, as evidenced by the non-overlapping confidence intervals, supporting our earlier observation regarding Figure 4.4. Moreover, the remaining self-interaction effects are, in general, significantly larger than the pairwise effects, with the exception of `CC Impact` in the symmetric model, and `CC Impact` and `CC Worry Others` in the asymmetric model.

Given the use of regularisation in model calibration, we must be careful not to draw conclusions about the *existence* of edges from this figure (Epskamp, Borsboom, et al., 2018). Instead, we may consider the proportion of bootstrapped models for which each edge is nonzero, shown in Figure 4.6. Relations which are selected in all models are not shown.

All edges excluded from the complete model, calibrated on the full dataset (Figure 4.4), have low selection probabilities in the bootstrapped models ($< 50\%$). The relation `CC Real` $\rightarrow$ `Politics` is excluded from the complete asymmetric model but included (bidirectionally) in the symmetric model. This is reflected in the corresponding selection probabilities—this edge is selected in 100% of bootstrapped symmetric models but only 23% of asymmetric models. Edges which *are* selected in the complete models

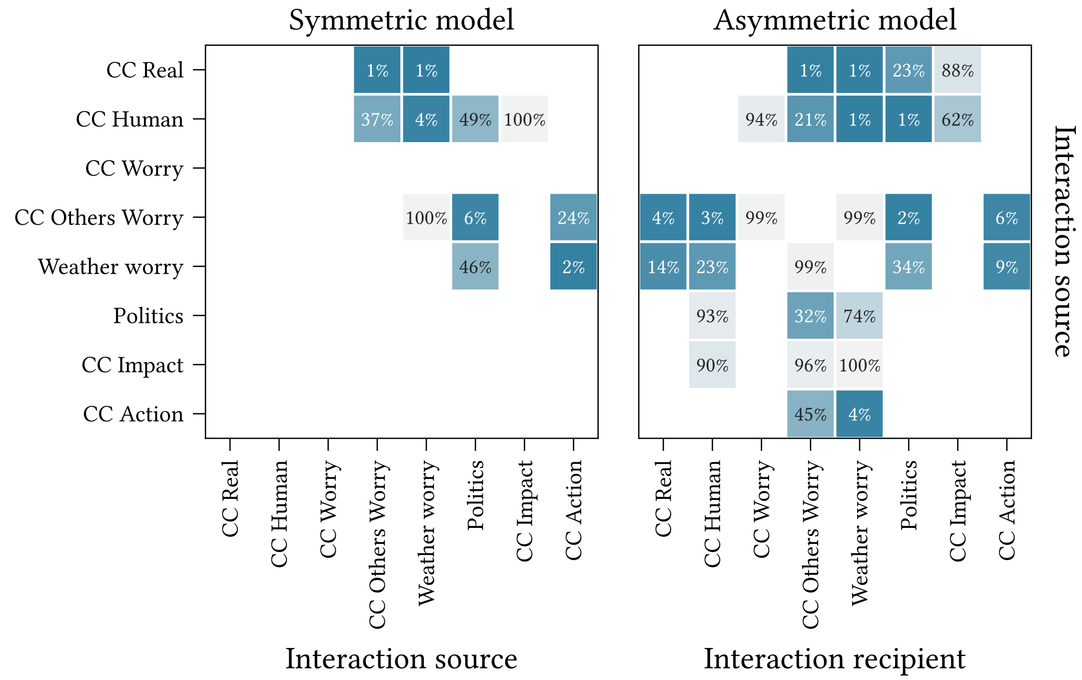


Figure 4.6: Edge selection probability for models calibrated to bootstrapped datasets sampled with replacement from the climate beliefs dataset (500 repeats). Edges are 'selected' if they survive regularisation with magnitude at least 0.01.

generally have a high selection probability. The notable exception in the asymmetric case, `CC Human` $\rightarrow$ `CC Impact`, has a relatively small effect size ($J_{i,j} \approx 0.04$).

We now evaluate the models' predictive capacities.[15] First, we examine the reliability of predicted transitions. Figure 4.7 shows the average binarisation probability for each belief (i.e., the probability that the corresponding observation in the second wave of the dataset is binarised to $+1$), binned by transition probability. Bins with no observations are not shown (e.g., extreme values for `CC Others Worry`). The two measurements are strongly correlated for most variables. The higher variation for `CC Real` and `CC Human` likely reflects the observed heavy skew in these variables toward larger values (see Figure 4.1). We observe no high/low probabilities for either `CC Others Worry` or `Weather Worry`, on account of the limited influence of other beliefs on these variables, as seen in the corresponding columns of Figure 4.4.

Next, we compare the calibrated model against a null model with only baseline activation (h_i) and self-influence ($J_{i,i}$) parameters. We measure the relative entropy of the binarised data from the second wave of the dataset, with respect to the probability distributions

[15]While we only show results for the asymmetric model here, the corresponding figures for the symmetric model are substantially similar, and all conclusions drawn regarding the asymmetric model apply also to the symmetric one.

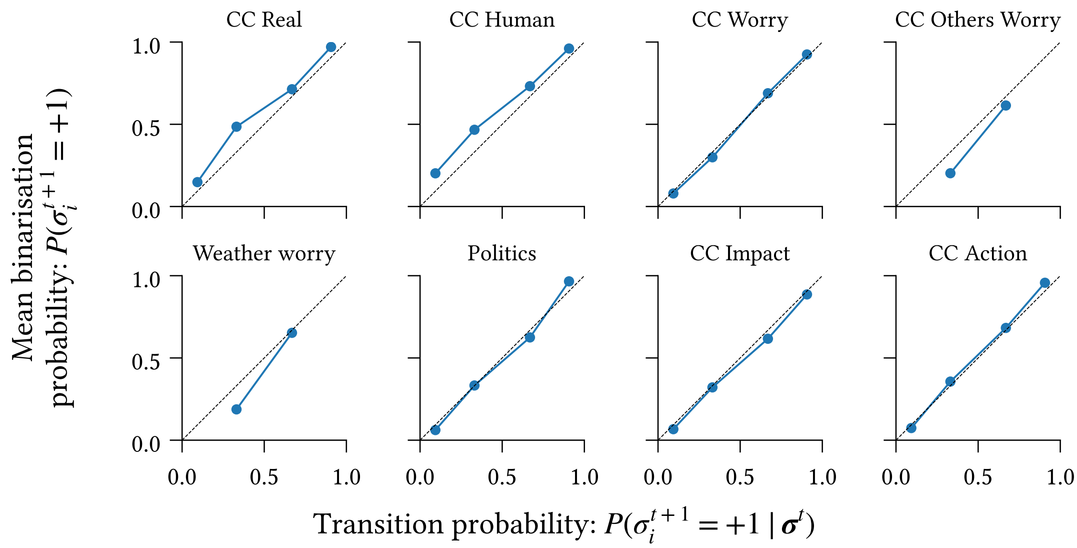


Figure 4.7: Reliability of transitions predicted by the model measured as the agreement between conditional probabilities under the model, $P_{\mathcal{M}}\left(\sigma_i^{t+1} = +1 \mid \boldsymbol{\sigma}^t\right)$, and expected proportion of $+1$ values after binarisation, calculated as the average transition probability for survey participants in a given conditional probability bin.

induced by each model given the first wave observations. For each participant and belief, we calculate the relative entropy as

$$D(P \parallel Q) := H_C(P, Q) - H(P) \tag{4.2}$$

Where $P := P\left(b_\xi\left(x^t_{(m),i}\right)\right)$ is the distribution over binarisations of the observation in the first timestep, and $Q := P_{\mathcal{M}}\left(\sigma^{t+1}_{(m),i} = +1\right)$ is the distribution over subsequent states according to the model, marginalising over the binarisation of the previous observation,

$$Q := P_{\mathcal{M}}\left(\sigma^{t+1}_{(m),i} = +1\right) = P_{\mathcal{M}}\left(\sigma^{t+1}_{i,(m)} = +1 \mid b_\xi\left(\boldsymbol{x}^t_{(m)}\right)\right) \cdot P\left(b_\xi\left(\boldsymbol{x}^t_{(m)}\right)\right) \tag{4.3}$$

The functions H and H_C denote the entropy and cross-entropy, respectively. The entropy term quantifies the uncertainty in the binarisation process, and the cross-entropy measures the 'expected surprise' when comparing model predictions with the true binarised dataset. Formally, these are defined as follows:

$$H(P) = -\sum_{s \in \pm 1} P(s) \log_2 P(s) \tag{4.4}$$

$$H_C(P, Q) = -\sum_{s \in \pm 1} P(s) \log_2 Q(s) \tag{4.5}$$

The relative entropy, $D(P \parallel Q)$, is small when either (i) the model accurately predicts

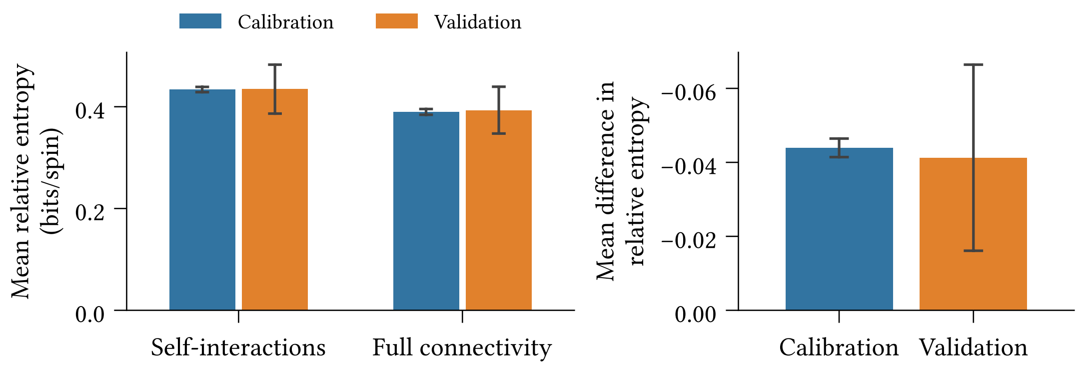


Figure 4.8: *(Left)* Mean absolute relative entropy of the binarised dataset under the calibrated model, for fully connected and null (only self-interactions) models, using 10-fold cross-validation. Average calculated across beliefs and participants. *(Right)* Mean difference in relative entropy. Confidence intervals are two standard deviations around the mean.

the observed binarised value such that the cross-entropy is low,[16] or (ii) the binarisation process is very uncertain. It follows that the relative entropy is high when the model fails to accurately predict the next state, despite the binarisation process being fairly deterministic.

To test the effect of including cross-interactions, we fit the fully connected model and null models using 10-fold cross-validation. For each survey participant in the validation (holdout) split, we calculate the mean difference in relative entropy between the fully connected and null models. Figure 4.8 shows the mean relative entropy across individuals and beliefs for the null and fully connected models (left), and the mean *difference* in relative entropy, averaged across survey participants (right).

We find, in the right-hand panel, that the fully connected model leads to a significant reduction in relative entropy averaged over survey participants ($p < 0.05$). While statistically significant, the reduction is small. Since the measurement is averaged across survey participants, this finding suggests that the self-interaction model is sufficient to explain the behaviour for most participants—reflecting the slow-moving dynamics of the observed belief system—but not all participants. We anticipate that individuals whose belief states change more between observations are better explained by the fully connected model than the null model. Figure 4.9 explores this hypothesis, showing the empirical probability density and cumulative distribution functions for the mean difference in relative entropy. We see substantial variation between individuals; the weak right tail and heavy left tail indicate cases that are better explained by the null and fully connected models, respectively.

[16]The relative entropy is strictly non-negative (Csiszar, 1975), so the binarisation entropy is a lower bound on the cross-entropy. This aligns with our intuition—the model cannot possibly be highly accurate when the binarisation process is very uncertain.

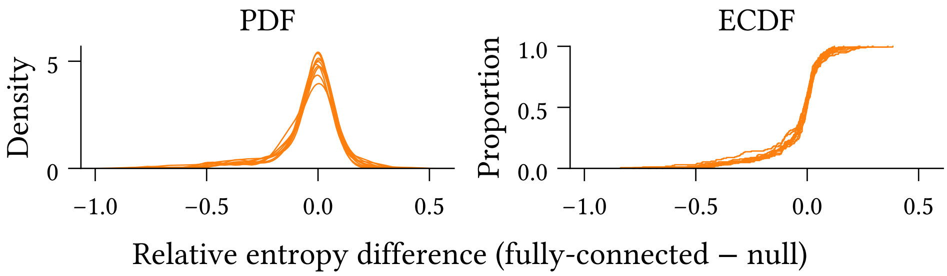


Figure 4.9: Probability density function (left) and empirical cumulative distribution function (right) of the mean difference in relative entropy between the fully connected and null (only self-interaction) models across survey participants. Negative values indicate lower relative entropy for the fully-connected model.

We examine a sample of cases from each tail in Figure 4.10. Each panel shows the change in binarisation probability between the survey waves for a sampled survey participant. In the first three panels of the top row, we observe scenarios in which most beliefs are initially aligned, and a subset of the remaining beliefs then update to align with this set, i.e., where the system shifts toward a more consistent state. Conversely, in the first, second, and fourth panels in the bottom row, we see the opposite scenario play out. That is, the system is initially well-aligned, yet some beliefs update to a less consistent state.

The observed behaviour aligns with our expectations, namely that the model with cross-interactions outperforms the null model in scenarios where the belief system state updates toward a more consistent state (according to the theory of cognitive dissonance), and is outperformed when participants' behaviour contradicts this theory.

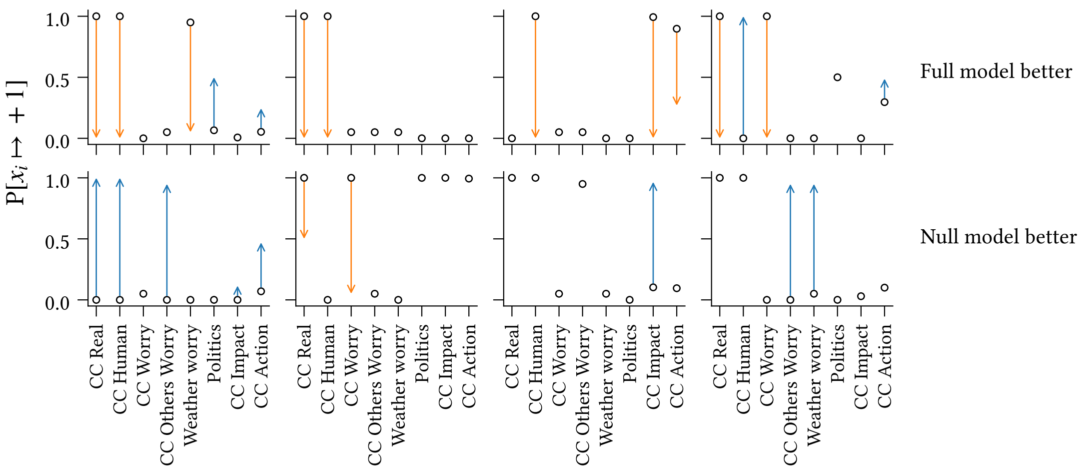


Figure 4.10: Observed transitions for sample survey participants in the left (*Top*) and right (*Bottom*) tails of Figure 4.9. Circles indicate the probability that participants' observations in the first wave are binarised to +1. Arrows denote the change in the second wave, colour-coded according to the direction of change.

Chapter 5

Experimental Methods

We now shift to the second half of this study, which addresses the research questions outlined in the first chapter. We will employ a combination of data-driven and simulation-based methods, using the models calibrated to the climate beliefs dataset (Chapter 11) in the previous chapter. This chapter outlines our experimental approach and the key quantities measured in the following chapters.

Note that our use of the climate beliefs dataset imposes a specific interpretive context of interpretation on our findings. In particular, we do not claim that the set of beliefs considered here comprehensively reflects the range of possible climate-related beliefs, and we expect that our results are likely to be sensitive to the (geographic and situational) contexts underlying the climate beliefs dataset.

5.1 Assessing asymmetric influence using the KBS model

To investigate the prevalence of asymmetric relations in the asymmetric model calibrated to the climate beliefs dataset, we examine differences in directional interaction effects between each pair of beliefs in the asymmetric models calibrated via bootstrapping in the previous chapter.

For each bootstrapped model, $\mathcal{M}_{(i)}$ with parameters $\langle \boldsymbol{J}_{(i)}, \boldsymbol{h}_{(i)} \rangle$, we obtain an estimate for the directional differential matrix:

$$\Delta_J^{(i)} = \boldsymbol{J}_{(i)} - \boldsymbol{J}_{(i)}^T \tag{5.1}$$

Recall that the k'th row of $\boldsymbol{J}_{(i)}$, for $k \in [N]$, describes the strength and direction of influence *from* the belief S_k *toward* each other belief. Hence for $k, \ell \in [N]$ we should interpret the element $\left(\Delta_J^{(i)}\right)_{k,\ell}$ of the directional differential matrix as the excess influence of belief S_k on S_ℓ. A positive value indicates that S_k exerts greater influence on S_ℓ than S_ℓ does on S_k.

Note that, while in the previous chapter (↩) we cautioned against using bootstrapped confidence intervals to test for the *existence* of edges by comparison with zero in regularised models, this caution does not apply to the *comparison* of edge weights via the mean difference (Epskamp, Borsboom, et al., 2018).

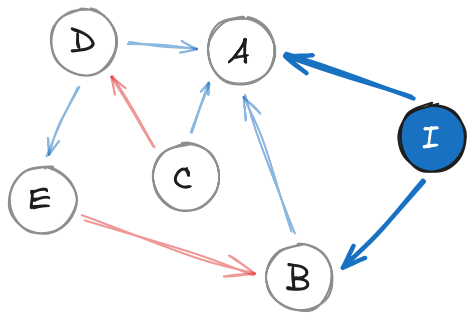
Degree centrality and strength centrality are commonly used to assess the *importance* or *position* of a belief within a belief system (Brandt et al., 2019; Brandt & Vallabha, 2023; Bringmann et al., 2019; Chambon et al., 2022). Degree centrality measures the number of edges adjacent to a node in a network. Strength centrality measures the sum of absolute edge weights. In directed networks, however, these centrality indices can be calculated using either incoming or outgoing edges, and the resulting values may differ. To understand the potential impact of symmetric-influence assumptions on judgements regarding belief importance, we therefore additionally compare the incoming and outgoing values of both centrality indices for each belief that exhibits significant asymmetric influence.

5.2 Modelling interventions

We now outline our approach to modelling interventions in the KBS model. In this study, we consider interventions which affect a belief system's *state*. Notably, this excludes interventions which affect the structure of a belief system, for instance, by changing the existence, sign, direction, or effect size of influence relations between beliefs.[17]

Let $\mathcal{M}$ be a belief system model with parameters $\langle \boldsymbol{J}, \boldsymbol{h} \rangle$. We can consider an intervention as an auxiliary node, I, in the belief system network, with state fixed at a particular value and outgoing edges toward a subset of beliefs:

[17]Although not considered here, we implement such structural interventions in the Ising Python package, which is published alongside this study. We discuss this topic in Chapter 8.



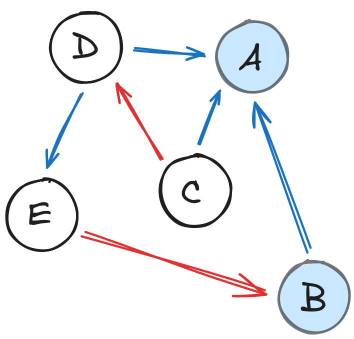
In the diagram above, the intervention node I influences the beliefs A and B. Consider the effective baseline activation for belief A at time $t+1$, as defined in Equation 2.4, given a previous configuration $\boldsymbol{s}^t$:

$$h_A^{\text{eff}}(\boldsymbol{s}^t) = h_A + \sum_{\text{node } j} J_{jA} s_j^t \tag{5.2}$$

Since the state of I is fixed (in this case, at $+1$), the intervention node's contribution to A's effective activation baseline is simply J_{IA}. We can then rewrite Equation 5.2, interpreting the intervention effect as an adjustment to the baseline activation h_A:

$$h_A^{\text{eff}}(\boldsymbol{s}^t) = (h_A + J_{IA}) + \sum_{\text{node } j \neq I} J_{jA} s_j^t \tag{5.3}$$

It follows that we can equivalently model interventions more simply as adjustments to the baseline activations of particular beliefs:



Let us define this more formally. For a belief system model $\mathcal{M}$ as defined above with $N \in \mathbb{N}$ nodes, an **intervention** is the function $\varphi_{\mathcal{M}} : \boldsymbol{\delta}_h \mapsto \mathcal{M}'$, where $\boldsymbol{\delta}_h \in \mathbb{R}^N$ is a vector of offsets to the baseline activations, and the model $\mathcal{M}'$ has parameters $\langle \boldsymbol{J}, \boldsymbol{h}' \rangle$, where

$$\boldsymbol{h}' = \boldsymbol{h} + \boldsymbol{\delta}_h \tag{5.4}$$

Our approach is analogous to the method described by Dalege et al. (2017) for modelling interventions in symmetric belief system models, variants of which have been used in other simulated-intervention studies (Bertero & Franetovic, 2025; Lunansky et al., 2022; Schlicht-Schmälzle et al., 2018; Wu et al., 2026).

The principles underlying this approach can also be used to model other exogenous factors that influence belief system dynamics but are not influenced by them. For instance, Dalege et al. (2025) use an analogous mechanism in a social network context, to represent how an individual's actual belief state infuences another individual's second-order belief about that state (i.e., their belief about what the first individual believes).

5.2.1 Intervention strength

Recall that in Chapter 2, we defined the probability distribution over states for a given belief as a non-linear (specifically, logistic) function of that belief's effective baseline activation (Equation 2.5). Since interventions adjust the baseline activation linearly, an intervention with fixed strength $\delta_h \in \mathbb{R}$ will affect individuals differently depending on their pre-intervention states.

Figure 5.3 illustrates how the probability of activation of an intervention belief (Equation 2.5) varies across different intervention scenarios as a function of the effective baseline activation. Since interventions constitute offsets to the effective baseline activation, they are reflected as horizontal shifts in the logistic curve.

The solid line denotes the null scenario in which no intervention is applied. Consider two individuals, positioned at different locations on this base curve:

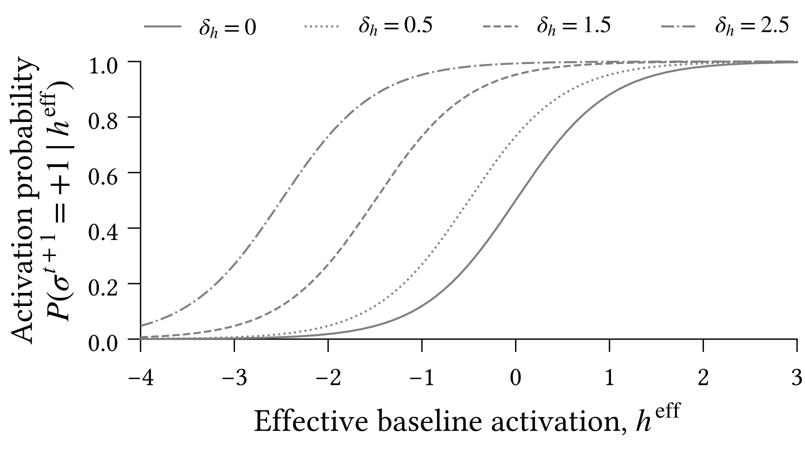


Figure 5.3: Impact of different intervention strengths ($\delta_h \in \{0, 0.5, 1.5, 2.5\}$) on activation probability for the intervention belief. Activation probability is calculated using Equation 2.5.

1. $h_{\text{eff}} = 0$: Ambivalent disposition toward the intervention belief, and

2. $h_{\text{eff}} = -2$: Strong negative disposition toward the intervention belief.

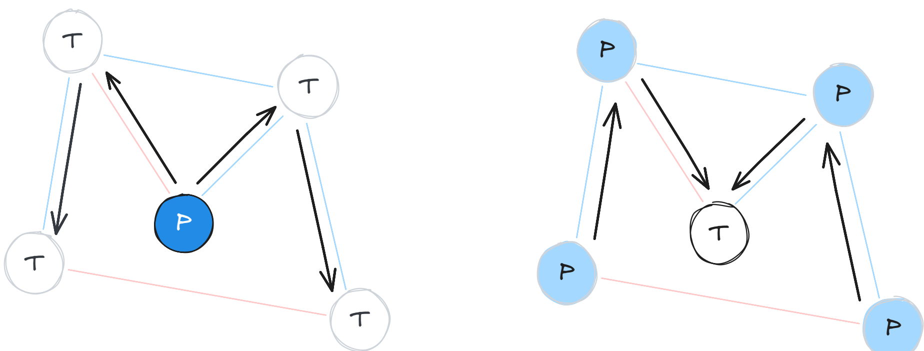
To understand the impacts of intervention strength on each individual, we draw a vertical line upward from the solid curve at each location, and read off the activation probability at each intersection with the other intervention curves. Under the null scenario ($\delta_h = 0$), the first individual exhibits an equal tendency toward each state, while the second is almost certain to adopt the state -1.

Notice that interventions are not experienced equally by the two individuals. For a weak intervention ($\delta_h = 0.5$), the first individual sees a substantial jump in activation probability, while the corresponding increase for the second individual is negligible. The stakes are reversed under a strong intervention ($\delta_h = 2.5$). Prior to intervention, the first's indifference makes them maximally susceptible to interventions, while the second's strong negative stance is only affected by large interventions. On the other hand, the second individual's state has the *potential* to change more than the first individual's. Consider that the interventions can, at best, shift the first individual from indifference to support, whereas the second individual could shift from opposition to support.

5.3 Outbound and Inbound interventions

For the experiments in the following chapters, we distinguish between **outbound** and **inbound** interventions, illustrated below with arrows denoting (a subset of) the pathways through which interventions may propagate.

Outbound **Inbound**



Outbound intervention experiments (left) examine how interventions on a particular belief propagate to other beliefs. On the other hand, inbound intervention experiments (right) consider how a focal belief's behaviour is differently affected by interventions elsewhere in the network. We refer to the belief on which an intervention is applied as the **point of intervention** (marked 'P' in the diagram) and the belief whose resulting state is measured as the **target** (marked 'T' in the diagram).

5.4 Choice of points-of-intervention and targets

Since our outbound intervention experiments are intended to assess differences in intervention propagation between symmetric and asymmetric belief systems, we select points of intervention that are likely to highlight these differences. So that interventions can propagate, we prioritise beliefs that are reasonably influential in both belief system models. Additionally, we attempt to select beliefs with varying degrees of asymmetry, conditional on the models featuring such beliefs.

For the inbound experiments, our intention is to assess differences in intervention strategy and the relative effectiveness of different interventions. In this case, ideal target beliefs are those which are reasonable targets in actual interventions. We select `CC Action` (attitudes toward climate action) for all inbound intervention experiments. Compared with the other variables in the climate beliefs dataset (for instance, beliefs about the existence of climate change or climate-related worry), we expect this belief to correlate most directly with climate-related behavioural choices, and hence be a justifiable intervention target.

5.5 Counterfactual experiments with Common Random Numbers

Most of our experiments involve comparing simulated behaviour across different KBS models, either to examine the impact of an intervention by comparing with a null (i.e., no-intervention) model, or to assess the impact of interaction symmetry assumptions by comparing symmetric and asymmetric models. The KBS model dynamics are inherently stochastic, with different trajectories reflecting differences in experimental conditions characterised by unmeasured, exogenous influencing factors.

We use Common Random Numbers (CRN) (Law, 2015, pp. 588–604) to ensure that measured differences in such experiments reflect differences between the compared models, rather than differences in the stochastic conditions under which they are simulated. Specifically, in every situation where we compare observables between two models, we: (i) initialise the models using identical random seeds, and (ii) for each stochastic operation, we use the same random numbers for each model. The second point follows directly from the first in the KBS model, since the number of stochastic operations per simulation timestep is fixed, and these occur in a pre-specified order.

As a safeguard, we verify each use of CRN in our experiments by comparing final floating-point samples drawn for each model after each simulation. We write these to disk and compare the sampled values after all simulations are complete. An error is raised if any differences are found.

5.6 Measuring the effects of interventions and asymmetry

We now define three quantities of interest that we will use to assess the impacts of interventions and the impacts of assumptions about asymmetry. The first two are used in Chapter 6 to investigate population-level model behaviour, while the third is used in Chapter 7 to examine individual-level intervention impacts. All three quantities are computed using Common Random Numbers, as discussed above.

First, let us establish some shared notation. For a KBS model $\mathcal{M}$, we denote the result of simulating $\mathcal{M}$ for $t \in \mathbb{N}$ timesteps (as in Section 2.4) from an initial state $\boldsymbol{s}_0$, as $\mathcal{M}^t(\boldsymbol{s}_0)$. Also, we denote the activation probability (as defined in Equation 2.5) for a belief S_i following this simulation as $p_i^t(\boldsymbol{s}_0, \mathcal{M})$.

The **effect-of-intervention (on state)** and the **effect-of-asymmetry**, respectively, measure the impact of an intervention on observed belief system behaviour with reference to our expectations in a no-intervention scenario (Definition 5.6.1), and the difference in observed behaviour in the asymmetric KBS model, compared with our expectations in the symmetric model (Definition 5.6.2).

Definition 5.6.1 (Effect-of-intervention (on state)) For an intervention (KBS) model $\mathcal{M}_\delta$ with an arbitrary point of intervention, we define the **effect-of-intervention (on state)** for $\mathcal{M}_\delta$ as the change in outcome at time $t \in \mathbb{N}$, with respect to the null (no-intervention) model, $\mathcal{M}_0$:

$$f_{\text{state}}(\mathcal{M}_\delta) = \mathcal{M}_\delta^t(\boldsymbol{s}_0) - \mathcal{M}_0^t(\boldsymbol{s}_0) \tag{5.5}$$

Definition 5.6.2 (Effect-of-asymmetry) Let $\mathcal{M}_{\text{asym}}$ be a calibrated asymmetric KBS model with an arbitrary intervention applied, and $\mathcal{M}_{\text{sym}}$ be a corresponding symmetric KBS model calibrated to the same dataset. We define the **effect-of-asymmetry** for $\mathcal{M}_{\text{asym}}$ as the difference in the effect-of-intervention (on state) with respect to the symmetric model:

$$f_{\text{asym}}(\mathcal{M}_{\text{asym}}) = f_{\text{int}}(\mathcal{M}_{\text{asym}}) - f_{\text{int}}(\mathcal{M}_{\text{sym}}) \tag{5.6}$$

Finally, the **effect-of-intervention (on influence)** measures the impact of an intervention on the total influence experienced by a given belief, with reference to our expectations under the no-intervention scenario (Definition 5.6.3). Note that this is richer than the effect-of-intervention (on state), which derives from the activation probabilities. We can make use of this richness in individual-level experiments, while in population-level experiments, Definition 5.6.1 is more straightforwardly applicable.

Definition 5.6.3 (Effect-of-intervention (on influence)) For an intervention (KBS) model $\mathcal{M}_\delta$ with an arbitrary point of intervention, we define the **effect-of-intervention (on influence)** for $\mathcal{M}_\delta$ as the change in activation probability at time $t \in \mathbb{N}$, with respect to the null (no-intervention) model, $\mathcal{M}_0$:

$$f_{\text{influence}}(\mathcal{M}_\delta) = p_i^t(\boldsymbol{s}_0; \mathcal{M}_\delta) - p_i^t(\boldsymbol{s}_0; \mathcal{M}_0) \tag{5.7}$$

5.7 Identifying the conditions for effective interventions

In the simulated interventions described above, the only distinguishing factor between individuals is their pre-intervention state. It follows that any difference in belief system dynamics between distinct individuals—after accounting for simulation stochasticity—is the result of their different initial states. Therefore, to identify the conditions under which an intervention is effective, it then suffices to characterise the set of *initial states* which yield effective interventions and distinguish them from those that do not.

Our goal is to find a function which maps from an intervention effect to sets of (unbinarised) states that yield that effect, $g : \mathbb{R} \to 2^X$, where $X = [-1, +1]^N \subset \mathbb{R}^N$. We will call g the **effect characterisation function**. We measure intervention effects here using the effect-of-intervention on influence (Definition 5.6.3).

Consider a function which does the opposite, mapping initial states to a measure of intervention effect, $f : X \to \mathbb{R}$. Given such a function, we can straightforwardly construct the corresponding effect characterisation function as:

$$g : y \mapsto \{x \in \pm 1^N \subset \mathbb{R}^N \mid f(x) = y\} \tag{5.8}$$

However, while technically satisfying the definition of the effect characterisation function, such a function would likely be useless for qualitatively determining the kinds of initial states which yield effective interventions. The elements of the codomain have (potentially) infinite cardinality, and are not immediately interpretable. For our purposes, we are less interested in the specific states that yield a given intervention effect, and more interested in a *concise description* of that set of states.

5.7.1 Method: Shallow regression decision tree

Regression decision tree models can provide such descriptions, using inequality bounds to partition the initial state space into regions, each of which is assigned a predicted effect. Given a parameterised regression decision tree, we may construct a concise approximation to the effect characterisation function by identifying, for each predicted effect, the combination of inequalities which define the corresponding infinite set of initial states. When the parameter estimation algorithm is restricted to shallow trees (e.g., depth 3 or 4, where depth refers to the number of inequality bounds defining each region), these combinations can also be interpreted as rules or *personas*.

Since the regression decision tree produces a full tree, all personas for a tree with depth $d \in \mathbb{N}$ have size d by default. However, these can often be compressed. When two personas differ in only one feature dimension, split at the same value, and both predict high-effect interventions, we can combine them into a single persona that omits that feature (i.e., spans the entire feature dimension).

Chapter 6

Existence and impact of asymmetry in belief systems

In this chapter we address the primary research questions, presented in Chapter 1:

RQ1. To what extent are belief-level influences *symmetric* or *asymmetric* in climate change belief systems in the United States?

RQ2. How do symmetric-influence assumptions impact expectations regarding population-level intervention strategy and effectiveness in climate change belief systems? To what extent do the impacts depend on where, in the belief system, an intervention is applied?

6.1 Existence of asymmetric relations

To assess the prevalence of asymmetric influence relations among climate-related beliefs, we examine the estimated sampling distribution of directional differentials, as defined in Section 5.1, obtained from the set of asymmetric KBS models calibrated to the bootstrapped climate beliefs dataset ($n = 500$) in Chapter 4.

Figure 6.1 shows the median directional differential (defined in Section 5.1) for each pair of beliefs in the asymmetric model, ordered from smallest to largest. The 90% confidence intervals are calculated as the 5th and 95th percentiles of the directional differential matrix elements across bootstrap samples. For each pair of beliefs, we display only the directional differential for which the median is positive (since Δ_J is symmetric), and we exclude diagonal entries (which are zero by definition).

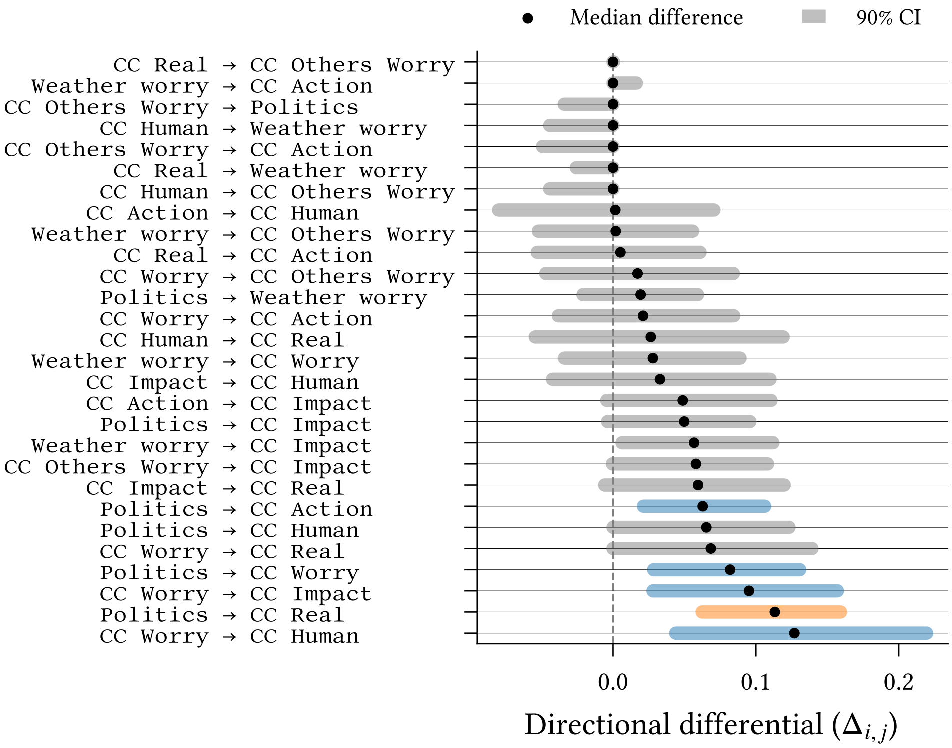


Figure 6.1: Directional differentials—the difference between directed interaction effects for a pair of beliefs—in reverse order of median value. Confidence intervals are calculated using the percentile method on models calibrated using bootstrapping ($n = 500$). Grey confidence intervals indicate insignificant asymmetry. For cases with significant asymmetry, the colour indicates that one (**orange**) or both (**blue**) of the directed interaction effects are nonzero.

Most relations appear to be inconclusively asymmetric (characterised by confidence intervals containing zero), with a small number of asymmetric relations. We can partition the asymmetric relations into two groups, according to whether directional interactions exist in only one direction, indicated using orange confidence intervals, or both directions with different strengths, indicated with blue confidence intervals (see the interaction matrix in Figure 6.2; cell entries are identical to the matrix shown in Figure 4.4). Pairs with two nonzero interactions comprise `Politics` $\rightarrow$ {`CC Action`, `CC Worry`} and `CC Worry` $\rightarrow$ {`CC Impact`, `CC Human`}; the only pair with a unidirectional interaction is `Politics` $\rightarrow$ `CC Real`.

Comparing directional strength and degree centrality indices for `CC Worry` and `Politics` across bootstrapped models, we find significant differences ($p < 0.05$) for strength centrality on `CC Worry`, and both strength and degree centrality on `Politics`. The difference in degree centrality for `Politics` arises from the difference in selection probability between outbound and inbound interactions with `CC Human` and `Weather Worry` (Figure 4.6). In particular, most bootstrapped models contain only outbound interactions from `Politics` to these beliefs.

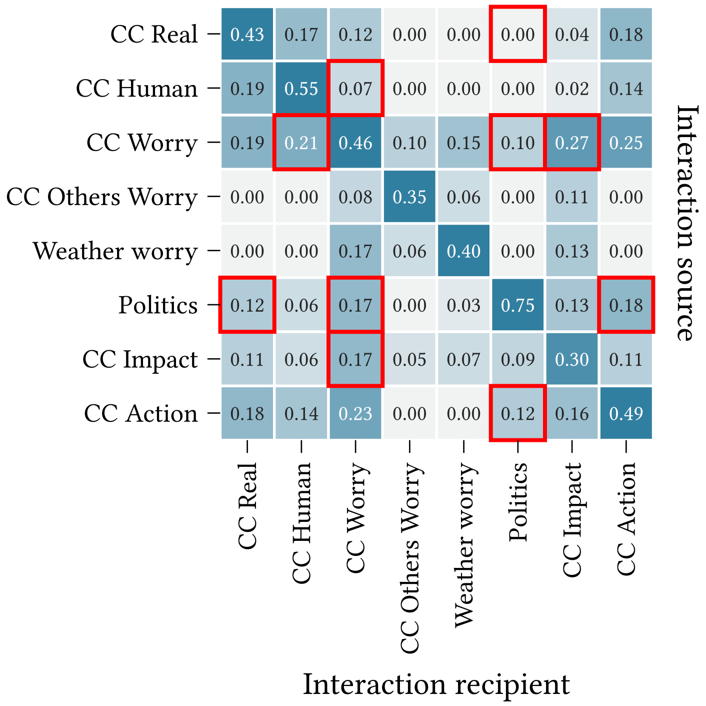


Figure 6.2: Interaction effect matrix ($\boldsymbol{J}$) for the asymmetric belief system model calibrated to the full climate beliefs dataset. Red borders indicate significant asymmetric pairs. The matrix is otherwise identical to the asymmetric interaction matrix in Figure 4.4.

Most directional differentials display substantial uncertainty (confidence interval width > 0.1 on average), reflecting the parameter accuracy discussed in Chapter 4 (Figure 4.5). This results in several inconclusive cases, where the median bootstrap-estimate directional differential is nonzero but the confidence interval contains zero. In other cases, we observe apparent symmetric or near-symmetric relations (for example, `Weather worry` $\longrightarrow$ `CC Others Worry`).

At the top of Figure 6.1, we see several apparently skewed confidence intervals. These all correspond to cases where both interactions are zero in the complete model (Figure 6.2), so the skews likely reflect differences in edge-selection probabilities between the two directional interactions, with one interaction effect occurring more frequently than the other. For instance, `CC Others Worry` $\longrightarrow$ `CC Human` exhibits a larger apparent skew than `CC Others Worry` $\longrightarrow$ `CC Real`, and this is reflected in observed differences in selection probability (Figure 4.6).

Due to the large confidence intervals on most directional differentials, we are less likely to observe significant asymmetric relations between variables with small interaction effects (since the possible difference in effects is smaller by definition). Indeed, all observed significant cases of asymmetry include at least one directional interaction effect with a magnitude exceeding 0.1. This raises the question of whether asymmetry is explained by a high total interaction influence in combination with sampling error. However, this is disputed by `CC Action`, which has several strong interaction effects, with outbound effects typically larger than those of `Politics`, yet does not exhibit significant asymmetric influence over any other variables. On the contrary, it is asymmetrically *influenced* by `Politics`.

6.2 Asymmetry affects intervention dynamics

We now investigate differences in belief system behaviour under intervention in symmetric and asymmetric models calibrated to the climate beliefs dataset. In this section, we focus on *collective* impacts (for instance, the proportion of individuals whose behaviour shifts due to an intervention). These reflect population-level intervention scenarios, such as media campaigns or policies which affect most individuals. We will return to *individual* impacts of intervention in the next chapter to understand who is most affected by interventions targeting attitudes toward climate action.

We consider both outbound and inbound interventions (defined in Section 5.3) to examine differences in how interventions propagate in general between the symmetric and asymmetric models, and to investigate how intervention strategy may differ between the two models. For outbound intervention experiments, we examine the effects of intervening on the following beliefs:

- `Weather worry`: Level of worry about future extreme weather events
- `CC Worry`: Level of worry about current and future climate change
- `Politics`: General political party identification and ideology

These choices are informed by the results of the previous section and the selection criteria outlined in Section 5.4. All three variables have (at least some) nontrivial outbound influence, `CC Worry` and `Politics` both exhibit significant asymmetric influence over other beliefs, and `Weather worry` has no asymmetric interactions.

For inbound intervention experiments, as discussed in Section 5.4, we consider interventions aimed at influencing the `CC Action` belief, which captures individuals' general support for pro-environmental action on climate change (e.g., support for specific policies and increased regulation, and affirmative attitudes toward individual responsibility on climate change).

In each of the intervention experiments described below, we use the models calibrated to the full climate beliefs dataset in Chapter 4. For a given experiment, we initialise the associated model (symmetric/asymmetric; intervention/null) with the binarised observations from the final wave of the climate beliefs dataset. We then draw consecutive samples from the model using parallel Glauber dynamics (Section 2.4) until $t = 5$, before measuring the target belief state.[18] To account for stochasticity in both the simulation procedure and binarisation of individuals' initial states, we perform 500 repeats for each intervention.

We first consider the impact of intervention strength on intervention effectiveness. Since, as discussed in Section 5.2.1, the change in behaviour at the point of intervention is sensitive to intervention strength, it follows that intervention strength may also affect the absolute or relative effects of an intervention, elsewhere in the belief system. Figure 6.3 compares the mean effect of intervention on state (Definition 5.6.1) for each possible target belief, across 'weak' and 'strong' interventions ($\delta_h \in \{0.5, 2.5\}$), for each of the outbound points of intervention listed above.

For each point of intervention, we observe a strong linear relationship between the intervention effects for the two strengths, indicating that—at the population-mean level—the impact of intervention strength on effect is largely a matter of scale, applying similarly across all target beliefs. For the following experiments, it then suffices to consider only a single intervention strength. As our current focus is on intervention propagation and the effects on beliefs beyond the point of intervention, we set $\delta_h = 2.5$,

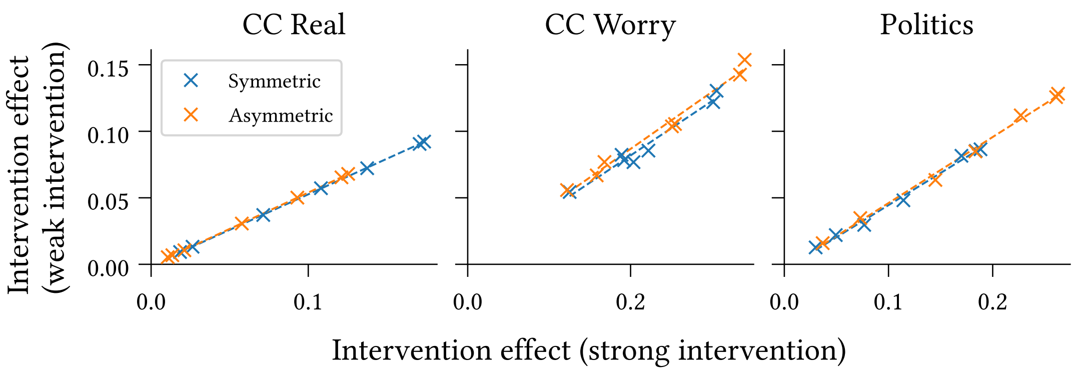


Figure 6.3: Weak ($\delta_h = 0.5$) and strong ($\delta_h = 2.5$) interventions exhibit strong positively correlated effects of intervention on belief state, consistently across points of intervention (panels) and intervention targets (scatterplots). This finding holds for both symmetric and asymmetric models calibrated to the climate beliefs dataset.

[18]Since the duration between waves in the climate beliefs dataset is approximately six months, in the calibrated model this corresponds to measuring the belief states roughly two and a half years after intervening.

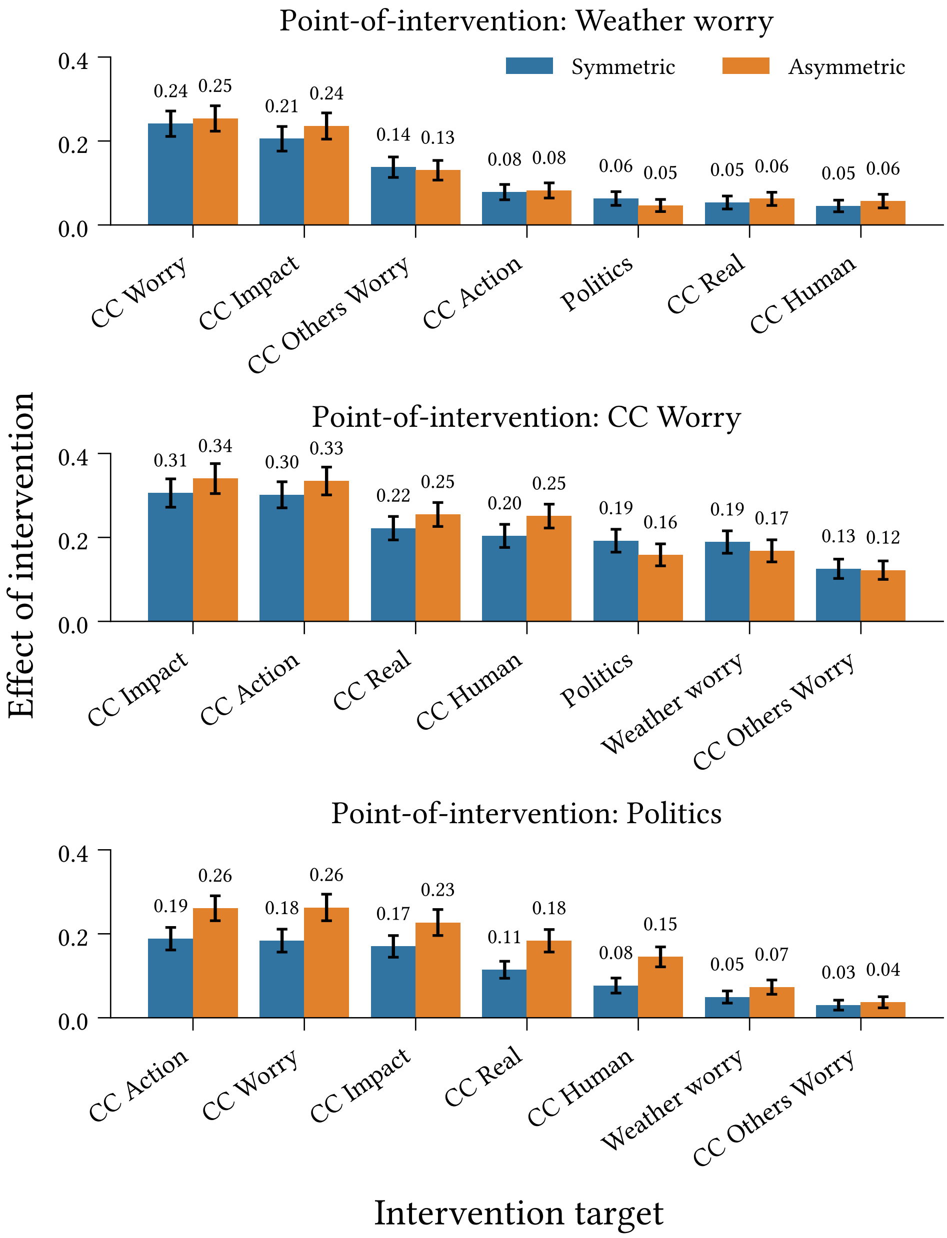


Figure 6.4: Outbound effect-of-intervention on belief state (Definition 5.6.1) at each target belief for interventions targeting `Weather Worry`, `CC Worry`, and `Politics`, in symmetric and asymmetric belief system models calibrated to the climate beliefs dataset. Intervention strength: $\delta_h = 2.5$. Confidence intervals display 1.96 standard deviations around the mean effect, measured across 500 repeated simulations.

so that the interventions are considered generally effective at shifting behaviour at the point of intervention.

Figures 6.4 and 6.5 show the mean effect of intervention on state (Definition 5.6.1) and the mean effect of asymmetry (Definition 5.6.2), respectively, across individuals

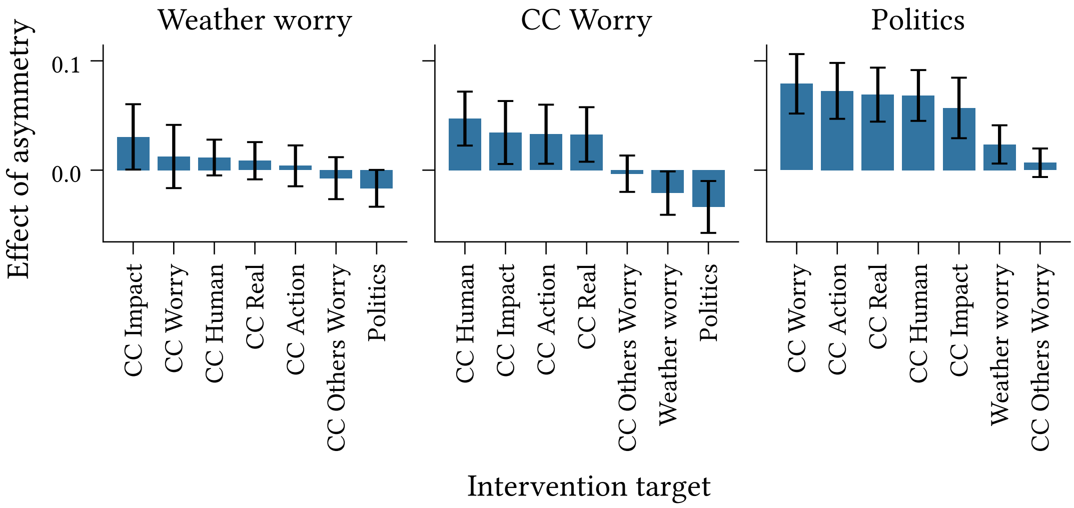


Figure 6.5: Outbound effect-of-asymmetry (Definition 5.6.2) at each target belief for interventions on `Weather Worry`, `CC Worry`, and `Politics`, for models calibrated to the climate beliefs dataset. Intervention strength: $\delta_h = 2.5$. Confidence intervals display 1.96 standard deviations around the mean effect-of-asymmetry ($n = 500$).

for the points of intervention listed above. The confidence intervals show two standard deviations around the mean values, as calculated across repeats.

We observe in Figure 6.4 that different targets exhibit different effects of intervention. The effect generally decreases with the magnitude of the interaction from the point of intervention to the target belief; however, we also note that all effects are nonzero ($p < 0.05$). This suggests that for the measurement time examined here, interventions act primarily through direct interactions with the target belief, with some indirect influence propagating through intermediary interactions. For instance, while `Weather Worry` has no direct influence on `CC Action`, we observe a nonzero effect-of-intervention due to indirect propagation, e.g., via `CC Worry`. We see further evidence of this through comparison with measurements taken at time $t = 10$ (approximately five years in the calibrated model; Figure B.1 in Appendix B), which show larger increases in the effect-of-intervention for variables with stronger incoming interactions from non-intervention beliefs.

Note that Figure 6.4 does not provide an accurate reflection of the difference in intervention effects between symmetric and asymmetric models, as overlapping confidence intervals around *mean values* do not, in general, imply insignificant differences for individual repeats, particularly when using Common Random Numbers (Section 5.5). The effect-of-asymmetry, shown in Figure 6.5, provides a cleaner comparison between the symmetric and asymmetric models. We observe no significant differences between the two models for the `Weather Worry` scenario. For the remaining two scenarios, however, we see several significant differences. In particular, all pairs of variables with asymmetric direct relations (that is, `Politics` $\rightarrow$ {`CC Action`, `CC Worry`, `CC Real`} and `CC Worry` $\rightarrow$ {`CC Impact`, `CC Human`}) display significant differences.

Comparing `Politics` and `CC Worry`, we observe that while `CC Worry` exhibits greater effects of intervention (Figure 6.4), `Politics` exhibits larger mean effects of asymmetry (Figure 6.5), although we note that the significance of the latter differences is inconclusive. In both cases, we see that cross-interaction effect values, J_{ij}, in the symmetric model are typically between the corresponding inbound and outbound interaction effects in the asymmetric model. We contrast this with `CC Impact`, which has mostly symmetric relations. While `CC Impact` and `Politics` have extremely similar outbound interaction strengths in the asymmetric model, `CC Impact` is considerably more influential than `Politics` in the symmetric model (i.e., it has higher average interaction effects).

Notice that the asymmetric model includes outbound (but not inbound) interactions from `Politics` to both `CC Human` and `Weather Worry`, while the symmetric model omits interactions with these beliefs entirely.[19] In contrast, `CC Worry` has inbound and outbound interactions with all other beliefs in both models. This explains the observed differences in the effects of intervention and asymmetry for `CC Worry` and `Politics`: `CC Worry` has higher effects of intervention in both models on account of its strong outbound interaction effects, which are also mostly retained in the symmetric model, while the removal of interactions for `Politics` in the symmetric model results in lower influence and fewer (direct and indirect) paths through which interventions can propagate, compared with the asymmetric model.

Turning our attention to inbound intervention dynamics, we now consider the effects of different interventions aimed at influencing the state of `CC Action`. The top panel of Figure 6.6 shows the mean effect of intervention on the state of `CC Action` for each possible point of intervention.

The mean effect-of-intervention, calculated separately for each point of intervention, erases information about the expected *relative* effectiveness of different interventions. To assess relative effectiveness, we also estimate the expected ranking over points of intervention (the bottom panel of Figure 6.6). For each repeat, we order the points of intervention in increasing order of the average collective effect of intervention. Higher values are assigned higher ranks.[20] In the case of a tie, we assign all tied beliefs the

[19] Recall that this difference in connectivity was found to be significant in the previous section (↩).

[20] We take ranks to be ordered by index, such that rank 1 is the lowest and (in this model, with seven possible points of intervention) rank 7 is the highest.

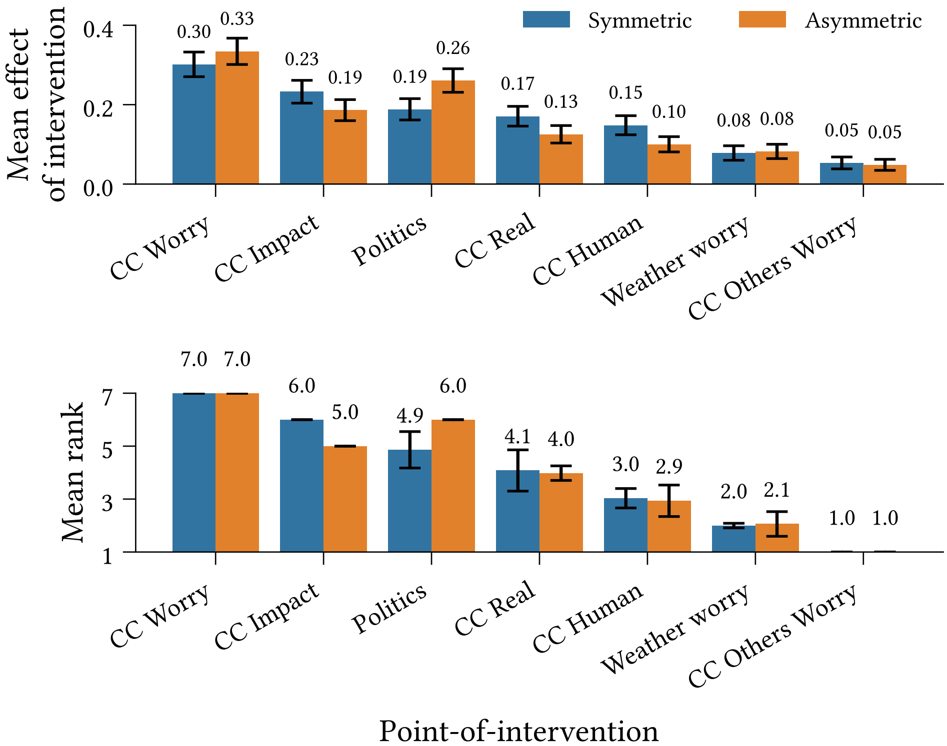


Figure 6.6: Inbound mean effect-of-intervention on belief state (Definition 5.6.1) (*Top*) and mean rank (*Bottom*) for interventions targeting `CC Action`. Ranks are calculated per-repeat with respect to the average effect-of-intervention. Higher values are assigned higher ranks. Confidence intervals display 1.96 standard deviations ($n = 500$).

minimum rank which would have been assigned to the group had they been distinct.[21]

As with the outbound experiments, we find that the effect-of-intervention on `CC Action` varies depending on where we intervene. Points of intervention (along the horizontal axes) are sorted in decreasing order of mean values. We observe identical orderings in the effect-of-intervention and ranking plots, indicating good correspondence between the two measures. We observe a clear inversion in this ordering for the asymmetric model, with `Politics` exhibiting the second-highest rank, and (at least) second-highest mean effect-of-intervention ($p < 0.05$).

Unlike in the outbound experiments, the order of effect sizes does not align well with the corresponding strengths of direct interactions into the target variable. For instance, `CC Real` and `CC Human` have the second- and third-strongest influence on `CC Action` in the symmetric model, yet the fourth- and fifth-highest effect-of-intervention. This likely reflects the low outbound connectivity of these variables compared with `CC Impact`

[21]For instance, suppose we have five variables, $\{A, B, C, D, E\}$ with average collective effects $\{0.2, 0.2, 0.6, 0.4, 0.1\}$. Then the assigned ranking would be $\text{Rank}(A, B, C, D, E) = [2, 2, 5, 4, 1]$. The variables B and C are both assigned the rank 2. No variable is assigned rank 3.

and `Politics`, despite the latter two beliefs having a smaller direct influence on the target. This suggests that indirect influence may play a critical role in the dynamics of interventions for some beliefs.

Most confidence intervals around the mean ranks are small, indicating negligible (if any) variation between simulations or possible binarisations of the dataset. However, we do observe three larger, overlapping confidence intervals for each model. These arise from variation in the mean effect-of-intervention, causing occasional inversions of the ranking order. Note that we only observe inversions between points of intervention which are adjacent in the sorted order.

Chapter 7

Heterogeneous belief systems and intervention effects

In this chapter, we address the final two research questions presented in Chapter 1:

RQ3. How does individual-level intervention effectiveness, measured as the resulting shift in behaviour toward a desired belief state, depend on an individual's pre-intervention beliefs, in asymmetric climate change belief systems?

RQ4. How do asymmetric climate change belief systems for liberal and conservative subpopulations compare structurally, with respect to sparsity, strength of belief-level interactions, and presence of particular interactions?

7.1 Heterogeneity in intervention effects

While the previous chapter was primarily concerned with the *collective* effects of intervention, Figure 7.1 shows that the expected effects of interventions targeting attitudes toward climate action can vary substantially among survey participants.

Figure 7.2 shows the conditions (or, personas) for which interventions on different nodes in the belief system are expected to be effective for targeting attitudes toward climate action. These are obtained using the regression decision tree approach to approximating the effect characterisation function, as described in Section 5.7. We regard an intervention as *effective* for a given individual if the expected effect-of-intervention on influence (Definition 5.6.3), estimated using the mean across $n = 500$ simulations, is greater than or equal to the population upper quartile. Since the threshold for an intervention to be considered effective is specific to each point-of-intervention, we exclude points-of-

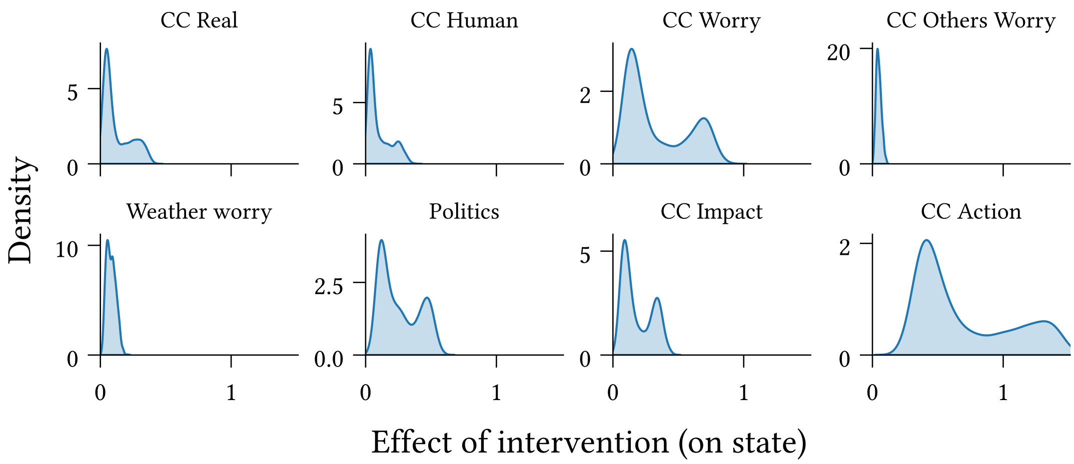


Figure 7.1: The expected effect-of-intervention on state (Definition 5.6.1) for strong interventions ($\delta_h = 2.5$) targeting attitudes toward climate action exhibits substantial variation between survey participants.

intervention whose upper quartile is below 0.1 (`CC Human`, `CC Others Worry`, and `Weather Worry`), which exhibit no meaningfully effective interventions.

Note that the identified personas are not necessarily complete. For instance, suppose that a pair of variables are highly correlated and are 'Low', in the initial state, whenever the intervention effect is high. A complete characterisation includes both variables; however, a decision tree is likely to include only one, since after splitting on one of the two variables, the other is redundant. This is especially true for shallow decision trees.

The expected difference in activation probabilities between the intervention and null models exhibits a clear bimodal distribution across survey participants for each scenario, with the higher mode contained within the upper quartile (indicated by the darker shaded regions). For each point-of-intervention we observe a small set of personas. In each case these personas exhibit high prevelence among individuals with high intervention effects, and considerably lower prevalence for other individuals, indicating that the identified personas accurately characterise the conditions for effective interventions. While prevalence among the high-effect individuals is generally high, this does vary across points-of-intervention. For instance, 20% of individuals predicted to be in the high-effect category for interventions on `CC Worry` are not represented by the identified persona, while for interventions on `Politics` only 3% of high-effect individuals are not represented.

For the present analysis, it is important to recognise that the intervention effect measured here does not, in general, reflect the change in belief state with respect to the initial state. It is entirely possible that both intervention and null models lead to an increase,

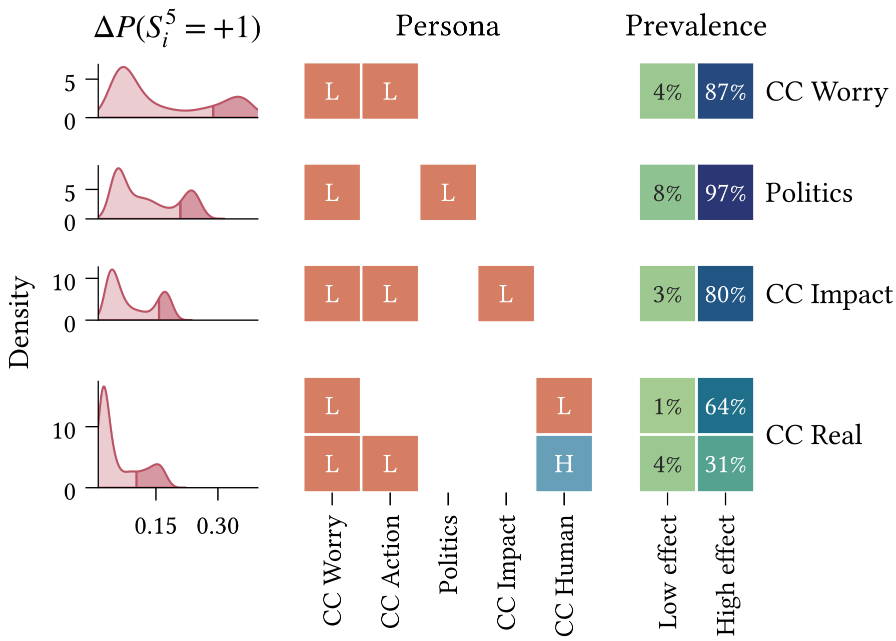


Figure 7.2: Pre-intervention states, or 'personas', for which interventions on different beliefs (rows) are most effective for changing attitudes toward climate action in the asymmetric KBS model calibrated to the climate beliefs dataset. The **left** panel shows the distribution of effects of intervention on influence (Definition 5.6.3) for the `CC Action` belief at $t = 5$ (approximately two and a half years) using a strong intervention ($\delta_h = 2.5$), with respect to the no-intervention scenario. The **center** panel shows the personas corresponding to those individuals in the upper quartile of effects (shaded region in the left panel). Rows are distinct personas. Cells correspond to intervals in the initial state space dimensions, i.e., the different beliefs, with $L = [-1, 0]$ and $H = [0, +1]$. The **right** panel shows each persona's prevalence within the upper quartile ('High-effect') and below the upper quartile ('Low-effect') of effects. Personas are identified using depth-3 regression decision trees fit with a mean-squared error objective and mean effect of intervention on influence as the response variable ($n = 500$). $R^2 > 0.9$ for all decision trees, measured using 10-fold cross validation.

or decrease, in the probability of desired behaviour for the target belief. Hence it is most appropriate to view 'high-effect' interventions as those which achieve substantially more desirable states than would be observed given no intervention.

With this in mind, we now consider several specific features of interest in the identified personas. Firstly, we observe that all personas require a low initial state for `CC Worry`. This may result from this variable's low inertia and high connectivity—in particular, its large outbound interaction effect toward the target variable (see Figure 6.2). These factors result in `CC Worry` being relatively influential and influentia*ble*, and therefore an effective indirect pathway for various interventions targeting `CC Action`. The significance of the requirement that `CC Worry` be *low* is evident when comparing the implications for the null and intervention models. Due to `CC Worry`'s considerable outbound interactions, an intervention which successfully changes this belief has increased potential for propagation. In the null model, however, these interactions work against the desired result—if `CC Worry` remains low, it exerts this influence on all other beliefs.

Second, we observe that for each point-of-intervention (with the exception of `CC Real` on account of high correlation with `CC Human`, relating to the remark earlier in this section ↩), a necessary condition for high effect is that the initial state of point of intervention itself be low. That is, for an intervention on X to be successful, X must not already be too high. This aligns with our prior expectations regarding the varied effects of interventions with respect to pre-intervention state (see Section 5.2).

Finally, the `CC Real` scenario is unique in that it includes two prevalent personas. The more prevalent persona corresponds to situations in which individuals are worried about the current/future effects of climate change, but do not believe that climate change is human-caused.[22] In the case where an individual does believe that climate change is human-caused, this intervention may still be effective, so long as they do not already have a positive attitude toward climate action.

7.2 Heterogeneity in belief system structure

Until this point, we have considered belief systems as common to a population of individuals. However, the relations between beliefs are inherently individual in nature. The existence, direction, and degree of relation between two beliefs is dependent on an individual's own beliefs regarding their relatedness.

Here we investigate the extent to which belief systems may vary between groups of individuals with different self-reported political ideologies. While this is still far from representative of the differences between individuals (Brandt & Morgan, 2022), it will allow us to examine general structural differences between these subpopulations. We use the models calibrated, in Chapter 4, to the subsets of the climate beliefs dataset comprising individuals who consistently (that is, in both waves) report their political ideology as being either 'conservative' or 'very conservative' ($n = 507$) or 'liberal' or 'very liberal' ($n = 375$).[23] We exclude the `Politics` variable from the model, since this is captured by the partitioning of the dataset.

[22] The negative state for `CC Human` is an aggregation of the beliefs that climate change is a natural phenomenon, and that climate change is not real (has no causes). See Chapter 11 for further details.

[23] Note that the total number of samples across the conservative and liberal subsets is smaller than the number of samples in the complete climate beliefs dataset, since we only retain data from participants whose ideology is with consistent across survey waves.

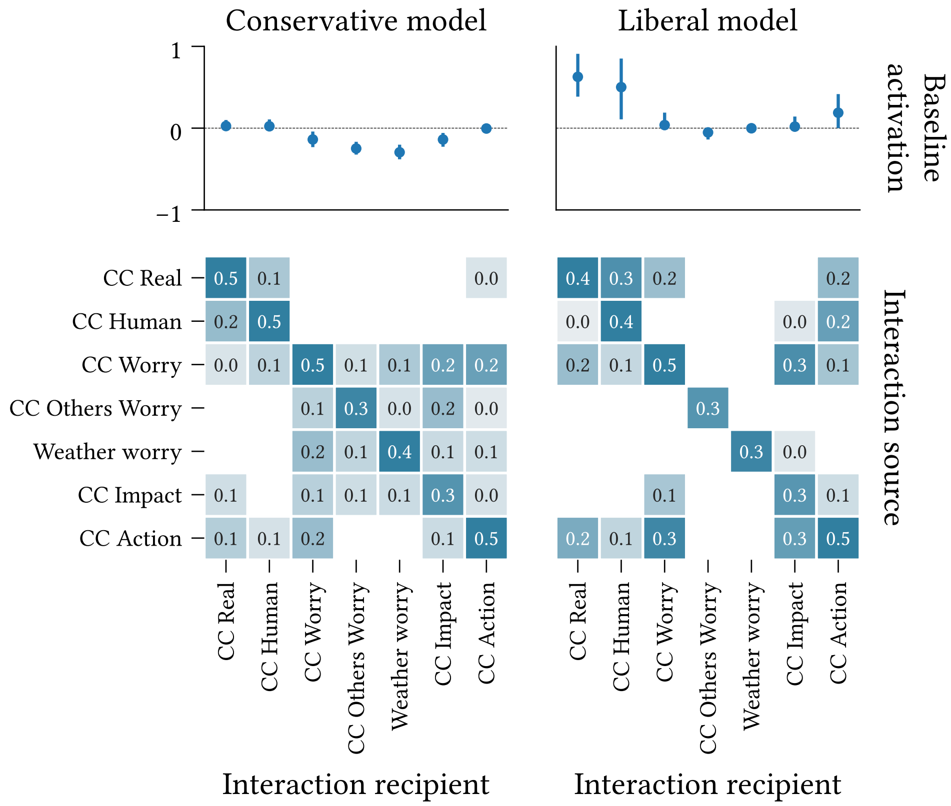


Figure 7.3: (*Top*) Baseline activations, $\boldsymbol{h}$, and (*Bottom*) interaction effect matrices ($\boldsymbol{J}$) for asymmetric belief systems calibrated to the conservative ($n = 507$) and liberal ($n = 375$) subsets of the climate beliefs dataset (Chapter 11) using the parameter estimation method in Section 3.2.

Figures related to model calibration can be found in Appendix B. The estimated parameters exhibit considerably higher uncertainty than in the model calibrated to the complete dataset due to the smaller sample sizes (Figure B.2). We find only one significant case of asymmetry, on `CC Worry` $\longrightarrow$ `CC Impact` in the conservative model (Figure B.3), and only a minority of significant differences ($p < 0.05$) between interaction strengths, all for edges which are stronger in the conservative model (Figure B.4).

Figure 7.3 shows the baseline activation parameters, $\boldsymbol{h}$, and interaction effect matrices, $\boldsymbol{J}$, for the conservative and liberal subpopulations. The model calibrated to the liberal subpopulation exhibits much higher uncertainty in the estimated baseline activations for belief in the existence and human-causes of climate change than the conservative subpopulation model; however, in the liberal model these are reliably positive. The conservative model shows negative baseline activations for worry about climate change and extreme weather, beliefs about others' worry, and beliefs about the general impacts of climate change. While these are generally low-magnitude, in several cases their sizes

Data subset	Sparsity	Mean nonzero interaction effect
Full	0.30	0.13
Conservative	0.38	0.10
Liberal	0.60	0.17

Table 7.1: Sparsity (proportion of missing cross-interactions) and mean interaction effect (over cross-interactions) for asymmetric models calibrated to the conservative and liberal subsets of the climate beliefs dataset, compared with the model calibrated on the complete dataset.

are comparable with the interaction effects influencing these variables (e.g., for `CC Worry Others` and `Weather Worry`).

We observe several structural differences between the two models, as well as with comparison to the model calibrated on the full dataset (Figure 6.2). Notably, the model calibrated to the liberal subpopulation has higher sparsity (proportion of missing cross-interaction edges) and mean interaction effect over cross-interactions than either the conservative model or the complete model (Table 7.1). Among the interactions present in the liberal model, 82% also occur in the conservative model, contrasting the 54% of conservative interactions which are also in the liberal model.

The conservative model displays broad (yet mostly weak) reinforcing interactions between climate-related worry, beliefs about others' worry, and climate-impact beliefs. This contrasts with the liberal model, in which, among these beliefs, only `CC Worry` and `CC Impact` are non-trivially related. Moreover, we observe that `CC Others Worry` and `Weather Worry` in fact have *no* incoming cross-interactions in the liberal model.

We also note differences in the interaction effects influencing `CC Action` between the two models. While the liberal model expects attitudes toward climate action to be influenced substantially by individuals' beliefs regarding the existence and nature of climate change, these influences are absent or trivial in the conservative model. Instead, we observe that worry about climate change (`CC Worry`) is the only large cross-interaction toward `CC Action` in the conservative model.

Comparing now with the complete model (calibrated to the full dataset), we find that interactions absent from the complete model are generally also absent from both smaller models. The exceptions are $\{$`CC Others Worry`, `Weather Worry`$\}$ $\longrightarrow$ `CC Action`, which are present in the conservative model, albeit with small effect sizes.

The converse does not hold; in several cases edges are absent from the smaller models, yet included in the complete model (e.g., `CC Human` $\longrightarrow$ `CC Action` in the conservative model, `CC Worry` $\longrightarrow$ `Weather Worry` in the liberal model). While it is tempting to interpret these as differences in the ideology-specific belief systems, this inference is not necessarily valid. Given the smaller datasets used to calibrate these models—and the slow-moving dynamics of the measured variables—these edges may be excluded on the basis that we do not observe their effects. This is less likely for effects which are more substantial in the complete model, and thus supported better by the dataset.

Chapter 8

Discussion

In this chapter we review the research questions posed back in Chapter 1 (restated below) in light of the results presented in the previous two chapters, and we discuss their place in the broader context of belief system dynamics. We then conclude the chapter by discussing limitations of our findings and implications for future work on belief system dynamics and belief-level interventions.

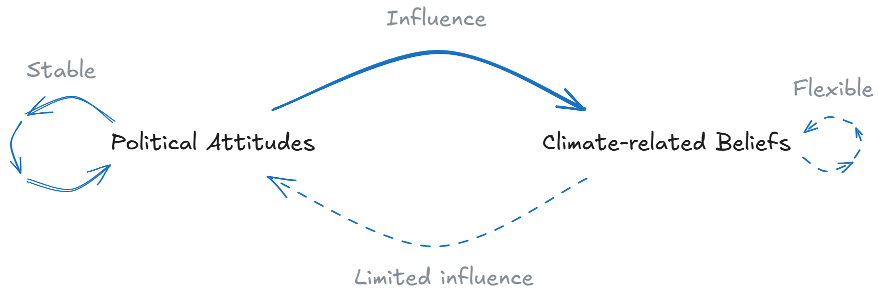
RQ1: *To what extent are belief-level influences* symmetric *or* asymmetric *in climate change belief systems in the United States?*

On the question of existence, Section 6.1 demonstrated asymmetric relations between several pairs of beliefs, while also finding that asymmetry may not *always* be exhibited. Two beliefs—political ideology/alignment and climate-related worry—displayed significant excess influence over several other beliefs.

How should we interpret these asymmetric relations? Recall that relations in the KBS model reflect temporal influence—how much one belief constrains the future state of another (Chapter 2).[24] An asymmetric relation reflects a constraint differential, where one belief has greater influence on the other than vice versa. More generally, the asymmetric model distinguishes between two forces: influence and influentiability. Influence (the strength of outbound interactions) determines how the extent to which one belief constrains others, while influentiability (the strength of inbound interactions) determines the extent to which its own behaviour is constrained by others.

[24]Note that this interpretation (in particular, the use of the verb 'constrains') depends on the assumption that the calibrated model captures within-person associations. We discuss this matter in some depth later in this chapter (↪).

Political identity being more influential than influentiable in this context is consistent with previous studies on its mutual influences with climate-related beliefs in the USA. First, political identity has been shown to substantially impact both climate-related beliefs (Whitmarsh, 2011), as well as support for climate policies (Bumann, 2021), with individuals tending to support or oppose specific policies on the basis of partisan identification rather than policy content (Unsworth & Fielding, 2014; Van Boven et al., 2018).



At the same time, while public beliefs toward climate change in the USA have shifted significantly over the past two decades (Hamilton et al., 2015; Marlon et al., 2022), political identity has been shown to be highly stable over similar timeframes (Brandt & Morgan, 2022; Green & Platzman, 2024). This suggests that while political identity contributes meaningfully to climate-related beliefs, changes in these states are less likely to incite changes in political identity.

Climate-related worry exhibits asymmetric influence over beliefs regarding the anthropogenic nature and current impacts of climate change. Both relationships have received comparatively less attention in prior research than those of political political with climate beliefs. One possible explanation for these asymmetric relations is that climate worry and climate beliefs influence one another through different mechanisms. For instance, Mead et al. (2012) suggest that climate worry may promote information-seeking behaviour, leading to changes in climate-related beliefs.

For both variables, we found significant differences between inbound and outbound centrality indices (strength centrality for climate worry; strength and degree centrality for political ideology/alignment). The undirected-network versions of these centrality indices are often used to assess belief importance, influence, or position in belief systems. Our findings indicate that such assessments may be misleading for beliefs involved in several asymmetric relations.

RQ2: *How do symmetric-influence assumptions impact expectations regarding population-level intervention strategy and effectiveness in climate change belief systems? To what extent do the impacts depend on where, in the belief system, an intervention is applied?*

The consequences of asymmetry for belief system dynamics were subsequently addressed in Section 6.2, where interventions on political ideology/alignment in the asymmetric model were found to be almost universally more effective than in the symmetric model. Moreover, for interventions targeting attitudes toward climate action, political ideology/alignment ranked higher with regards to collective effect in the asymmetric model than the symmetric model, displacing a mostly-symmetric belief with similar outbound interactions. Interestingly, while interventions on climate-related worry typically outperformed those on political ideology/alignment in terms of absolute effect of intervention, the differences between the asymmetric and symmetric models for this variable were comparatively less pronounced.

It is important to note that the effectiveness of interventions on political ideology/alignment is *despite* this variable's high inertia (i.e., strong self-interaction effect). When this variable is negative, high inertia leads to a lower pre-intervention effective baseline activation, making it harder to intervene. The effectiveness of interventions on this variable may be explained by fact that if an intervention successfully 'flips' political ideology/alignment, the high inertia works in favour of the intervention.

For the measurement timescale used in these experiments (approximately 2.5 years), interventions appeared to act primarily through direct interactions, though smaller indirect effects were also present; this is consistent with empirical results obtained by Chambon et al. (2022) in an experimental study concerning beliefs relating to COVID-19. Additional experiments included in Appendix B found that longer timeframes led to increased indirect intervention effects. The presence of indirect effects is broadly expected given the pre-defined model dynamics; however, the observed magnitudes suggest that while state changes may propagate beyond direct connections, this process is typically slow. We return to this point shortly in our discussion on **RQ3** (↪), which finds that indirect propagation may nonetheless contribute meaningfully to intervention effectiveness at an individual level.

The distinction in behaviour between political ideology/alignment and climate-related worry likely relates to differences in influentiality and influentiability between the two beliefs. Climate-related worry is more influential than political ideology/alignment, on account of its increased outbound interactions and interaction strength. The key difference, however, lies in the beliefs' influentiability; while climate-related worry has inbound and outbound interactions with all other beliefs, political ideology/alignment has considerably fewer *inbound* than *outbound* edges.[25] That is, several beliefs influenced by this variable have no influence on it themselves. Since the symmetric model must include or exclude both directional interactions between a pair of beliefs, cases where the asymmetric model specifies an interaction in only one direction necessarily lead to differences in model behaviour. This draws attention to a broader issue regarding relational symmetry assumptions.

One of the symmetric model's key strengths is its small(er) parameter count. With fewer degrees of freedom than the asymmetric model, it can achieve more accurate parameter estimates when calibrated to the same dataset (i.e., smaller confidence intervals in Figure 4.5). However, this comes at the cost of model misspecification when a subset of the true relations are asymmetric. We might expect the inferred symmetric influence relations to be similar to the average of the corresponding directed effects, yet this reasoning turns out to be flawed for at least two reasons. Firstly, interaction effects are not estimated in isolation, but jointly with all other parameters which influencing the same spin (or *both* spins in the symmetric case). Hence, when one interaction parameter changes—for instance, if we replace a pair of asymmetric interactions by their average —the rest are likely to change as well. Secondly, regularisation is often used to obtain sparse network representations and reduce overfitting, and distorts parameter values non-linearly in the process. In reality we find that while most symmetric interactions lie between their asymmetric analogues, this is not always the case, and those that are do not fall predictably near the middle. As observed in the case of political ideology/ alignment, the symmetric model may also exclude pairwise interactions altogether, or create a (bi-)directional interaction where influence actually flows unidirectionally.

RQ3: *How does individual-level intervention effectiveness, measured as the resulting shift in behaviour toward a desired belief state, depend on an individual's pre-intervention beliefs, in asymmetric climate change belief systems?*

In Section 7.1 we found that intervention effectiveness depends predictably on individuals' pre-intervention belief states. Most high-effect interventions targeting attitudes toward climate action required low initial values for both the point-of-intervention and target.

Surprisingly, pre-intervention climate-related worry was required to be low for *all* of the personas identified as characteristic of effective interventions, even when this was neither the point-of-intervention, nor the target. This finding may be explained by the combination of high influence and influentiability associated with climate-related worry,

[25] Note that this point pertains to the model calibrated on the complete climate beliefs dataset. This model omits several interactions (due to regularisation) which are inconsistently excluded from the bootstrap models used to assess the significance of asymmetric relations.

thereby making this variable an effective indirect route for various interventions (not restricted to this particular target). This stands in contrast with political ideology/ alignment, which has similarly high influence on the target variable, but is harder to influence, as discussed above.

At first glance this finding appears to contradict our earlier discussion on **RQ2** (↩), which found that indirect interventions have limited impact on collective effects over short timeframes. However, recall that the corresponding experiments measured indirect intervention effects by examining beliefs with no direct influence from the point-of-intervention. In other words, all propagation of intervention effects to these beliefs is indirect. Compare this with the present analysis, in which the target belief (attitudes toward climate action) is directly influenced, nontrivially, by all of the considered points-of-intervention—this implies that effective interventions on these points-of-intervention, targeting this belief, are possible *in principle*. Therefore the apparent discrepancy in our findings is likely explained by low pre-intervention levels of climate-related worry marginally increasing the effectiveness of interventions which were already effective.

In general, the identified personas characterise the conditions for effective interventions accurately. The rare instances where individuals with these traits fell outside the high-effect region may be attributable to the rough-and-ready use of the upper quartile to classify high-effect interventions. On the other hand, while the personas capture *most* cases where interventions are effective, this varies between points-of-intervention. This directly reflects the limited representational capacity of effect characterisation functions based on shallow decision trees; a tree depth of three permits at most eight personas to describe the full range of intervention effectiveness, each of which comprises at most three conditions.[26]

The results also highlighted a separate issue with our regression decision tree approach to characterising intervention effectiveness, namely that the descriptions may be incomplete when important features are highly correlated. This is a result of the decision tree optimisation procedure, which selects the ‘splits’ which best account for unexplained variance. We saw this reflected in the characterisation of effects for interventions on beliefs about the existence of climate change. While effective interventions here generally require that the initial point-of-intervention state be low, this was omitted due to high correlation between this variable and beliefs regarding the causes of climate change.

Importantly, the findings from this experiment must be interpreted in the context of the belief system model calibrated to the complete dataset. Individual differences in

[26]The personas are given by the paths from the root to each leaf; a binary tree with depth three has $2^3 = 8$ leaves, and thus eight personas.

belief systems (as discussed in a moment) are likely to result in heterogeneous responses to interventions beyond those characterised here. Potentially influential factors include political identity and social norms. Both vary individually (Laursen & Faur, 2022) and socially (Brown & Enos, 2021; Laursen & Faur, 2022; Webster et al., 2022), and both have been demonstrated as moderating the effects of interventions in the USA targeting support for climate policy and related beliefs (Allcott, 2011; Gromet et al., 2013; Unsworth & Fielding, 2014; van Valkengoed et al., 2022).

RQ4: *How do asymmetric climate change belief systems for liberal and conservative subpopulations compare structurally, with respect to sparsity, strength of belief-level interactions, and presence of particular interactions?*

Finally, Section 7.2 showed (potentially) substantive differences between asymmetric belief system models calibrated separately to conservative and liberal subpopulations; however, given the smaller sample sizes used to calibrate these models—and consequently, higher parameter uncertainty—these results must be interpreted with caution.

When calibrated on the non-bootstrapped data subsets, each model featured high-magnitude interactions not present in the other, yet while the liberal model was considerably sparser than the conservative model, most interactions in the former were also present in the latter. In comparison with the asymmetric model calibrated on the complete dataset, used in the prior experiments, the ideological models were both sparser, yet together had high overlap in connectivity with the complete model. We found only one case of significant asymmetry, which was also present in the model fit to the complete dataset.

These findings demonstrate both significant differences and similarities between relational belief system structures, conditional on political ideology, in line with earlier studies comparing belief systems on partisan identity (Lee et al., 2024). The observed differences suggest the presence of higher-order relationships between beliefs, consistent with the view that belief system relations, themselves, correspond to beliefs (e.g., belief that two states of affairs are related) (Fishbein & Ajzen, 1977, p. 219).

Sparse networks imply a restricted set of commonly-observed transitions. This is, to an extent, expected for the liberal model, which is calibrated using fewer observations than the conservative model; however, the higher sparsity is also corroborated by significantly lower edge selection for several interaction in the liberal model across bootstrap samples. This finding may suggest a greater degree of consistency in the observed dynamics between individuals who self-identify as liberal.

Previous studies by Gregersen et al. (2020) and Lind et al. (2024) identified beliefs about the anthropogenic causes of climate change as more strongly associated with various climate-related beliefs for non-right-leaning individuals. These findings were not replicated in our results; instead we found limited association (inbound or outbound) with this variable in both models. There are several possible explanations for this apparently contrary result, including parameter uncertainty in the present study. Firstly, both studies are based in a European, as opposed to US, context; Lee et al. (2025) showed cross-national differences in climate belief systems including related variables. Second, both measure association using cross-sectional correlational measures, which may differ considerably from the time-lagged interaction parameters used in the present study.

8.1 Limitations

When beliefs are fairly stable—as in the climate beliefs dataset—cross-sectional methods for inferring belief system structure tend to identify *between*-person associations more so than the *within*-person associations which are typically desired (Brandt & Morgan, 2022). We partially mitigate this problem by calibrating to longitudinal data, and using self-interaction terms to capture persistence in belief states. However, our model does not account for belief stability due to the presence of stable traits (Hamaker et al., 2015; Usami et al., 2019).

Hamaker et al. suggest modelling individual random intercepts (in our case, individual baseline activations) which are fixed across waves and account for some of the effects of stable traits. However, at least three waves of data are required for the random intercept model to be identifiable, while the climate beliefs dataset comprises only two waves.[27] In unreported experiments we tried a similar approach, modelling baseline activations as linear functions of demographic factors (e.g., age, education, rural/urban status), which requires only two waves to be identifiable. While the resulting baseline activations varied substantially between individuals, indicating that they captured some differences in stable traits, we found little-to-no impact on either the inferred interactions or the intervention experiment results. Hence further investigation is required to determine the extent to which the models calibrated in Chapter 4 capture within-person associations.

[27]For a given individual, each pair of consecutive waves constitutes a single observation (given we model conditional transition probability). For a set of $N \in \mathbb{N}$ beliefs the asymmetric model comprises $N^2 + N$ parameters, with M additional parameters when modelling individual baselines. Hence for $M \in \mathbb{N}$ individuals we require at least three observations per-individual for the problem to be determined. Note that more waves may be required when N is large, such that the number of non-intercept model parameters is larger than M.

Relatedly, since the climate beliefs dataset comprises only two waves, we cannot distinguish between belief system dynamics arising due to endogenous factors (the states of other beliefs) and exogenous factors (current events which affect beliefs). We implicitly assume that (i) exogenously-driven dynamics at an individual level are not substantially correlated between individuals, and thus not reflected in the model, and (ii) the time between measurements (approximately six months) is sufficiently short that widespread exogenous factors are negligible.

The second assumption has questionable validity, since for instance, both the 2020 US presidential election and January 6 Capitol attack occurred during this timeframe. However, we observed highly stable political beliefs over this period, consistent with prior studies (Green & Platzman, 2024). Given the significance and considerable media coverage of both events, as well as their highly-political nature, we would expect political beliefs to be affected more so than climate-related beliefs. The fact that we do not see this reflected may suggest minimal exogenous impacts, more generally, on the beliefs considered in this study.

Our present focus on endogenous dynamics also does not discount the substantial role of social interaction in belief change (Converse, 1964; DeGroot, 1974; Galesic et al., 2021; Karashiali et al., 2023) or stability (Brown & Enos, 2021; Prentice & Miller, 1996), including in intervention contexts (Brewer et al., 2017). Rather, this decision reflects the fact that the dataset used in this study did not include social network information. The matter of integrating social and cognitive forces in similar belief system models has been explored in several accounts (Aiyappa et al., 2024; Dalege et al., 2025; Rodriguez et al., 2016; van der Maas et al., 2020), generally assuming individuals have dual objectives of cognitive consistency and social coherence (Festinger, 1962; Gawronski & Strack, 2012; Heider, 1946). The Networks of Belief theory, proposed by Dalege et al. (2025), builds on the Causal Attitude Network model which also serves as the foundation for our proposed model. Since our work is mostly orthogonal to theirs, we anticipate that it would be straightforward to incorporate such social influences in our model.

To ensure computational tractability and model simplicity we have defined the KBS model dynamics as progressing via discrete timesteps. However, this means that the model parameters inferred during calibration are a function of the time between observations, known as the *time-interval dependency problem* (Gollob & Reichardt, 1987). Gollob & Reichardt (1987) demonstrate that in certain contexts, inferred effect size, sign, and existence can be sensitive to interval duration. This is particularly evident in cases where the true dynamics play out at a much faster timescale than the measurements.[28] Furthermore, Ryan & Hamaker (2022) argue that when several state updates can occur between observations, the inferred effects should be treated as *total effects* rather than *direct effects*. This is potentially critical for the present study, which fundamentally assumes that the interaction effects relating beliefs are direct effects.

However, we do not expect these problems to be especially consequential in the context considered in this study. In particular, per the results of previous studies (Green & Platzman, 2024; Kiley & Vaisey, 2020; Marlon et al., 2022), we expect most of the examined beliefs to be quite stable over the measurement timeframe (six months). As such, we do not expect any given belief to exhibit multiple significant changes in state between observations. We may, however, fail to detect causal dynamics between beliefs which play out on a quick timescale. For instance, suppose that following natural disasters, individuals typically become increasingly concerned about impending extreme weather, causing potentially rapid shifts in their beliefs regarding climate change. If both changes occur between measurements, our calibration method will attempt to explain both transitions with respect to individuals' belief states prior to the natural disaster, thereby missing the fast causal influence of extreme weather concerns on climate-related beliefs.

We now discuss three representational limitations of the model used in the present study, pertaining to (i) the use of pairwise relations, (ii) representation of belief states as Ising model spins, $s \in \{-1, +1\}$, as opposed to binary variables, $s \in \{0, 1\}$, and (iii) the non-dependence of interacton effect magnitude on belief states.

Consider the following (hypothetical) motivating example. Suppose that parents' attitudes toward childhood vaccination typically depend, in a simplistic way, on their perceived relative risk of a vaccine, and the disease it protects against. Positive attitudes prevail when the disease is considered more risky or dangerous than the vaccine. Values around family wellbeing conceivably influence the relationship between perceived risk and vaccination attitudes, serving to amplify the existing effect.

[28]In their original paper, Gollob & Reichardt (1987) illustrate this with respect to the effects of aspirin over time. See Ryan & Hamaker (2022) for extended discussion on this example.

Suppose that in an attempt to promote increased childhood vaccination rates, we undertake a media campaign appealing to family wellbeing values. The expected outcome is evident: attitudes improve for individuals who are relatively more concerned about the disease; for other individuals the opposite effect ensues.

This relational structure can be seen as a triplet-interaction extension to the belief system model used in the present study. In a belief system model with triplet interactions, the *effective* influence relations between pairs of beliefs can vary in accordance with the specific belief state. Findings from Brewer et al. (2017) suggest that such interventions, which selectively leverage or amplify existing beliefs may be more effective than interventions which attempt to change belief states. However, such relational structures cannot be captured in the pairwise model adopted here.

Next, we consider the implications of our decision to model belief states as spins, with values $s \in \{-1, +1\}$. Consider the following belief:

Believes that climate change is happening.

Which comprises both an epistemic position (*Believes that*) and a state-of-affairs (*climate change is happening*). Negating each of these components yields two reasonable choices for the 'opposite' state:

Does not *believe that climate change is happening.*

Believes that climate change is ***not*** *happening.*

As a consequence of our decision to model belief states as spins, we are assuming that beliefs always exert influence over associated variables, regardless of their state. However, it is not clear that this assumption is always reasonable, in particular for beliefs where the opposite state is a negation of the epistemic position (the first example), or in the case of neutral or ambiguous states (van der Maas et al., 2026).

Finally, while the present model assumes that interaction effects in asymmetric belief systems are independent of the 'influencing' variable's state, we argue that there is reason to think that this may not always be true. Consider two variables:

- **Happening:** The belief that climate change is happening.
- **Action:** General attitude toward climate action.

An individual who *does not* believe in climate change logically should not support climate action. In the asymmetric belief system model, this corresponds to a large positive interaction, such that **Action** aligns with **Happening**. However, individuals who *do* believe in climate change may nonetheless oppose climate action for other reasons (e.g., cost or prioritisation), suggesting that **Action** may be less constrained when **Happening** is high than when it is low.

8.2 Implications for future work

Our findings suggest that while the symmetric assumption may often be valid, or at least a reasonable approximation, asymmetric relations between beliefs are nonetheless possible. When true relations are asymmetric, not accounting for this when modelling belief interactions is a case of model misspecification, and can lead to incorrect inferences regarding the relative influence of different beliefs. The primary issue here is that symmetric models do not account for differences between a belief's influence (how much it affects the states of other beliefs) and influentiability (how much its own state is affected by other beliefs).

Despite the promising results of this study, several questions remain. The limited number of waves in the climate beliefs dataset prohibits us from distinguishing within-person and between-person effects, which is required to make strong claims regarding causal influences. We note that the broader CCCV survey from which the climate beliefs dataset used for calibration in this study does contain several additional waves, which are usable if we drop our requirements regarding the number of variables and inclusion of specific beliefs.

However, this presents a separate issue regarding the intervals between measurements, as inter-response times between different pairs of waves can differ significantly, violating the model requirements. Further consideration is required to determine whether, for instance, the additional waves can be used *only* to estimate individual baselines, while evenly-spaced observations are used to estimate interaction effects. Ryan & Hamaker (2022) propose continuous-time models as an alternative solution, enabling calibration with heterogeneous inter-response times, while also reducing the potential issues associated with the *time-interval dependency problem* (discussed under Limitations, ↩).

As discussed above, the regression decision tree approach to modelling the effect characterisation function can produce incomplete descriptions when important features are highly correlated (↩), and may not identify all effective-intervention conditions due to the somewhat arbitrary nature of tree depth to limit complexity (↩). The first issue can be resolved post-hoc (by assessing correlations with identified variables). One promising direction for the second is to identify more flexible rulesets, allowing arbitrary quantity and size, while penalising characterisation complexity directly, e.g., using description length (Aoga et al., 2018; Proença & van Leeuwen, 2020).

The models calibrated in Chapter 4 are assumed to reflect 'natural' belief system dynamics (i.e., minimal exogenous influence). By simulating interventions on these models, we are therefore assuming transferrability to situations *with* exogenous influences in the form of interventions. However, prior studies have demonstrated that belief system dynamics may differ in such situations, for instance due to increased salience of certain beliefs (Unsworth & Fielding, 2014). As such, experimental validation—and, ideally, calibration to data collected in controlled intervention scenarios—is a natural continuation to the present study.

Our findings also suggest and support several broader directions for future research. Firstly, we posited two explanations for the asymmetric relations observed with regards to political beliefs and climate-related worry ($\hookleftarrow$). These are retrospectively applied to the findings, so arguably have minimal evidential weight (Popper, 1963). However, they demonstrate how the asymmetric KBS model may be used to test (as opposed to generate) hypotheses about the general mechanisms by which asymmetric belief relations may occur.

Secondly, while belief system structure is typically expected to vary on an individual basis (Brandt & Morgan, 2022; Morgan & Wisneski, 2017), calibrating individual belief system models is generally considered infeasible due to the associated data requirements (Brandt, 2022). Moreover, due to the potential stability of belief dynamics, as observed in the present study, even substantial individual-level data is likely to reflect only a small set of possible belief states, limiting counterfactual analysis in individual models. However, our findings demonstrate structural similarities, as well as differences, across ideological groups. This suggests that a middle-ground approach, between individual-level and population-level networks may be effective in capturing both the ways in which belief systems are different, and the ways in which they are similar. This is akin to the notion of partial-pooling in multi-level Bayesian models, which assumes individuals in a group vary with respect to one another, but are constrained by common parameters (Gelman et al., 2013; Gelman & Hill, 2007; McElreath, 2020). Note that this is distinct from the estimation of individual *baseline activations* for the purpose of inferring within-person associations.

In our experiments we considered interventions as attempts to change the state of a belief directly. However, previous studies indicate that in certain situations, interventions targeting belief system *structure* may be more effective (Aggarwal & Hughes, 2026; Brewer et al., 2017). Rather than attempting to affect beliefs directly, structural interventions can be seen as amplifying or changing the influence between beliefs. Although we have not considered such interventions in this study, the open source *Ising* Python package released alongside this study already supports these, inviting future work on this topic.

Finally, we adopt a simplistic view of the interface between belief systems and the external world. This is most evident, for instance, in our treatment of interventions as acting directly and identically on individual beliefs, and the assumption that current events (e.g., the 2020 US presidential election) have minimal impact on observed belief dynamics. In reality, some beliefs may be easier or harder to affect directly than others. Communication of interventions or current events may be noisy (e.g., subject to interpretation), potentially impacting the magnitude, direction, and set of affected beliefs, also on an individual basis. Clearly, a more realistic treatment of this interface is required for the proposed methods to be useful in the context of actual interventions. From a modelling perspective, Dalege et al. (2025) consider a related problem, namely the interface between belief systems belonging to different individuals, modelling peer influence as indirect, via social beliefs which may only be indicative of the true external state.

Chapter 9

Conclusions

Most studies on belief system structure or dynamics acknowledge the likelihood that some beliefs may be more causally influential than others. Despite this, current research deals, almost universally, in belief system models which assume symmetric influence between beliefs. Theoretical models which capture asymmetric relations remain scarce, as do empirical studies on the existence of such asymmetry and its potential implications for belief system dynamics.

In this study we have addressed both theoretical and empirical shortcomings. We first introduced the Kinetic Belief System model (KBS) as a theory-based model of belief system dynamics based on the Causal Attitude Network model (Dalege et al., 2016), which: (i) *does not* assume symmetric influence and (ii) *does not* assume equilibrium dynamics. We then used the KBS model, calibrated to a longitudinal dataset comprising beliefs about climate change, collected in the US, to address the four research questions outined in the first chapter.

Our empirical investigation identified several clear asymmetric influence relations which appeared to be structured around a small set of highly influential beliefs (political views and climate-related worry). This suggests that (i) asymmetry may be the exception rather than the norm, and (ii) asymmetry may be better characterised in terms of the difference between a belief's influence and influentiability *in general*, than as a property of specific relations. Furthermore, our simulated intervention experiments demonstrated that asymmetry in a belief system can meaningfully affect intervention effectiveness, in some cases potentially changing conclusions about where to intervene. These effects were most pronounced when targeting or intervening on variables with several asymmetric influence relations.

Our secondary investigation examined how intervention effectiveness varies among individuals, and potential differences between liberal and conservative belief systems. In the first case, intervention effectiveness was found to depend predictably on individuals' pre-intervention belief states. Interestingly, *all* highly effective intervention scenarios featured a low initial level of climate-related worry, suggesting that beliefs which are both highly influential and highly influentiable may serve as effective indirect pathways for various interventions.

Second, despite relatively smaller sample sizes, conservative and liberal belief systems differed in sparsity and specific belief relations, while also displaying broad structural similarities. Both findings align with those of previous studies using symmetric networks (Lee et al., 2024). The similarities suggest that while belief systems likely vary between individuals, they also likely display common structural features (e.g., based on shared experiences or world-views). The observed differences in pairwise relations when stratifying by political ideology suggest the presence of higher-order relations between beliefs. This is to say that the existence, strength, or direction of influence between two beliefs may depend on the specific state of a third (in this case, political ideology).

As we draw to a close, our findings invite continued investigation in several respects. Firstly, though we have used the KBS model here to identify asymmetric influence, it is equally amenible to testing pre-existing hypotheses about beliefs which are more (or less) influential than they are subject to influence. Secondly, in the previous chapter we discussed a hypothetical example in which higher-order interactions result in complex intervention dynamics. Though not currently captured by the KBS model, our comparison of the conservative and liberal belief systems suggests this may be a worthwhile extension. Relatedly, although we have only considered interventions targeting belief *states*, prior studies suggest that amplifying certain *associations* between pre-existing beliefs (rather than trying to change them directly) may also be effective (Aggarwal & Hughes, 2026; Brewer et al., 2017). As such, studies considering structural interventions targeting influence relations themselves are a natural progression to the intervention experiments described here.

Finally, while our empirical findings are promising, they would be considerably strengthened by investigation into the extent to which the inferred belief systems reflect within-person effects, as opposed to between-person effects. Disentangling these effects has received increased attention in recent years (Brandt, 2022; Brandt & Morgan, 2022), yet remains difficult without substantial individual-level data. Although not unique to the present study, we consider addressing this limitation critical to establishing the relevance of our empirical findings.

Returning to the primary topic of our investigation, our findings support the existence of asymmetric relations between beliefs about climate change. Crucially, we have also demonstrated that this *matters*, both for studies concerning belief system structure, and those concerning belief system dynamics. When true relations are asymmetric, the use of symmetric models amounts to misspecification. This can cause incorrect assessments of relation existence and strength, which are frequently used to assess belief importance via node centrality indices. Moreover, this has immediate and obvious dynamic implications, since variables deemed non-influential in the symmetric model may simply be non-influentia*ble* in reality.

Chapter 10

Ethics and Data Management

I acknowledge that the thesis adheres to the ethical code and research data management policies of UvA and IvI.[29,30]

Where our findings relate to real phenomena we have clarified the context of interpretation, as is necessary in all research using computational modelling to address questions about the natural world. During the production of this thesis we have been committed to the standards outlined in the Netherlands Code of Conduct for Research Integrity.[31] In addition, generative AI has not been used for any aspect of this thesis.

The following table lists the data used in this thesis (including source code). I confirm that the list is complete and the listed data are sufficient to reproduce the results of the thesis. If a prohibitive non-disclosure agreement is in effect at the time of submission "NDA" is written under "Availability" and "License" for the concerned data items.

Description	Availability	License
Ising Python package, KBS model implementation	DOI: 10.5281/zenodo.21933349	MIT
Repository with experiment code, thesis documents	DOI: 10.5281/zenodo.21933642	MIT
Typst thesis template	DOI: 10.5281/zenodo.21933474	MIT
Longitudinal Panel of Perceptions About Climate Change and Covid survey	Available on request	–

[29]Ethics in Research at UvA: https://student.uva.nl/en/topics/ethics-in-research

[30]Research data management policies at UvA: https://rdm.uva.nl/en

[31]Netherlands Code of Conduct for Research Integrity: https://www.nwo.nl/en/netherlands-code-of-conduct-for-research-integrity

Chapter 11

Climate beliefs dataset

In this chapter we outline the context and construction of the dataset used for model calibration in Chapter 5. We will first describe the broader context and complexities of the dataset underlying this study, before outlining our data validation and cleaning methods in Section 11.2. Finally, in Section 11.3 we detail the construction of the targeted dataset used for model calibration (Chapter 5).

While many empirical studies on belief systems rely on cross-sectional data (Lee et al., 2025; Powell et al., 2023; van Noord et al., 2025), longitudinal data is necessary to capture changes in individuals' internal cognitive states over time. In this study we are fortunate to have had access to data from the **Longitudinal Panel of Perceptions About Climate Change and Covid** (**CCCV**), a representative longitudinal survey comprising six waves of responses from individuals residing in the United States, collected between April 2020 and March 2023 (Constantino et al., 2022). The assessed dimensions include general demographic information (e.g., age, gender, education, and financial status), beliefs, attitudes, and experiences relating to concurrently-salient topics such as COVID-19, climate change, or the 2020 US presidential election, and support for hypothetical policies. In addition to a wide-form table of survey response data, the dataset includes a codebook which specifies, for each survey item, the question text, occurrence in survey waves, and conditional display logic where applicable. At present, the codebook is limited to Waves 1—5, i.e., excluding the sixth (final) wave.

However, the dataset is not without complexities. Survey participation varies, with each participant responding to a (possibly non-contiguous) subset of waves. Figure 11.1 shows the number of participants who have responded to each combination of survey waves, with a minimum count of 1500. While each individual wave has a relatively high response rate (roughly 4000–5000), this rapidly drops off when other waves are

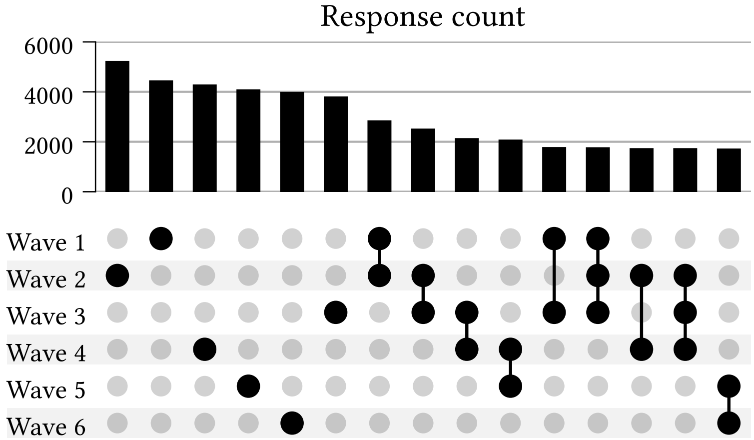


Figure 11.1: Number of repeat participants for different wave combinations in the Longitudinal Panel of Perceptions About Climate Change and Covid dataset, prior to dataset cleaning. Excludes combinations with fewer than 1500 responses.

considered. For instance, only ~2500 individuals responded to Waves 1 and 2, and only ~1900 of these individuals also responded to Wave 3.

Survey content also varies between waves and participants, both with regards to *which* questions are included, and *how* they are presented. Certain questions are displayed only in particular waves, or only to either repeating or new participants. Some questions are shown conditionally based on an individual's responses to prior questions (within the same wave). Others vary according to survey treatment conditions, such that individuals in different treatment groups are presented different variants of a question. This survey logic is not always executed correctly; in some cases participants are shown survey questions incorrectly (e.g., 'new' participants shown questions intended only for 'repeating' participants). We discuss this issue further in Section 11.1.

Question format, text, and response schemas also occasionally change between survey waves. Changes in response schema are less common, however, and typically affect items with categorical responses. With a view to constructing the calibration dataset for our experiments, we note that most response schemas are not binary.

Casting our attention to the survey timing, we examine the survey response dates for each wave (Figure 11.2) and the distribution of interval durations between consecutive-wave responses across individuals (Figure 11.3), i.e., the 'inter-response time'. In Figure 11.2 we observe that the survey waves occur with irregular spacing and duration. Some consecutive pairs of waves are much closer than others. For instance, notice that:

(i) The duration from the start of Wave 1 until the end of Wave 3 is shorter than the duration *between* Waves 5 and 6, and

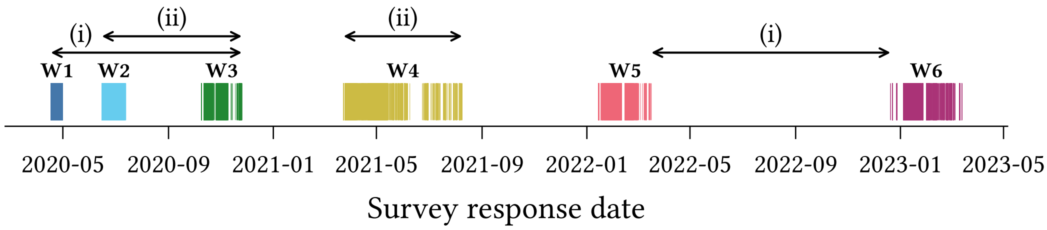


Figure 11.2: Per-wave survey response dates for the Longitudinal Panel of Perceptions About Climate Change and Covid dataset (prior to dataset cleaning). Annotations illustrate differences in intervals spanned by/between survey waves.

(ii) The duration spanned by Wave 4 alone exceeds that of the interval spanned by both Waves 1 and 2.

This poses a potential problem for model calibration, since the Kinetic Belief System model (defined in Chapter 2) operates on the assumption that samples are equispaced.

We also see the irregular spacing reflected in the inter-response time distributions. For each pair of consecutive waves, we observe substantial variation in the time between responses for different participants.

Finally, we note that the time interval spanned by the longitudinal dataset includes several notable events which could reasonably be expected to influence—and confound—the dynamics of beliefs in myriad contexts. These include the COVID-19 pandemic, which arrived in the US only three months prior to the first survey wave (Holshue et al., 2020), the 2020 US Presidential Election which occurred during Wave 3, and the January 6 United States Capitol Attack, which occurred between Waves 3 and 4.

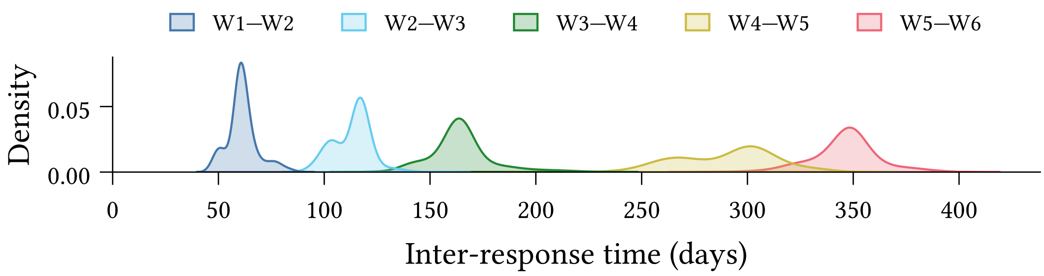


Figure 11.3: Inter-response time distribution between each pair of consecutive waves in the Longitudinal Panel of Perceptions About Climate Change and Covid dataset (prior to dataset cleaning). The inter-response time between two waves is calculated, for individuals present in both waves, as the number of days between their responses.

11.1 Data Validation

The complexity of the CCCV survey, as outlined above, necessitates a rigorous approach to data validation, both to identify errors in the expected schema and to ensure that the data matches our expectations.

We have implemented a general validation pipeline comprising three stages:

1. **Type-level validation:** The set of columns is exactly as expected, and all columns have the correct data type.
2. **Response-value validation:** All *non-null* responses are valid according to the schema specified in the codebook.
3. **Null-value validation:** Responses are null if, and only if, we expect them to be null.

11.1.1 Type-level validation

Type-level validation ensures that the *observed* data schema matches the *prescribed* schema. Since response data types vary between questions (see Table 11.1), the type-checking must be flexible and capable of handling complex data types.

For instance, categorical multiple-choice responses are represented using a `list[Enum]`, where `Enum` is a question-specific enum-type.[32] Type validation for a multiple-choice question thus requires checking: (i) that the response column comprises `list`'s, and (ii) that all list elements belong to the set of values defined by the `Enum`.

Response format	Raw type	Coerced type
Text	`str`	`str`
Single response (ordinal)	`int`	`int`
Single response (categorical)	`int`	`Enum`
Multiple response	`str`	`list[Enum]`
Numeric	`float`	`float | int`

Table 11.1: Data type coercion mapping for different question types in the Longitudinal Panel of Perceptions About Climate Change and Covid survey.

[32]An enum is a type defined by a finite set of allowable values. In our case the values are human-readable strings. For instance, the `dem_urban` survey question, which asks 'What kind of area do you live in?' has responses with data type described by the enum $\{\mathrm{Urban}, \mathrm{Suburban}, \mathrm{Rural}\}$.

11.1.2 Response-value validation

Response-value validation then ensures that all non-null response values are valid according to the survey codebook. For most variables this is straightforward. Numeric and single-response ordinal variables typically have a defined range (e.g., $18 \leq \text{age} \leq 99$, or $1 \leq x \leq 5$ for a 5-point Likert scale variable). Text-entry responses are always considered valid.

Single-response categorical questions have `Enum` type, so type-level validation is sufficient to ensure that the responses do not contain any values outside the set defined by the `Enum`. However, the allowable values occasionally change between survey waves. Hence response-value validation is also required for categorical single-response and multiple-response variables, and in general must handle schema variation between waves.

11.1.3 Null-value validation

Finally, null-value validation ensures that responses are null if, and only if, they are expected to be null. For a question Q, the response of a participant P in wave W is permitted to be null, iff, at least one of the following four conditions is true:

N1. Q is not asked in wave W,

N2. In wave W, Q is only shown to *new* participants, but P is a *repeating* participant (or vice versa),

N3. In wave W, Q is only shown to a treatment group of which P is not a member, or

N4. In wave W, Q is conditionally shown to participants based on their response to a distinct question Q', and P's response to Q' does not satisfy the condition.

Formally we can write the condition for P's response R being null as:

$$\text{null}(R) \iff (\text{N1} \lor \text{N2} \lor \text{N3} \lor \text{N4}) \tag{11.1}$$

Null-value validation then consists in identifying the set of responses $\boldsymbol{R}$ such that for $R \in \boldsymbol{R}$, exactly one of $\text{null}(R)$ and $(\text{N1} \lor \text{N2} \lor \text{N3} \lor \text{N4})$ is true. Validation succeeds if $\boldsymbol{R}$ is empty. Otherwise the members of $\boldsymbol{R}$ are counterexamples to the above conditions.

11.1.4 Implementation

We implement type-level and simple response-value validation using the Pandera Python library for DataFrame validation (Bantilan et al., 2022). We implement manual validation checks for more complex response-value cases, such as questions whose response schema varies between waves.

The null-value validation logic, i.e., the process of testing for contradictions Equation 11.1, is also manually implemented. Conditions N1 and N2 are derived automatically from the survey codebook. Conditions N3 and N4 must be manually specified, as the codebook currently does not have a standardised method for describing conditional display logic.

11.1.5 Validation results

At the time of writing, approximately 25% of the complete survey question set for Waves 1–5 has been validated, including all survey questions which were considered for the targeted calibration dataset (Section 11.3). Since Wave 6 is currently undocumented in the survey codebook we can reasonably validate neither the data schema, nor the presence of null-values. We have therefore not yet validated any of the data from Wave 6, and exclude this wave from the present study.

All type-level and response-value validation checks succeed, providing a strong guarantee that the data schema matches our expectations per the codebook. We do, however, encounter several problems during null-value validation. Sara Constantino's guidance was instrumental in the process of diagnosing these problems.

In some cases, these were due to errors in the codebook itself. These errors are relatively straightforward to identify from the null-value validation results, since they often affect all individuals in a particular wave. For instance, if the codebook specified that a question is not presented in Wave 1, yet all responses are non-null, then this most likely indicates an error in the codebook. While some cases are more subtle, such as where a treatment class is misspecified, when arising due to a codebook error we still expect the contradictions to affect a well-defined subset of the population (in this case, the individuals in a particular treatment class).

In other cases we identify contradictions which are due to errors in the survey process, which caused certain survey questions to be displayed to some individuals in conditions which did not satisfy the specified survey logic. Figure 11.4 shows examples observed in Waves 2 and 3. Some cases were systematic, affecting all individuals until the survey process was updated (*left*); however, in other cases we have not been able to identify the

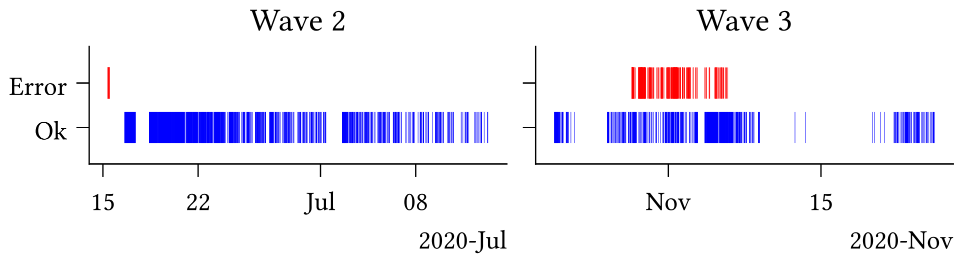


Figure 11.4: Systematic (*left*) and unresolved (*right*) null-value validation errors identified in Waves 2 and 3 of the Longitudinal Panel of Perceptions About Climate Change and Covid survey.

source of the error (*right*). We exclude all responses which fail the null-value validation from further analysis in the data cleaning stage (Section 11.2.2).

The described validation process is primarily concerned with ensuring that the data schema — comprising types and values — aligns with our expectations as per the survey codebook. While comprehensive in this regard, the process does not account for *all* possible forms of errors or inconsistencies. One category which is currently unaccounted for, but is worth mentioning, comprises inconsistencies in the responses from a given individual over multiple waves. For instance, the dataset includes several cases in which individuals presented the question:

"Were you born in the United States?"

respond with 'Yes' in one wave, but 'No' in another. Validating data for inconsistencies such as this requires careful consideration on a question-by-question basis to assess response types for conflicts. We consider this category beyond the scope of this study, describing it here only to illustrate the bounds of the above validation process.

11.2 Normalisation, cleaning, and transformations

We apply pre-processing steps to make the dataset amenible to downstream analysis, which are divided into three categories:

- **Normalisation:** Ensuring the data schema (variable names, types, and values) conform to consistent standards.
- **Cleaning:** Filtering erroneous or otherwise-problematic data.
- **Transformation:** Structural dataset enrichments, including constructed data variables and improved organisation of information.

11.2.1 Normalisation

Variable names in the original dataset are subject to some inconsistency, featuring a mixture of `snake_case`, `CamelCase`, `dot.case`, and `kebab-case`. We therefore first convert the names of all variables specified in the codebook to a standard format, making all names lowercase, and replacing hyphens and periods with underscores.

The dataset also contains several columns which are not described in the codebook, often containing survey metadata, the functions of which are typically indicated using a variable name prefix. For instance, the `Group_` prefix indicates that a column describes treatment group membership. We do not normalise these variable names here, so as to maintain identiability for later transformations (Section 11.2.3).

Secondly, we coerce all variable data types to a standard set. In most cases this is straightforward. Dates and times represented as strings are coerced to timezone-free `datetime` types. Non-floating-point numeric values (which constitute most survey response types) are coerced to 64-bit integers.

Categorical variable responses are originally represented as integeres, which presents three issues. Firstly, it imposes an ordinal structure which does not, in general, make sense for categorical variables. For instance, consider the first four response options to a survey question asking participants about their recent experience with different forms of extreme weather:

1. Severe winter storm, 2. Hurricane, 3. Tornado, 4. Heat wave

One may devise any number of reasonable ordinal interpretations over this set (e.g., 'highest average risk of property damage', or 'annual emergency response cost'), yet such interpretations are context-dependent and not inherent to the categories themselves. Recording categorical responses using numeric types therefore risks spurious interpretations. Second, integer values provide no information about the categories to consumers of the dataset, requiring that they cross-reference manually with the codebook. Thirdly, in some cases the mapping from categories to integers for a given question varies between survey waves. We account for all three of these issues by coercing categorical variables to `Enum` types, which preserve information about the underlying categories, and do not assume an ordinal structure.

Multiple-selection survey items are more complex still. These are represented as strings comprising comma-separated integers, whose values typically refer to non-ordinal categories. We coerce these to `list[Enum]` types, which have all the benefits of the categorical enums discussed above, and enable simpler programmatic analysis (e.g., testing for set membership, determining differences in response sets between waves).

Finally, we perform a series of value updates to ensure consistency between responses to different questions. This consists in replacing all empty string responses with `null` values,[33] and re-mapping integer columns (both numeric and ordinal) such that the minimum value is zero.

11.2.2 Cleaning

Following schema normalisation, we then perform a cleaning stage, which primarily includes filtering out invalid or problematic data. We only filter out entire survey responses, rather than answers to specific survey items. Survey responses may be removed for any one of three reasons.

Firstly, we remove responses where the `participant_id` column is `null`, such that the individuals cannot be traced across survey waves. Secondly, we remove any responses from participants who fail the survey validity check in *any* of their participation waves. Thirdly, we remove any responses which fail either of the response-value or null-value validation checks described in the previous section (Section 11.1). We recognise that these issues are not necessarily problematic in every research context (e.g., cross-sectional studies are unconcerned with tracing an individual's responses across waves), and thus have made each condition optional in the dataset construction code.

11.2.3 Transformation

Finally, we apply several transformation steps with the goal of simplifying manipulation and analysis of the existing dataset, or enriching it with additional information.

The original dataset includes two survey items querying participants' alignment with the two largest US political parties (i.e., the Republicans and the Democrats). The first item asks whether the participant whether they consider themselves as *Republican*, *Democrat*, or *Independent*. Those who respond *Independent* are then presented a follow-up question asking whether they *lean* toward either (or neither) party. We replace these two survey items with a single variable combining the responses, with possible values:

$$\text{Democrat} \prec \text{Leaning Democrat} \prec \text{Independent} \prec \text{Leaning Republican} \prec \text{Republican}$$

[33] Note that this particular value mapping must happen *after* null-value validation (described in Section 11.1), otherwise empty strings will be considered errors.

While this supplements the first item with additional information, we note two potential limitations of this transformation. Firstly, some participants who respond *Republican* or *Democrat* to the first question may more accurately describe themselves as only leaning in a given direction, if they had been aware of the (conditional) second question. Therefore the ‘Leaning’ responses are perhaps better interpreted as ‘Independent with Republican/Democrat tendencies’.

Secondly, our implicit assumption that *Independent* is a midpoint between *Republican* and *Democrat* would misrepresent individuals who consider themselves ‘right-of-Republican’ or ‘left-of-Democrat’. However, the available data does not allow us to distinguish such cases. Furthermore, this issue is also present in the original first question.

Our second transformation concerns survey questions with different variants, which are conditionally displayed based on treatment group membership. The original dataset includes, for each treatment group associated with a given question, a separate response column, as well as a binary indicator column describing group membership. Since group membership is always mutually exclusive in the dataset, we coalesce these various columns, replacing them with singular response and group membership columns. The group membership column contains integer indexes as opposed to the original binary indicators. In some cases, different treatment groups have an associated numerical value. For instance, the `ccSolveX` items query support policies compensating communities affected by climate change, with the treatment group specifying the cost (X) of such a policy to the participant. In these cases we supplement the dataset with an additional column containing the associated numerical value, simplifying downstream analyses.

Finally, we improve the dataset organisation by constructing a database comprising tables for survey item and question metadata and responses, as well as participants themselves (Figure 11.5). The requirement for such as database arose during the construction of the climate beliefs dataset used for model calibration (Section 11.3). The original data, comprising a codebook (a spreadsheet) and an RData file containing rows of participant responses, is not immediately amenible to our task, which requires reasoning about both survey logic and participation in tandem. The constructed database primarily serves to link survey metadata (extracted from the codebook) with survey response data.

Recognising the hierarchical structure of the longitudinal dataset, in which a single survey item may vary in question text or response schema between waves or treatment conditions, we separate the concepts of *survey items* (the underlying concept being assessed) and *survey questions* (the particular mode of assessment). Survey questions are associated with particular waves, participant types (i.e., new or repeating), and treatment conditions, but may be related under a common survey item. *Survey responses*

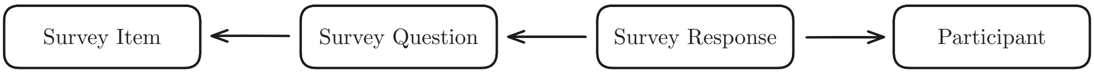


Figure 11.5: The constructed survey database comprises tables for survey items (underlying concepts assessed in the survey), survey questions (the particular mode in which an item is assessed in a given wave or treatment condition), survey responses, and participants. Arrows denote data hierarchy, e.g., each response belongs to a participant, and is associated with a particular survey question.

are associated with a particular question as well as a *survey participant*. We include a separate table for participants which records wave participation as well as the initial wave in which a participant has responded.

The database is stored on disk as a collection of parquet files, which preserve complex data type information, including those in Table 11.1, and are supported in both R (e.g., using Arrow) and Python (e.g., using Pandas or Polars).

11.3 Climate beliefs dataset

In light of the discussed complexities and breadth of content of the CCCV survey, we construct a smaller, targeted dataset of beliefs relating to climate change, which we expect—on theoretical grounds—to exhibit interdependent behaviour. This serves as the calibration dataset for the experiments described in this study. We will refer to this targeted dataset as the **climate beliefs dataset**.

We aim for 7–10 variables in the final set, to keep both the number of model parameters and parameter uncertainty acceptably small. We first filter the survey items to consider only those which assess beliefs (in the inclusive sense described by Galesic et al. (2021), which includes both epistemic positions about states of affairs, as well as views, opinions, and preferences), and which either relate directly to climate change, or which are expected to influence climate-related beliefs (e.g., political alignment). The parameter estimation method used to calibrate the Kinetic Belief System model (outlined in Chapter 3) imposes several additional requirements on the constructed dataset, in particular, the dataset should comprise at least two waves, with approximately equispaced observations per-individual, and with no null values. We additionally constrain the dataset to include three specific variables of interest:

1. **CC Worry**: Level of worry regarding current and future climate change.

2. **CC Others Worry**: Belief about how worried *others* are about climate change.

3. **CC Human**: Belief that climate change is caused by human activities.[34]

Climate-related worry is generally considered an influential factor for other beliefs relating to climate change, including support for climate policy (Bouman et al., 2020; Bumann, 2021; Goldberg et al., 2021; Mead et al., 2012; Whitmarsh et al., 2022), and is expected to be influenced by beliefs others' level of worry, as a descriptive norm (Gavrilets et al., 2024). The third variable serves to provide a more complex view of individuals' beliefs about the nature of climate change; while most Americans believe in the existence of climate change, there is greater variation in beliefs about its causes (Hamilton et al., 2015).

To satisfy these requirements while maximising the number of observations (for model calibration purposes), we limit the climate beliefs dataset to waves 3 and 4, which contain 1693 repeat observations, excluding survey errors. After removing items with no substantial correlations, and small groups with no substantial external correlations, we identify 17 relevant survey items (see Appendix C for the full set of items).

However, this set of items still exceeds our target range of 7–10. We therefore proceed to identify groups of similar or redundant variables which may be removed or combined into interpretable index variables. Note that cross-sectional methods should not be used for this analysis, as they fail to capture temporal relatonships, which are fundamental to the kinetic belief system model (defined in Chapter 2). Instead, we examine the temporal and contemporaneous networks obtained using lag-1 vector autoregression (VAR). These networks are derived from the solution to the following regression problem, described in Epskamp, Waldorp, et al. (2018):

$$\begin{aligned} \boldsymbol{y}^t &= \boldsymbol{B}\boldsymbol{y}^{t-1} + \varepsilon^t \\ \varepsilon^t &\sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Theta}) \end{aligned} \tag{11.2}$$

Where $\boldsymbol{y}^t$ is the vector of observed variable values for each individual at time t. The *temporal network* matrix $\boldsymbol{B}$ comprises linear next-state predictors, while the *contemporaneous network* matrix $\boldsymbol{K} = \Theta^{-1}$ describes the pairwise conditional correlations which remain after accounting for temporal effects. The entries of $\boldsymbol{K}$ are analogous to partial correlations, constituting the conditional correlation between X and Y given the remaining variables.

[34]In the CCCV survey, the item `CC Human` has four possible (categorical) responses, reflecting all combinations of 'human activities' and 'natural causes' as the causes of climate change. We re-map these values to 'human-caused' and 'not human-caused', such that the variable is instead binary, and thus both amenible to the analysis described below and interpretable in the kinetic belief system model in Chapter 4.

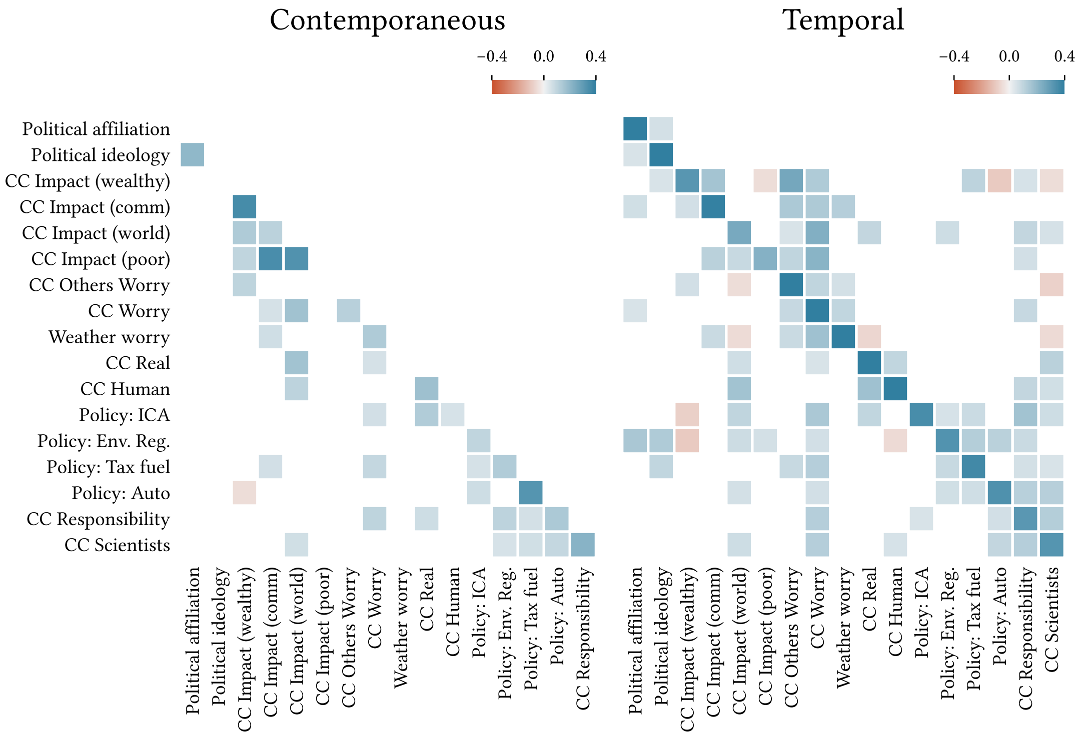


Figure 11.6: Contemporaneous (left) and temporal (right) network matrices for the filtered candidate set. Entries in row i of the temporal matrix are the regression predictors for the subsequent value of variable X_i. Rows and columns are ordered using hierarchical clustering on the temporal matrix: $d_{\boldsymbol{K}}(X, Y) = 1 - |\boldsymbol{K}|$.

Figure 11.6 displays the contemporaneous and temporal networks for the filtered set of 17 candidate variables, with rows and columns ordered using hierarchical clustering on the temporal matrix to highlight groups of related items.[35] Values with magnitude less than 0.05 are excluded, as is the (symmetric) upper triangle of the contemporaneous network. The values in each row of the temporal matrix are the regression predictors for that row's variable. Therefore we interpret row i as describing the variables which *influence* item i, and column i as describing the variables which are influenced *by* item i.

We note three apparent clusters of variables: (i) political views, (ii) beliefs about climate change impacts, and (iii) positions on climate policy, and other beliefs relating to climate action. In the first case, political affiliation (partisan identity) and ideology (conservative $\longleftrightarrow$ liberal) cluster together in both networks, yet appear to have minimal external associations. We retain this group on theoretical grounds, since political beliefs are often key determinants for climate-related beliefs and policy positions (Bumann, 2021; Palm et al., 2017). Secondly, beliefs regarding the impact of climate change on different population groups, `CC Impact (X)`, exhibit strong internal contemporaneous associations, relatively weaker temporal associations. We choose here to combine these

[35] We have re-coded the survey items such that the matrix values are predominantly positive.

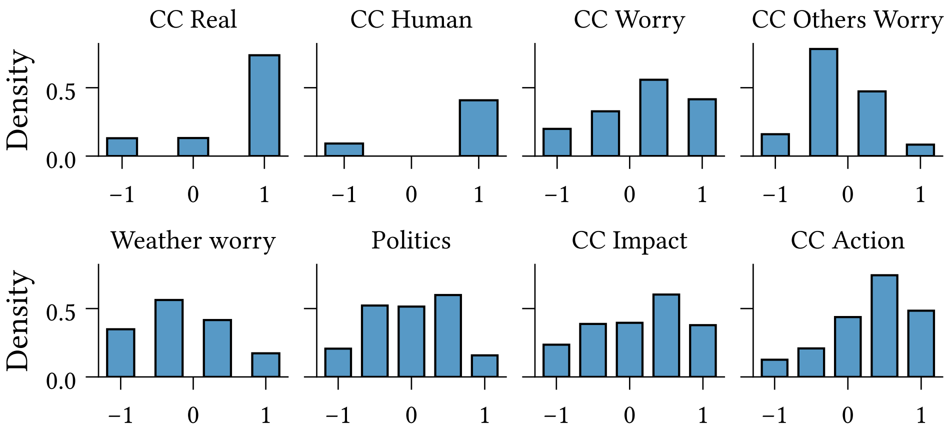


Figure 11.7: Marginal distribution for each of the eight variables in the climate beliefs dataset (Table 11.2).

into an index variable, although an alternative strategy would be to retain only `CC Impact (World)` and/or `CC Impact (Wealthy)`, noting that these variables each have several (and distinct) temporal associations. Finally, the right-most six variables form clear clusters in both networks. Four of these relate directly to climate policy, while the remaining two assess participants views on the importance of individual action in managing climate change, and the role of scientists in guiding the climate change response. We combine these into a third cluster, which we interpret as reflecting an individual's general attitude toward climate action.

We construct the index variables by first re-scaling each constituent item to the interval $[-1, 1]$ and then taking the average. All other variables are also re-scaled in the same way. The final dataset comprises eight variables, described in Table 11.2. Figures 11.7 and Figure 11.8 show the marginal distributions, and the contemporaneous and temporal networks, respectively, for the climate beliefs dataset.

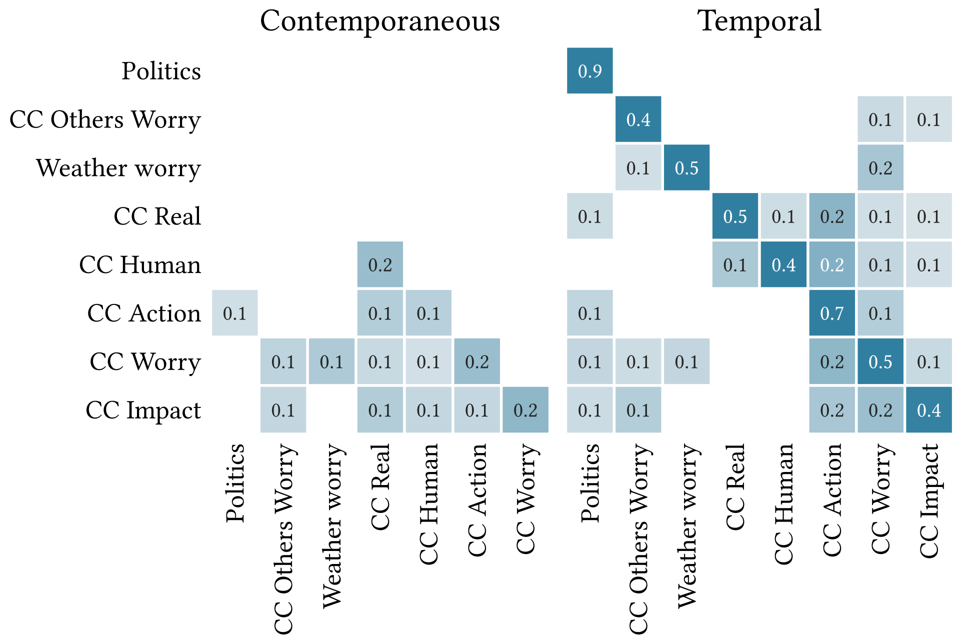


Figure 11.8: Contemporaneous and temporal network matrices for the climate beliefs dataset. Entries in row i of the temporal matrix are the regression predictors for the subsequent value of variable X_i. Rows and columns are ordered using hierarchical clustering on the temporal matrix: $d_{\boldsymbol{K}}(X, Y) = 1 - |\boldsymbol{K}|$.

Item	Index	Interpretation
CC Real	No	Belief that climate change is/is not real.
CC Human	No	Belief that climate is/is not caused (at least partly) by human activities.
CC Worry	No	Level of worry about current and future climate change.
CC Others Worry	No	Belief regarding *others'* level of worry about current/future climate change.
Weather Worry	No	Level of worry about near-term extreme weather events/natural disasters.
Politics	Yes	Political views, a combination of political alignment and ideology.
CC Impact	Yes	Belief about current climate change impacts' severity, in general.
CC Action	Yes	General attitude toward action on climate change.

Table 11.2: Variables included in the climate beliefs dataset. Index variables are constructed by taking the average of their constituent columns, after re-scaling to the interval $[-1, 1]$.

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Appendix A

Derivations

A.1 Possibility of internal consistency

Conjecture 1.1.1

Internal consistency in a calibrated Kinetic Belief System model is possible to achieve, if, and only if, it is possible to re-code the calibration dataset variables such that there are no negative edges in the resulting model.

♡

Proof. First, we note the following fact, which is simply the definition of internal consistency in the Kinetic Belief System model (KBS).

"Internal consistency if possible in a KBS model, iff, there exists a configuration of belief states such that every pair of positively associated beliefs have the same state, and every pair of negatively associated beliefs have opposite states."

As a corrolary, have that internal consistency is *not* possible in a KBS model, iff, for *every* configuration of belief states there exist a pair of beliefs which are in conflict, i.e., which are positively associated but have opposing states, or which are negatively associated but have the same state.

Now, let D be a calibration dataset and $\mathcal{M}$ be a KBS model calibrated to D, with parameters $\boldsymbol{\theta} = \langle \boldsymbol{J}, \boldsymbol{h} \rangle$.

For a pair of beliefs S_i, S_j within the model, where there exists an interaction term J_{ij}, the sign of this interaction term describes whether the variables representing these beliefs in the dataset D are aligned ($J_{ij} > 0$) or misaligned ($J_{ij} < 0$). Call this result **A**.

For any belief, if we were to reverse the scale of the associated dataset variable, this has the effect of multiplying all (inbound and outbound) interaction effects connecting to this belief by -1.

Notice that there is a bijective mapping between possible re-codings of the dataset and possible model configurations. Specifically, if we let the configuration $[+1, +1, ..., +1]$ correspond to the original dataset coding, then for any re-coding which reverses the scales for a subset of the variables, this corresponds uniquely to the configuration in which the associated belief states are also inverted.

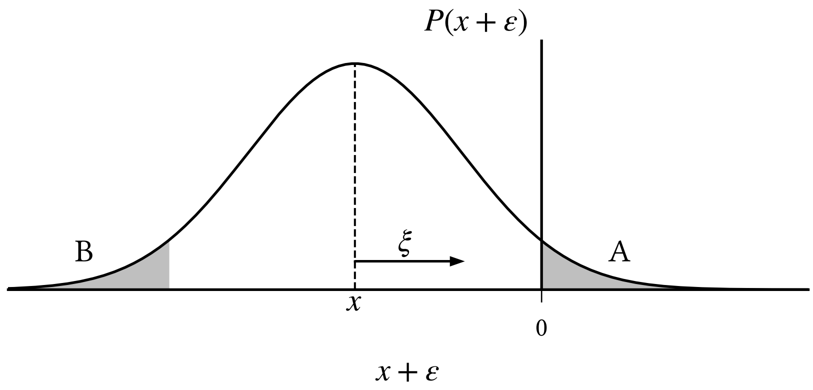
We now prove the main result. Suppose that for a given re-coding of the dataset, all edges in the calibrated model are positive. Then the corresponding configuration under the bijective mapping is an example of internal consistency in the original model. On the other hand, suppose that for every re-coding there exists at least one negative edge. Then it follows that in the original model, all possible belief configurations result in at least one pair of beliefs which are misaligned, and hence internal consistency is not possible. Furthermore, these results apply not only to the original model, but to all models calibrated under re-codings of the dataset, due to the bijection between re-codings and configurations. $\square$

A.2 Smooth thresholding probability

Let b_ξ be a soft thresholding function for $\xi \in \mathbb{R}_{>0}$, and let $x \in \mathbb{R}$. We now prove the statement from Equation 3.2, which states that b_ξ maps x to $+1$ with probability described in terms of the standard normal cumulative distribution function:

$$P(x \mapsto +1) = \Phi\left(\frac{x}{\xi}\right) \tag{1}$$

Recall the following diagram from Chapter 3:



In this example, $x < 0$, but the following argument holds in general. Notice that in this diagram, x is mapped to $+1$ if the sampled noise term, $\varepsilon \sim \mathcal{N}(0, \xi)$, is such that $x + \varepsilon$ is in region 'A', i.e.,

$$P(x \mapsto +1) = P(x + \varepsilon > 0) \tag{2}$$

Or equivalently,

$$P(x \mapsto +1) = P(\varepsilon > -x) \tag{3}$$

Using the symmetry of the Gaussian distribution, this reduces to the standard normal cumulative distribution, as desired:

$$\begin{aligned} P(x \mapsto +1) &= P(\varepsilon > -x) \qquad (4)\\ &= P(\varepsilon < x) \qquad (5)\\ &= \Phi\left(\frac{x}{\xi}\right) \qquad (6) \end{aligned}$$

A.3 Parameter estimation

Let D be a dataset comprising $T \in \mathbb{N}$ observations for each of $M \in \mathbb{N}$ individuals, where each observation measures the (not necessarily binary) state of $N \in \mathbb{N}$ beliefs, and let b_ξ be a soft thresholding function, as defined in Equation 3.2, for $\xi \in \mathbb{R}_{>0}$. We assume that observations from different individuals are independent.

A.3.1 Single-belief transition probability

Here, we show that the transition probability for a belief S_i, as defined in Chapter 2 in terms of the activation probability and the logistic function (Equation 2.5), has the following equivalent form:

$$P\big(\sigma_i^{t+1} = s^{t+1} \mid \boldsymbol{\sigma}^t = \boldsymbol{s}^t\big) = \frac{\exp\big(s^{t+1} \cdot h_i^{\text{eff}}(\boldsymbol{s}^t)\big)}{2\cosh h_i^{\text{eff}}(\boldsymbol{s}^t)} \tag{7}$$

Recall that the conditional probability that S_i takes on the value $+1$ given the previous state is defined as:

$$P\big(\sigma_i^{t+1} = +1 \mid \boldsymbol{\sigma}^t = \boldsymbol{s}^t\big) = \text{logistic}\big(2h_i^{\text{eff}}(\boldsymbol{s}^t)\big) \tag{8}$$

where $h_i^{\text{eff}}(\boldsymbol{s})$ is the effective baseline activation as defined in Equation 2.4. To derive the explicit form of $P\big(\sigma_i^{t+1} \mid \boldsymbol{\sigma}^t\big)$, we first rewrite Equation 8 equivalently as:

$$\begin{aligned} P\big(\sigma_i^{t+1} = +1 \mid \boldsymbol{\sigma}^t = \boldsymbol{s}^t\big) &= \frac{1}{1 + \exp\big(-2h_i^{\text{eff}}(\boldsymbol{s}^t)\big)} && (9) \\ &= \frac{\exp\big(h_i^{\text{eff}}(\boldsymbol{s}^t)\big)}{\exp\big(h_i^{\text{eff}}(\boldsymbol{s}^t)\big) + \exp\big(-h_i^{\text{eff}}(\boldsymbol{s}^t)\big)} && (10) \end{aligned}$$

from which we obtain the complimentary probability as:

$$P\big(\sigma_i^{t+1} = -1 \mid \boldsymbol{\sigma}^t = \boldsymbol{s}^t\big) = \frac{\exp\big(-h_i^{\text{eff}}(\boldsymbol{s}^t)\big)}{\exp\big(h_i^{\text{eff}}(\boldsymbol{s}^t)\big) + \exp\big(-h_i^{\text{eff}}(\boldsymbol{s}^t)\big)} \tag{11}$$

It follows that the general probability distribution $P\big(\sigma_i^{t+1} \mid \boldsymbol{\sigma}^t = \boldsymbol{s}^t\big)$ is given by:

$$\begin{aligned} P\big(\sigma_i^{t+1} = s^{t+1} \mid \boldsymbol{\sigma}^t = \boldsymbol{s}^t\big) &= \frac{\exp\big(s^{t+1} \cdot h_i^{\text{eff}}(\boldsymbol{s}^t)\big)}{\exp\big(h_i^{\text{eff}}(\boldsymbol{s}^t)\big) + \exp\big(-h_i^{\text{eff}}(\boldsymbol{s}^t)\big)} && (12) \\ &= \frac{\exp\big(s^{t+1} \cdot h_i^{\text{eff}}(\boldsymbol{s}^t)\big)}{2\cosh h_i^{\text{eff}}(\boldsymbol{s}^t)} && (13) \end{aligned}$$

A.3.2 Expected log-likelihood

Here, we derive the explicit form of the expected log-likelihood for a Kinetic Belief System model (KBS) with respect to an observational dataset.

The expected log-likelihood for a parameterisation, $\boldsymbol{\theta} \in \mathbb{R}^p$, of the KBS model, with respect to D, is given by:

$$\begin{aligned}
\mathcal{L}_D(\boldsymbol{\theta}) &:= \mathbb{E}\Big[\log P\big(b_\xi(D) \mid \boldsymbol{\theta}\big)\Big] && (14)\\
&= \sum_{m=1}^{M} \mathbb{E}\Big[\log P\Big(\boldsymbol{\sigma}^1_{(m)}, ..., \boldsymbol{\sigma}^T_{(m)}\Big)\Big] && (15)\\
&= \sum_{m=1}^{M} \sum_{t=1}^{T-1} \mathbb{E}\Big[\log P\Big(\boldsymbol{\sigma}^{t+1}_{(m)} \mid \boldsymbol{\sigma}^t_{(m)}\Big)\Big] && (16)
\end{aligned}$$

where we use our assumptions regarding the independence of individuals in the dataset D and the KBS model's Markov assumption, as well as the linearity of the expectation. Substituting the conditional transition probability from Equation 2.6, we may rewrite this in terms of the transition probabilities for individual beliefs, as:

$$\mathcal{L}_D(\boldsymbol{\theta}) = \sum_{m=1}^{M} \sum_{t=1}^{T-1} \sum_{i=1}^{N} \mathbb{E}\Big[\log P\Big(\sigma^{t+1}_{(m),i} \mid \boldsymbol{\sigma}^t_{(m)}\Big)\Big] \tag{17}$$

Using the alternative form for the conditional probability in Equation 7, we may expand Equation 17 as follows:

$$\begin{aligned}
\mathcal{L}_D(\boldsymbol{\theta}) &= \sum_{m=1}^{M} \sum_{t=1}^{T-1} \sum_{i=1}^{N} \mathbb{E}\left[\log\left(\frac{\exp\Big(\sigma^{t+1}_{(m),i}\Big)}{2\cosh h^{\text{eff}}_i\Big(\boldsymbol{\sigma}^t_{(m)}\Big)}\right)\right] && (18)\\
&= \sum_{m=1}^{M} \sum_{t=1}^{T-1} \sum_{i=1}^{N} \mathbb{E}\Big[\sigma^{t+1}_{(m),i} \cdot h^{\text{eff}}_i\Big(\boldsymbol{\sigma}^t_{(m)}\Big) - \log 2\cosh h^{\text{eff}}_i\Big(\boldsymbol{\sigma}^t_{(m)}\Big)\Big] && (19)\\
&= \sum_{m=1}^{M} \sum_{t=1}^{T-1} \sum_{i=1}^{N} \mathbb{E}\Big[\sigma^{t+1}_{(m),i} \cdot h^{\text{eff}}_i\Big(\boldsymbol{\sigma}^t_{(m)}\Big)\Big] - Z(\boldsymbol{\theta}; D) && (20)
\end{aligned}$$

where $Z(\boldsymbol{\theta}; D)$ is the expected log partition function:

$$Z(\boldsymbol{\theta}; D) = \sum_{m=1}^{M} \sum_{t=1}^{T-1} \sum_{i=1}^{N} \mathbb{E}\Big[\log 2\cosh h^{\text{eff}}_i\Big(\boldsymbol{\sigma}^t_{(m)}\Big)\Big] \tag{21}$$

A.3.3 Partial derivatives of the objective function

We now derive the partial derivatives of the objective function (f in Equation 3.6) with respect to each parameter in the KBS model. Let θ be an arbitrary parameter in a KBS model parameterisation $\boldsymbol{\theta} \in \mathbb{R}^p$.

Taking the partial derivative of $\mathcal{L}_D(\boldsymbol{\theta})$ with respect to θ, using the form specified in Equation 19, we find:

$$\frac{\partial}{\partial\theta}\mathcal{L}_D(\boldsymbol{\theta}) = \sum_{m=1}^{M}\sum_{t=1}^{T-1}\sum_{i=1}^{N}\frac{\partial}{\partial\theta}\mathbb{E}\Big[\sigma_{(m),i}^{t+1}\cdot h_i^{\text{eff}}\big(\boldsymbol{\sigma}_{(m)}^t\big) - \log 2\cosh h_i^{\text{eff}}\big(\boldsymbol{\sigma}_{(m)}^t\big)\Big] \tag{22}$$

$$= \sum_{m=1}^{M}\sum_{t=1}^{T-1}\sum_{i=1}^{N}\mathbb{E}\left[\sigma_{(m),i}^{t+1}\cdot\frac{\partial}{\partial\theta}h_i^{\text{eff}}\big(\boldsymbol{\sigma}_{(m)}^t\big) - \frac{\partial}{\partial\theta}\log 2\cosh h_i^{\text{eff}}\big(\boldsymbol{\sigma}_{(m)}^t\big)\right] \tag{23}$$

$$= \sum_{m=1}^{M}\sum_{t=1}^{T-1}\sum_{i=1}^{N}\mathbb{E}\left[\Big(\sigma_{(m),i}^{t+1} - \tanh\Big[h_i^{\text{eff}}\big(\boldsymbol{\sigma}_{(m)}^t\big)\Big]\Big)\cdot\frac{\partial}{\partial\theta}h_i^{\text{eff}}\big(\boldsymbol{\sigma}_{(m)}^t\big)\right] \tag{24}$$

We now determine the partial derivatives of the effective baseline activation. First, for a baseline activation parameter $\theta = h_i$, we have:

$$\frac{\partial}{\partial h_i}h_k^{\text{eff}}(\boldsymbol{s}) = \begin{cases}1, & \text{if } k = i\\ 0, & \text{otherwise}\end{cases} \tag{25}$$

Similarly, if the KBS model has asymmetric interaction parameters, then for $\theta = J_{ji}$, the corresponding partial derivative of the effective baseline activation $h_k^{\text{eff}}(\boldsymbol{s})$ will be zero if $k \neq i$, and otherwise is given by s_j:

$$\frac{\partial}{\partial J_{ji}}h_k^{\text{eff}}(\boldsymbol{s}) = \begin{cases}s_j, & \text{if } k = i\\ 0, & \text{otherwise}\end{cases} \tag{26}$$

If the KBS model has symmetric interaction parameters, then each interaction parameter, $\theta = J_{ij}$, contributes to the behaviour of both S_i and S_j. Therefore the partial derivative of the effective baseline activation $h_k^{\text{eff}}(\boldsymbol{s})$ is nonzero if $k \notin \{i, j\}$:

$$\frac{\partial}{\partial J_{ij}}h_k^{\text{eff}}(\boldsymbol{s}) = \begin{cases}s_j, & \text{if } k \in \{i, j\}\\ 0, & \text{otherwise}\end{cases} \tag{27}$$

We must also account for the partial derivative of the regularisation term. For an arbitrary parameter θ and regularisation hyperparameters $\lambda \in \mathbb{R}_{\geq 0}, \varepsilon \in \mathbb{R}_{>0}$, this is:

$$\frac{\partial}{\partial\theta}\lambda\sum_{\theta'\in\boldsymbol{\theta}}\sqrt{\theta'^2+\varepsilon} = \lambda\frac{\partial}{\partial\theta}\sqrt{\theta^2+\varepsilon} \tag{28}$$

$$= \frac{\lambda\theta}{\sqrt{\theta^2+\varepsilon}} \tag{29}$$

Finally, the complete partial derivative of the objective function f, with respect to a parameter θ is:

$$\frac{\partial}{\partial \theta} f(D;\boldsymbol{\theta}) = \frac{\partial}{\partial \theta} \mathcal{L}_D(\boldsymbol{\theta}) - \frac{\partial}{\partial \theta} \lambda \sum_{\theta' \in \boldsymbol{\theta}} \sqrt{\theta'^2 + \varepsilon} \tag{30}$$

Substituting the above results, we obtain the following partial derivatives of f with respect to each parameter type:

$$\frac{\partial}{\partial h_i} f(D;\boldsymbol{\theta}) = \sum_{m=1}^{M} \sum_{t=1}^{T-1} \mathbb{E}\Big[\sigma_{(m),i}^{t+1} - \tanh\Big[h_i^{\text{eff}}\Big(\boldsymbol{\sigma}_{(m)}^{t}\Big)\Big]\Big] - \frac{\lambda h_i}{\sqrt{h_i^2 + \varepsilon}} \tag{31}$$

$$\frac{\partial}{\partial J_{ji}} f(D;\boldsymbol{\theta}) = \sum_{m=1}^{M} \sum_{t=1}^{T-1} \mathbb{E}\big[\alpha_{m,t}(i,j)\big] - \frac{\lambda J_{ji}}{\sqrt{J_{ji}^2 + \varepsilon}} \qquad \text{(Asymmetric)} \tag{32}$$

$$\frac{\partial}{\partial J_{ij}} f(D;\boldsymbol{\theta}) = \sum_{m=1}^{M} \sum_{t=1}^{T-1} \mathbb{E}\big[\alpha_{m,t}(i,j) + \alpha_{m,t}(j,i)\big] - \frac{\lambda J_{ij}}{\sqrt{J_{ij}^2 + \varepsilon}} \qquad \text{(Symmetric)} \tag{33}$$

where

$$\alpha_{m,t}(i,j) = \Big(\sigma_{(m),i}^{t+1} - \tanh h_i^{\text{eff}}\Big(\boldsymbol{\sigma}_{(m)}^{t}\Big)\Big) \cdot \sigma_{(m),j}^{t} \tag{34}$$

Appendix B

Additional results

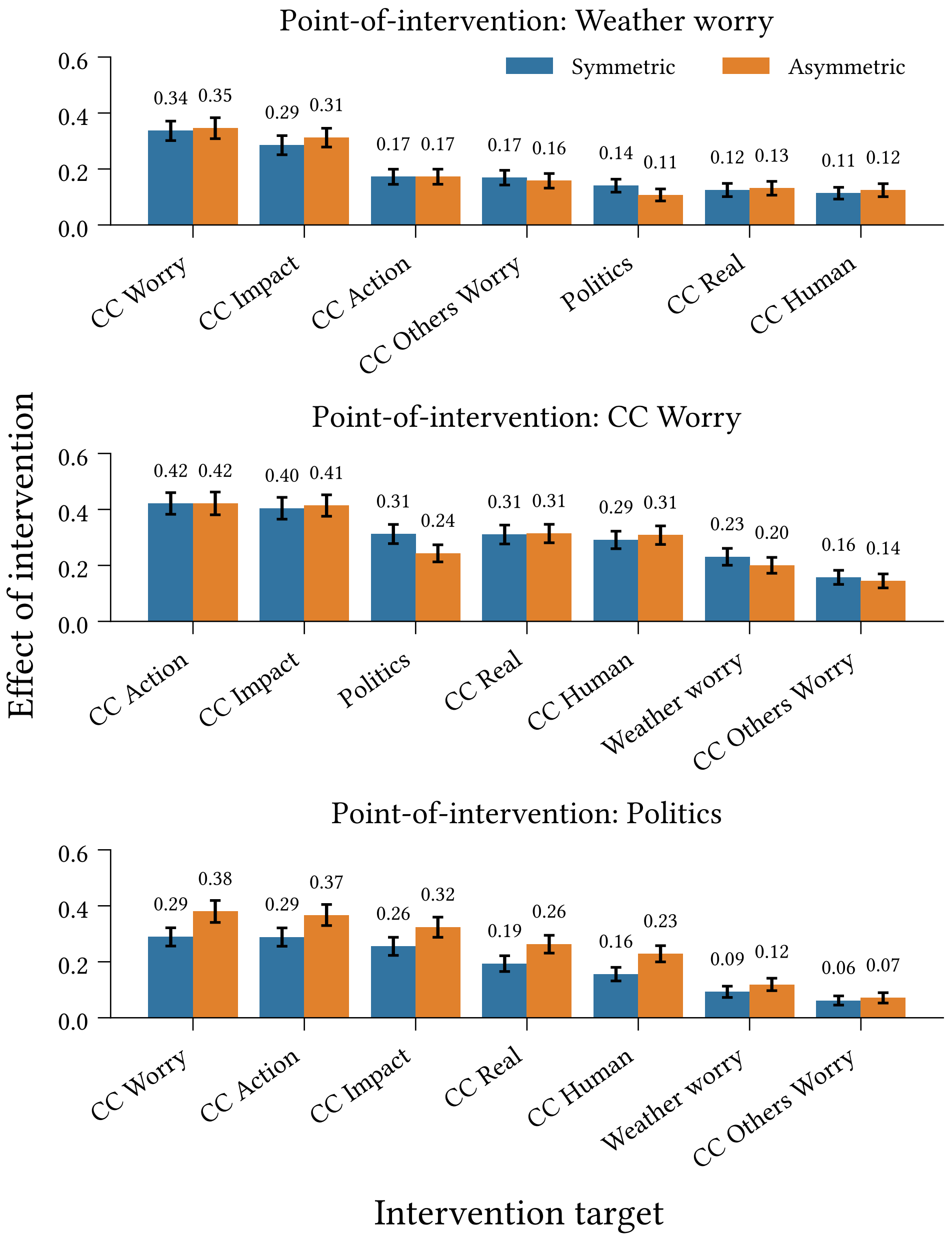


Figure B.1: Outbound effect of intervention (Definition 5.6.1) for interventions targeting `Weather Worry`, `CC Worry`, and `Politics`, in symmetric and asymmetric belief system models calibrated to the climate beliefs dataset. Measurements taken at $t = 10$ (approximately five years in the models' timescale). Intervention strength: $\delta_h = 2.5$. Confidence intervals display 1.96 standard deviations around the mean effect, measured across 500 repeated simulations.

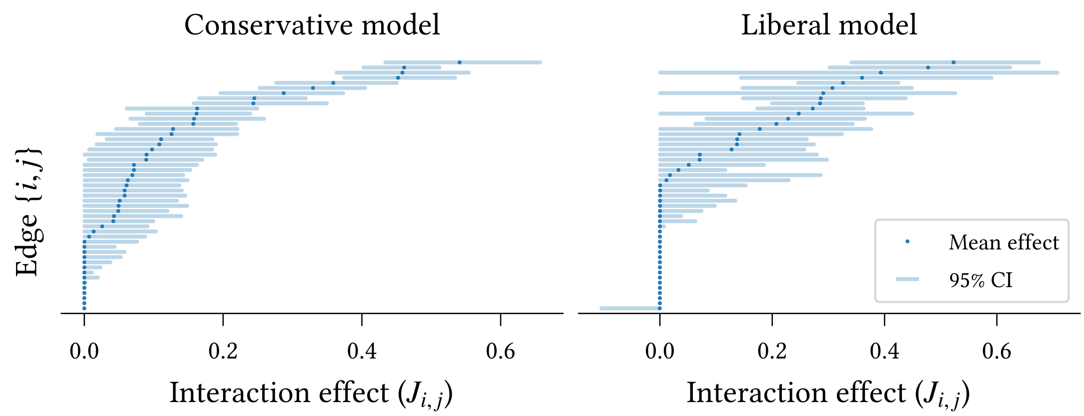


Figure B.2: Edge accuracy in ideology-specific models.

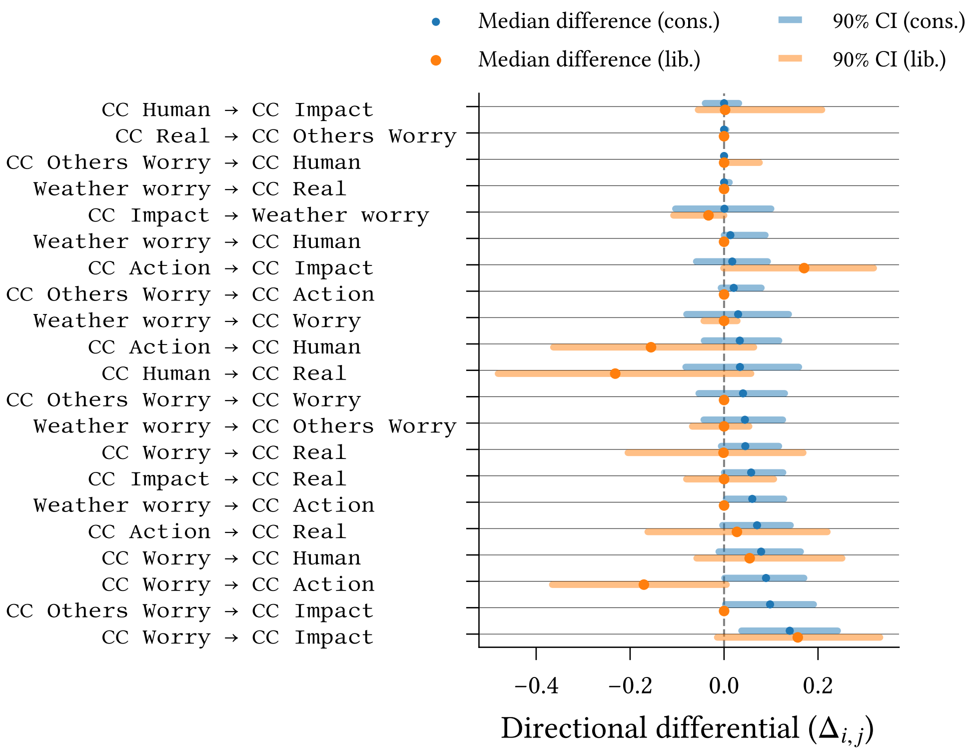


Figure B.3: Directional differentials for ideology-specific-models.

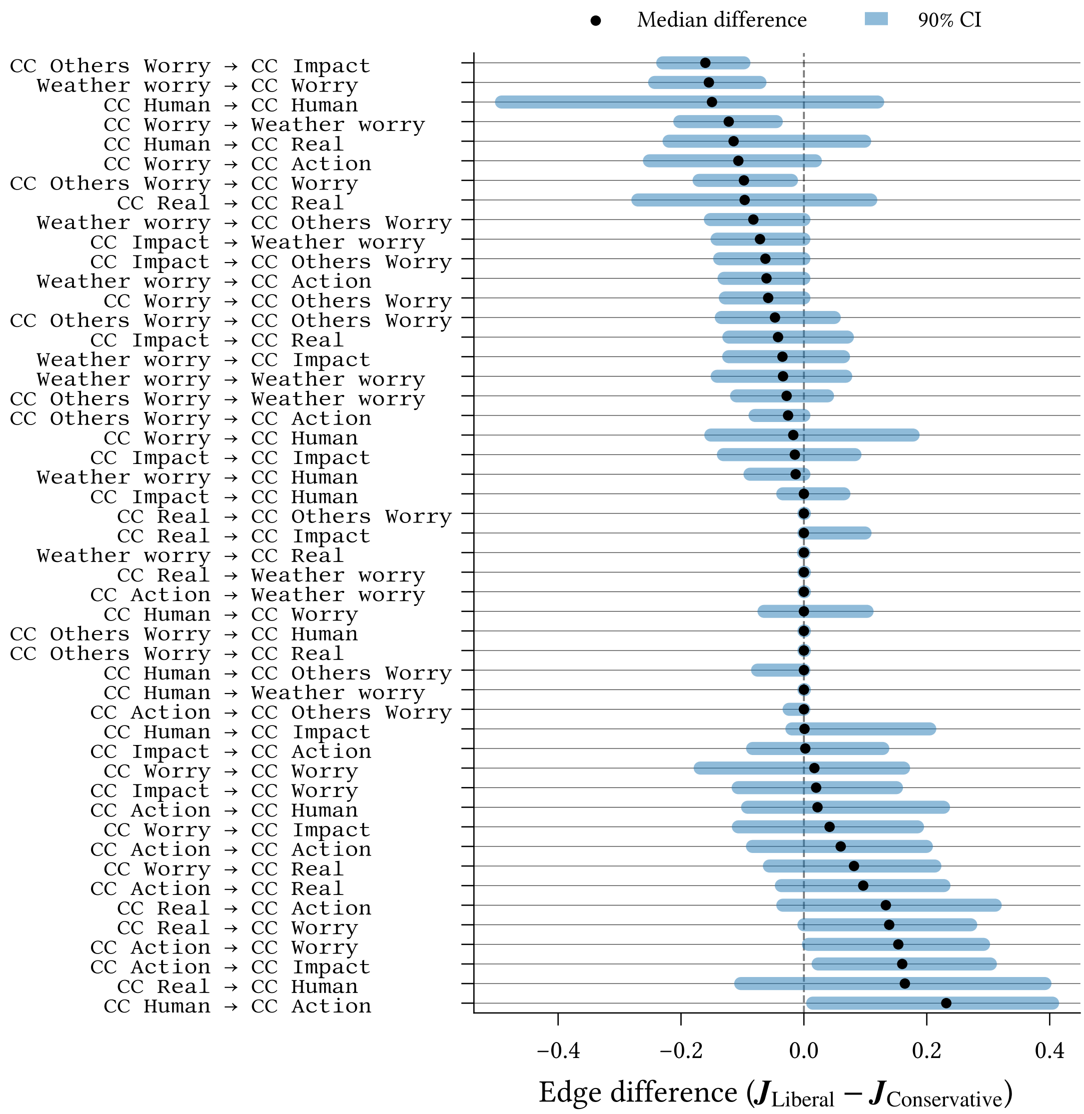


Figure B.4: Differences in interaction effect between ideological models.

Appendix C

Climate beliefs dataset variables

Item	Question text
CC Real	Do you think that climate change is happening?
CC Human	Do you think rising temperatures are a result of human activities, natural causes, or both?
CC Impact X	How much do you think climate change is currently harming X in general? Where X is one of: "the world", "wealthy communities in the United States", "poor communities in the United States", "your local community"
CC Others Worry	How worried do you think most Americans are about global warming/climate change these days?
CC Worry	How worried are you about current and future global warming/climate change?
Weather Worry	How worried are you about an extreme weather event or natural disaster happening to you personally in the next year?
CC Responsibility	It is important that individuals take action on issues of climate change.
CC Scientists	Scientists with appropriate expertise should guide how we respond to climate change
Policy: ICA	Do you favour an international agreement committing the USA and other countries to reduce their carbon emissions?
Policy: Tax fuel	How much do you support/oppose a tax on the production/distribution of carbon-based fuelds?
Policy: Auto	How much do you support/oppose stronger carbon emissions standard for car manufacturers?
Policy: Env. Reg.	Which statement comes closer to your views?
Political affiliation (**)	In politics today, do you consider yourself a:
Political ideology	What is your political ideology?

Table C.1: Complete set of filtered survey items used to construct the climate beliefs dataset (Section 11.3)